Sampling, DT signals, system properties

Answers and worked solutions · Day 02 · Sun 25 Oct · pack D1

5. Practice

From your course files

Sample Test 1 Q1 (sample test, 5 marks). Sampling.

(a) $T_s = 1/F_s = 0.01$ s. Substitute $t = nT_s$: $x(n) = 10e^{-0.0001n}\cos(4\pi n) = 10e^{-0.0001n}$, because $\cos(4\pi n) = 1$ for every integer n (the 200 Hz tone aliases to DC, since $F_s/2 = 50$ Hz). So $x(0) = 10$, $x(1) = 9.9990$, $x(2) = 9.9980$, $x(3) = 9.9970$.

(b) The window $u(n+1) - u(n-2)$ is 1 for $-1 \le n \le 1$, where 2^{-n} gives $2, 1, 0.5$. There is a spike of height 1 at $n = 3$. So $x(n) = \{2, \underset{\uparrow}{1}, 0.5, 0, 1\}$ for $n = -1, \ldots, 3$: stems of height 2, 1, 0.5 at $n = -1, 0, 1$, nothing at $n = 2$, and 1 at $n = 3$. Note $x(2) = 0$.

Sample Test 1 Q2 (sample test, 4 marks). DT Signal Operations.

The underlined 1 is $x(0)$, so the samples sit at $n = -2, \ldots, 3$.

- a. $x(n-3)$ occupies $n = 1..6$ and $x(n+1)$ occupies $n = -3..2$. They overlap only at $n = 1, 2$: at $n = 1$, $x(-2) + x(2) = 8 + 6 = 14$; at $n = 2$, $x(-1) + x(3) = 9 + 1 = 10$. So $x(n-3) + x(n+1) = \{8, 9, 1, \underset{\uparrow}{9}, 14, 10, 1, 9, 6, 1\}$ for $n = -3, \ldots, 6$.
- b. $x(n/2) = \{8, 0, 9, 0, \underset{\uparrow}{1}, 0, 9, 0, 6, 0, 1\}$ for $n = -4, \ldots, 6$.
- c. $x(n)(-1)^n = \{8, -9, \underset{\uparrow}{1}, -9, 6, -1\}$.
- d. $E = 64 + 81 + 1 + 81 + 36 + 1 = 264$. This is an energy signal.

Sample Test 1 Q3 (sample test, 4 marks). DT System Properties.

- a. Linear and causal. The n coefficient makes it time-varying, and it is dynamic.
- b. Linear, time-invariant and causal. $\cos(0.2\pi)$ is a constant.
- c. Time-invariant only. The -1 makes it nonlinear, and $x[n+1]$ makes it non-causal.
- d. Linear and time-invariant. $x[n+1]$ makes it non-causal.
- None of the four is static: each has $y[n-k]$ feedback, so the output has memory.

T1 Q1 (tutorial).

Hint: look for a leftover π. A DT sinusoid is periodic only if $f_0 = \omega_0/2\pi$ is rational.

- a. A CT sinusoid is always periodic: $T_0 = 2\pi/5$ s.
- b. $f_0 = 5/(2\pi)$ is not rational, so not periodic.
- c. $f_0 = 1/(12\pi)$ is not rational, so not periodic.
- d. $f_1 = 1/(16\pi)$ (irrational) and $f_2 = 1/16$. The first factor is not periodic, so $x(n)$ is not periodic.
- e. $f_1 = 1/4$, $f_2 = 1/16$, $f_3 = 1/8$, all rational, so $N_1 = 4$, $N_2 = 16$, $N_3 = 8$ and $N = \mathrm{LCM}(4, 16, 8) = 16$.

T1 Q3 (tutorial).

$x_{\max} = 6.35$ and $x_{\min} = -6.35$, so the range is 12.7.

- a. $L = \dfrac{x_{\max} - x_{\min}}{\Delta} + 1 = \dfrac{12.7}{0.1} + 1 = 128$, and $b \geq \log_2 128 = 7$, so **7 bits**.
- b. $L = 12.7/0.02 + 1 = 636$, and $b \geq \log_2 636 = 9.313$, so **10 bits**.

T1 Q4 (tutorial).

$L = 2^b = 2^8 = 256$ levels over $x_{\max} - x_{\min} = 1$ V, so $\Delta = \dfrac{1}{L-1} = \dfrac{1}{255} \approx$ **0.0039 V**.
Bit rate $= bF_s = 8 \times 20 =$ **160 bits/s**. $F_{\max} = F_s/2 =$ **10 Hz**.

T2 Q1 (tutorial).

Key: S/D = static/dynamic, L/NL = linear/nonlinear, TI/TV = time-invariant/time-varying, C/NC = causal/non-causal, St/Unst = stable/unstable.

- a. **S, NL, TI, C, St.** $x(n) = 0$ gives $y(n) = 1 \neq 0$, so nonlinear. $|y| \leq 1$ always.
- b. **D, L, TI, NC, Unst.** $x(n-d) \mapsto \sum_{k=-\infty}^{n+1} x(k-d) = \sum_{m=-\infty}^{n-d+1} x(m) = y(n-d)$. It uses the future input $x(n+1)$. With $x(n) = u(n)$ (bounded by 1), $y(n) \to \infty$.
- c. **S, L, TV, C, St.** $x(n-d) \mapsto x(n-d)\cos(\omega_0 n) \neq y(n-d) = x(n-d)\cos(\omega_0(n-d))$.
- d. **D, L, TV, NC, St.** $x(n-d) \mapsto x(-n-d+2) \neq y(n-d) = x(-n+d+2)$. It needs future inputs for $n \leq 0$ (e.g. $y(0) = x(2)$).
- e. **S, NL, TI, C, St.** $|a_1 x_1 + a_2 x_2| \neq a_1|x_1| + a_2|x_2|$.
- f. **D, L, TV, NC, St.** $x(n-d) \mapsto x(2n-d) \neq y(n-d) = x(2(n-d))$. It needs future inputs for $n > 0$ (e.g. $y(1) = x(2)$).

T2 Q2 (tutorial).

Since T is time-invariant, delaying pair 2 by one sample gives $x_2(n-1) = \{\underset{\uparrow}{0}, 0, 0, 3\} \xrightarrow{T} y_2(n-1) = \{\underset{\uparrow}{0}, 0, 1, 0, 2\}$. Now $x_3(n) = \frac{1}{3}x_2(n-1)$, so if T were linear the output would have to be $\frac{1}{3}y_2(n-1) = \{\underset{\uparrow}{0}, 0, \frac{1}{3}, 0, \frac{2}{3}\}$. But the observed $y_3(n) = \{1, \underset{\uparrow}{2}, 1\}$ is different. T fails superposition, so it is **nonlinear**.

New questions

N1 (easy) [3]

a. $F_1 = 150$ Hz and $F_2 = 450$ Hz, so the Nyquist rate is **900 Hz**. Since $F_s = 500$ Hz, the 450 Hz tone aliases.

b. $\omega_1 = 2\pi(150/500) = 0.6\pi$ and $\omega_2 = 2\pi(450/500) = 1.8\pi \to 1.8\pi - 2\pi = -0.2\pi$. So

$$x(n) = 4\cos(0.6\pi n) - 2\sin(0.2\pi n).$$

c. Substitute $n = 500t$: $y(t) = 4\cos(300\pi t) - 2\sin(100\pi t)$. The 450 Hz tone reappears at 50 Hz with its sign inverted. ($F_a = 450 - 500 = -50$ Hz.)

N2 (easy) [2]

a. The frequencies are $f_1 = 0.15 = 3/20$, $f_2 = 0.225 = 9/40$ and $f_3 = 1/5$, so $N_1 = 20$, $N_2 = 40$ and $N_3 = 5$. Then $N = \mathrm{LCM}(20, 40, 5) = \mathbf{40}$. A numerical check confirms that no smaller N works.

b. $f_0 = 0.3/(2\pi)$ is irrational, so the signal is **aperiodic**.

N3 (medium) [3]

a. $L = 5/0.001 + 1 = 5001$, and $\log_2 5001 = 12.29$, so $b =$ **13 bits**.

b. $\Delta = 5/(2^{13} - 1) = 5/8191 =$ **0.610 mV**, and $|e_q| \le \Delta/2 =$ **0.305 mV**.

c. The bit rate is $13 \times 8000 =$ **104 kbit/s**. The highest recoverable frequency is $F_s/2 =$ **4 kHz**.

N4 (medium, chained) [4]

The indices are $n = -2, \ldots, 2$, with $x(0) = 3$.

a. $x(n-2)$ occupies $n = 0..4$ and $x(n+1)$ occupies $n = -3..1$. Subtract column by column:

n	-3	-2	-1	**0**	**1**	**2**	**3**	**4**
$x(n-2)$				2	-1	3	1	4
$x(n+1)$	2	-1	3	1	4			
result	-2	1	-3	1	-5	3	1	4

b. Substitute directly. The index $-2n+1$ is odd for every integer n, so only $x(\pm 1)$ and $x(3)$ can appear. $n=0$ gives $x(1)=1$; $n=1$ gives $x(-1)=-1$; $n=-1$ gives $x(3)=0$. So $x(-2n+1)=\{\underset{\uparrow}{1},-1\}$, with values only at $n=0,1$.

c. $x_e(\pm 2)=\frac{1}{2}(2+4)=3$, $x_e(\pm 1)=\frac{1}{2}(-1+1)=0$ and $x_e(0)=3$. So $x_e(n)=\{3,0,\underset{\uparrow}{3},0,3\}$. This is even-symmetric, as expected.

d. $E=4+1+9+1+16=$ **31**. It is finite and non-zero, so this is an **energy signal** with $P=0$.

N5 (medium) [3]

a. **Static, nonlinear, TI, causal, stable.** It is nonlinear because $x=0$ gives $y=1\neq 0$. It is stable because $|y|<e^{M_x}$.

b. **Dynamic, linear, TI, non-causal, stable.** Substituting $x(k-d)$ and setting $m=k-d$ gives $\sum_{m=n-d-2}^{n-d+2}x(m)=y(n-d)$, so it is TI. It is non-causal because $y(n)$ uses $x(n+1)$ and $x(n+2)$. It is stable because $|y|\leq 5M_x$.

c. **Dynamic, linear, time-varying, non-causal, stable.**

- Dynamic: $y(2)=x(4)$.
- Time-varying: delaying the input gives $x(n^2-d)$, but $y(n-d)=x((n-d)^2)$. Concrete case: with $x=\delta(n)$, $y=\delta(n)$; with $x=\delta(n-1)$, $y=\delta(n-1)+\delta(n+1)$, which is not $\delta(n-1)$.
- Non-causal: $y(2)$ needs the future input $x(4)$.
- Stable: $|y|\leq M_x$.

N6 (hard, chained) [5]

a. From Example 4, $r_{xx}(l)=\{-1,0,\underset{\uparrow}{6},0,-1\}$ for $l=-2,\ldots,2$, and zero elsewhere.

b. $y(n)=\{0.5,\ 1,\ \ -0.5\}$ at $n=3,4,5$. Shift x right by l, multiply and add:

- $l=1$: the overlap is $y(3)x(2)=0.5(-1)=-0.5$.

- $l = 3$: full overlap,
  $y(3)x(0) + y(4)x(1) + y(5)x(2) = 0.5(1) + 1(2) + (-0.5)(-1) = 3$.
- $l = 5$: $y(5)x(0) = -0.5$.
- Every other lag gives 0 (for example, at $l = 2$: $y(3)x(1) + y(4)x(2) = 1 - 1 = 0$).

So $r_{yx}(l) = -0.5,\ 0,\ 3,\ 0,\ -0.5$ for $l = 1, \ldots, 5$, and zero elsewhere.

c. The peak magnitude is at $\hat{d} = \mathbf{3} = D$. Also $0.5\, r_{xx}(l - 3)$ gives $-0.5, 0, 3, 0, -0.5$ at $l = 1..5$, which matches. (With the other order, $r_{xy}(l) = r_{yx}(-l)$ peaks at $l = -3$. This is why you must state your definition.)

d. From $d = 2F_s r/c$: $r = dc/(2F_s) = 3(3 \times 10^8)/(2 \times 10^6) = \mathbf{450\ m}$.

6. Quick recall check

Question

1. No. $f_0 = 0.7/2\pi$ is irrational.
2. At 300 Hz, because $1300 - 1000 = 300$ Hz.
3. 9 bits, because $2^8 = 256 < 300 \leq 512 = 2^9$.
4. It is not linear ($x = 0$ gives $y = 2$). It is time-invariant (there is no explicit n).
5. The signal energy E_x.

LCCDE and convolution

Answers and worked solutions · Day 04 · Tue 27 Oct · pack D2

5. Practice

From your course files

T2 Q4 (tutorial).

> *Hint: simplify $h_2 - h_3 * h_4$ sample by sample before convolving.*
>
> **(a)** Cascade gives convolution and parallel paths add (the lower path enters the junction with a minus sign):
>
> $$h(n) = h_1(n) * \big(h_2(n) - h_3(n) * h_4(n)\big).$$
>
> **(b)** Using $x(n) * \delta(n-d) = x(n-d)$:
>
> $$h(n) = \{\underset{\uparrow}{2}, 1, 2\} * \big((n+1)u(n) - (n-1)u(n-2)\big) = \{\underset{\uparrow}{2}, 1, 2\} * \big(u(n) + u(n-1)\big)$$
>
> $$= 2\big(u(n)+u(n-1)\big) + \big(u(n-1)+u(n-2)\big) + 2\big(u(n-2)+u(n-3)\big) = \mathbf{2u(n) + 3u(n-1) + 3u(n-2) + 2u(n-3)}$$
>
> which is $\{2, 5, 8, 10, 10, \ldots\}$. (i) **Causal**, since $h(n) = 0$ for $n < 0$. (ii) **Unstable**, since $\sum_k |h(k)| = \infty$ ($h(n) \to 10$). See also Example 4.

T3 Q2 (tutorial).

> *Hint: substitute x directly. There is no feedback.*
>
> (a) There are no past outputs in the equation, so there is no feedback: **the characteristic modes are zero** (there are none).
>
> (b) By direct substitution:
>
> $$y[n] = \tfrac{1}{4}\big(5^{-n}u[n] + 2\cdot 5^{-(n-1)}u[n-1] + 5^{-(n-2)}u[n-2]\big) = \mathbf{\tfrac{1}{4}5^{-n}\big(u[n] + 10u[n-1] + 25u[n-2]\big)}$$

T3 Q4 (tutorial).

> (a) $\lambda^2 + \lambda + \frac{1}{4} = (\lambda + \frac{1}{2})^2 = 0$, a repeated root $\lambda = -\frac{1}{2}$. The modes are $(-2)^{-n}$ and $n(-2)^{-n}$, so $y_h[n] = C_1(-2)^{-n} + C_2 n(-2)^{-n}$.
>
> (b) Try $y_p[n] = K2^{-n}$: $K(1+2+1) = 1$, so $K = 0.25$. The ICs enter **only** through the recursion:
>
> $$y[0] = x[0] - y[-1] - \tfrac{1}{4}y[-2] = 1 - (-1) - \tfrac{1}{4}(2) = 1.5, \qquad y[1] = x[1] - y[0] - \tfrac{1}{4}y[-1] = 0.5 - 1.5 - \tfrac{1}{4}(-1) = -0.75.$$

Fit $y[n] = C_1(-2)^{-n} + C_2 n(-2)^{-n} + 0.25 \cdot 2^{-n}$: $n = 0$ gives $C_1 = 1.25$; $n = 1$ gives $-0.625 - 0.5C_2 + 0.125 = -0.75$, so $C_2 = 0.5$.

$$\mathbf{y[n] = 1.25(-2)^{-n} + 0.5\,n(-2)^{-n} + 0.25 \cdot 2^{-n}, \quad n \geq 0}$$

Note: the printed solution's line for $y[1]$ reads "$0.5 - 2.5 - \frac{1}{4}(-1)$". It should be $0.5 - 1.5 - \frac{1}{4}(-1)$, because $y[0] = 1.5$. The printed result, -0.75, is correct, and so is the final answer (verified in Python to $n = 7$).

T3 Q5 (tutorial).

(a) No feedback, so substitute $x[n] = \delta[n]$:
$h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-4] = \{\underset{\uparrow}{\mathbf{1}}, \mathbf{2}, \mathbf{3}, \mathbf{0}, \mathbf{4}\}$. Don't skip the zero at $n = 3$.

(b) $\lambda^2 + 0.2\lambda - 0.08 = 0$ gives $\lambda = 0.2, -0.4$, so $h[n] = C_1(0.2)^n + C_2(-0.4)^n$ for $n \geq 0$. Recursion: $h[0] = 1 = C_1 + C_2$ and $h[1] = -0.2h[0] = -0.2 = 0.2C_1 - 0.4C_2$. So $C_1 = \frac{1}{3}$, $C_2 = \frac{2}{3}$ and $\mathbf{h[n] = \left(\frac{1}{3}(0.2)^n + \frac{2}{3}(-0.4)^n\right)u[n]}$.

(c) $\lambda^2 - 2\lambda + 1 = 0$ gives $\lambda = 1$ repeated, so the modes are 1 and n: $h[n] = C_1 + C_2 n$. $h[0] = C_1 = 1$ and $h[1] = C_1 + C_2 = 2$, so $\mathbf{h[n] = (1 + n)u[n]}$.

T3 Q6 (tutorial).

(a) Long multiplication (Playbook E). x starts at $n = 0$ and h at $n = -2$, so y starts at $n = -2$:

$$y[n] = \{2 \;\; -3 \;\; \underline{-3} \;\; -5 \;\; 6 \;\; 5 \;\; 10 \;\; 12\}$$

(b)

$y[n] = \sum_k h[k]x[n-k] = \sum_{k=0}^{n} 2^{-k}3^{n-k} = 3^n \sum_{k=0}^{n} 6^{-k} = \mathbf{3^n\left(\frac{1 - 6^{-(n+1)}}{1 - 6^{-1}}\right)} = ($

.

(c) The same with $e^{j\omega_0}$ in place of 3: $\mathbf{y[n] = e^{j\omega_0 n}\left(\frac{1 - (2e^{j\omega_0})^{-(n+1)}}{1 - (2e^{j\omega_0})^{-1}}\right)u[n]}$. Further simplification is not necessary.

(d) $\cos(\omega_0 n) = \mathcal{R}\{e^{j\omega_0 n}\}$ and h is real, so the output is the real part of (c):
$y[n] = \mathcal{R}\left\{e^{j\omega_0 n}\left(\frac{1 - (2e^{j\omega_0})^{-(n+1)}}{1 - (2e^{j\omega_0})^{-1}}\right)\right\}u[n]$. (Alternatively write $\cos(\omega_0 n) = \frac{1}{2}(e^{j\omega_0 n} + e^{-j\omega_0 n})$ and use (c) twice.)

New questions

Q1 (easy) [2].

The output starts at $n = -1 + 0 = -1$ and has $3 + 4 - 1 = 6$ samples.

	$n = -1$	**0**	**1**	**2**	**3**	**4**
$1 \cdot h$	3	0	1	2		
$2 \cdot h$		6	0	2	4	
$-1 \cdot h$			-3	0	-1	-2
y	3	6	-2	4	3	-2

$$\mathbf{y(n)} = \{\mathbf{3}, \underset{\uparrow}{\mathbf{6}}, \mathbf{-2}, \mathbf{4}, \mathbf{3}, \mathbf{-2}\}$$

Check: $\sum y = 12 = (2)(6)$.

Q2 [3].

$y(n) = \sum_k x(k)h(n-k) = \sum_k 0.5^{n-k}$ over $0 \le k \le 3$ (from x) and $k \le n$ (from $u(n-k)$).

- $n < 0$: there is no overlap, so $y = 0$.
- $0 \le n \le 3$: $y(n) = 0.5^n \sum_{k=0}^{n} 2^k = 0.5^n(2^{n+1} - 1) = 2 - 0.5^n$.
- $n \ge 4$: $y(n) = 0.5^n \sum_{k=0}^{3} 2^k = 15(0.5)^n$.

$$\mathbf{y(n)} = \begin{cases} 0 & n < 0 \\ 2 - 0.5^n & 0 \le n \le 3 \\ 15(0.5)^n & n \ge 4 \end{cases}$$

Values: $1, 1.5, 1.75, 1.875, 0.9375, \ldots$ (Python-checked). At the joint, $n = 4$ gives $15/16 = 0.9375$, which matches $y(3) \cdot 0.5 = 0.9375$ (the pulse has ended, so the output just decays).

Q3 [3].

The mode is $\lambda = 0.5$. With $x = \delta$ and zero ICs: $h(0) = 1$ and $h(1) = 0.5(1) - 1 = -0.5$. For $n \ge 2$, $h(n) = 0.5h(n-1)$.

Here $m = k = 1$, so the mode form holds only from $n = 1$: $h(n) = C(0.5)^n$ for $n \ge 1$. From $h(1)$, $C = -1$. Write $h(n) = D\delta(n) - 0.5^n u(n)$. Then $n = 0$ gives $D - 1 = 1$, so $D = 2$.

$$\mathbf{h(n) = 2\delta(n) - (0.5)^n u(n) = \{\underset{\uparrow}{1}, -0.5, -0.25, -0.125, \ldots\}}$$

It is causal. It is IIR. It is stable: $\sum |h| = 1 + \sum_{n\geq 1} 0.5^n = 2 < \infty$ (equivalently $|\lambda| = 0.5 < 1$).

Alternative: by linearity and time invariance, $h = g(n) - g(n-1)$ with $g = 0.5^n u(n)$. This gives the same sequence.

Q4 [4].

1. $\lambda^2 - \lambda + 0.25 = (\lambda - 0.5)^2 = 0$, a repeated root, so $y_h = C_1 0.5^n + C_2 n 0.5^n$.
2. Try $y_p = K3^{-n}$. Since $1/3$ is not a root this is fine: $K(1 - 3 + 0.25 \cdot 9) = 1$, so $0.25K = 1$ and $K = 4$.
3. Recursion: $y(0) = 1$. $y(1) = \frac{1}{3} + 1 = \frac{4}{3}$.
4. $n = 0$: $C_1 + 4 = 1$, so $C_1 = -3$. $n = 1$: $0.5(-3) + 0.5C_2 + \frac{4}{3} = \frac{4}{3}$, so $C_2 = 3$.

$$\mathbf{y(n) = (-3(0.5)^n + 3n(0.5)^n + 4(1/3)^n)u(n)}$$

Check $n = 2$: the recursion gives $\frac{1}{9} + \frac{4}{3} - \frac{1}{4} = 1.19444$, and the formula gives $-0.75 + 1.5 + 0.4444 = 1.19444$.

Q5 [3].

a. $\lambda^2 - \lambda + 0.5 = 0$, so $\lambda = \frac{1 \pm j}{2} = \frac{1}{\sqrt{2}} e^{\pm j\pi/4}$. The modes are $(1/\sqrt{2})^n \cos(\pi n/4)$ and $(1/\sqrt{2})^n \sin(\pi n/4)$.

b. Recursion: $h(0) = 1$ and $h(1) = h(0) = 1$. Let $h(n) = (1/\sqrt{2})^n [A\cos(\pi n/4) + B\sin(\pi n/4)]$. At $n = 0$, $A = 1$. At $n = 1$, $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1 + B) = 1$, so $B = 1$.

$$\mathbf{h(n) = \left(\frac{1}{\sqrt{2}}\right)^n \left[\cos\frac{\pi n}{4} + \sin\frac{\pi n}{4}\right] u(n) = \{1, 1, 0.5, 0, -0.25, -0.25, \ldots\}}$$

Check $h(2)$: the recursion gives $1 - 0.5 = 0.5$, and the formula gives $0.5[0 + 1] = 0.5$.

c. **Stable**: $|\lambda| = 0.707 < 1$, so h decays geometrically. Numerically $\sum |h| \approx 3.33$.

Q6 (chained) [7].

a. There is one mode, $\lambda = 0.5$, and $h_1(0) = 1$, so $h_1(n) = (0.5)^n u(n)$.

b. $h = h_1 * h_2 = \dfrac{0.5^{n+1} - 0.25^{n+1}}{0.5 - 0.25} u(n)$, so

$$\mathbf{h(n) = (2(0.5)^n - (0.25)^n)u(n).}$$

Check: $h(0) = 1 = h_1(0)h_2(0)$.

c. The recursion with $x = \delta$ gives $h(0) = 1$ and $h(1) = 0.75$. The formula gives $2 - 1 = 1$ and $1 - 0.25 = 0.75$. The characteristic equation $\lambda^2 - 0.75\lambda + 0.125 = (\lambda - 0.5)(\lambda - 0.25)$ has one root from each stage. A cascade's modes are the union of the stages' modes.

d. $y = h * x = 2[0.5^n u * 0.5^n u] - [0.25^n u * 0.5^n u]$. The first term has equal bases, so it is $2(n+1)0.5^n$. The second is $\dfrac{0.5^{n+1} - 0.25^{n+1}}{0.25} = 2(0.5)^n - 0.25^n$. Subtracting:

$$\mathbf{y(n) = (2n(0.5)^n + (0.25)^n)u(n)}$$

Check: $y(0) = 1 = h(0)x(0)$.
$y(1) = 1 + 0.25 = 1.25 = h(0)x(1) + h(1)x(0) = 0.5 + 0.75$. All values were Python-checked.

6. Quick recall check

Question

1. Only the y coefficients: $\lambda^k + a_1\lambda^{k-1} + \cdots + a_k = 0$. An FIR LCCDE has no feedback, so it has no modes ("the modes are zero").
2. $C_1(-\frac{1}{2})^n + C_2 n(-\frac{1}{2})^n$, written in the course as $C_1(-2)^{-n} + C_2 n(-2)^{-n}$.
3. $\dfrac{a^{n+1} - b^{n+1}}{a - b}u(n)$; and $(n+1)a^n u(n)$.
4. No. $\sum |h(k)| = \infty$ because a unit-magnitude oscillation never decays.
5. It starts at $n = -3$, with $4 + 5 - 1 = 8$ samples ($n = -3$ to 4).

DTFT and its properties

Answers and worked solutions · Day 06 · Thu 29 Oct · pack D3

5. Practice

From your course files

T4 Q1 (tutorial).

(a) Finite geometric series with 6 terms (compare with the figure in Section 1):

$$X(\omega)=\sum_{k=0}^{5}e^{-j\omega k}=\frac{\mathbf{1-e^{-j6\omega}}}{\mathbf{1-e^{-j\omega}}}\qquad\left(=e^{-j5\omega/2}\frac{\sin 3\omega}{\sin(\omega/2)}\right)$$

(b) Substitute $n=-k$; the ratio is $\frac{1}{2}e^{j\omega}$:

$$X(\omega)=\sum_{k=-\infty}^{0}2^{k}e^{-j\omega k}=\sum_{n=0}^{\infty}\left(\tfrac{1}{2}e^{j\omega}\right)^{n}=\frac{\mathbf{1}}{\mathbf{1-\frac{1}{2}e^{j\omega}}}$$

(c) Change variable $n=k+4$; the constant is $3^4e^{j4\omega}$:

$$X(\omega)=\sum_{k=-4}^{\infty}\left(\tfrac{1}{3}\right)^{k}e^{-j\omega k}=3^4e^{j4\omega}\sum_{n=0}^{\infty}\left(\tfrac{1}{3}e^{-j\omega}\right)^{n}=\frac{\mathbf{81e^{j4\omega}}}{\mathbf{1-\frac{1}{3}e^{-j\omega}}}$$

(d) The ↑ is under the 0. Pair the terms and use Euler's identity:

$$X(\omega)=-2e^{j2\omega}-e^{j\omega}+e^{-j\omega}+2e^{-j2\omega}=\mathbf{-2j\big(2\sin(2\omega)+\sin(\omega)\big)}$$

Real and odd x gives purely imaginary, odd X.

T4 Q2 (tutorial).

The ↑ is under the -3, so $x(0)=-3$. Use the section 1.4 shortcuts only.

- a. $X(0)=\sum_k x(k)=-1+2-3+2-1=\mathbf{-1}$
- b. $\angle X(0)=\angle e^{\pm j\pi}=\pm\pi$ (a negative real number, not 0)
- c. $\int_{-\pi}^{\pi}X(\omega)\,d\omega=2\pi x(0)=\mathbf{-6\pi}$
- d. $X(\pi)=\sum_k x(k)e^{-j\pi k}=\sum_k x(k)(-1)^k=-1-2-3-2-1=\mathbf{-9}$
- e. $\int_{-\pi}^{\pi}|X(\omega)|^2d\omega=2\pi\sum_k|x(k)|^2=2\pi(1+4+9+4+1)=\mathbf{38\pi}$

T4 Q3 (tutorial).

(a) $H(\omega)=\dfrac{Y(\omega)}{X(\omega)}=\dfrac{\mathbf{1-\frac{1}{2}e^{-j\omega}}}{\mathbf{1-\frac{1}{8}e^{-j\omega}}}$

(b) $e^{j\omega n}$ is an eigenfunction of LTI systems, so $\mathbf{H(\pi/5) = 3e^{-j\pi/3}}$. It is not possible to determine $H(\omega)$ for other values of ω from this one pair.

(c) $u(n) - u(n-1) = \delta(n)$, so $y = h$: $h(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$. That is $\left(\frac{1}{5}\right)^{n} u(n)$ shifted by -2, so $\mathbf{H(\omega) = \dfrac{e^{j2\omega}}{1 - \frac{1}{5}e^{-j\omega}}}$.

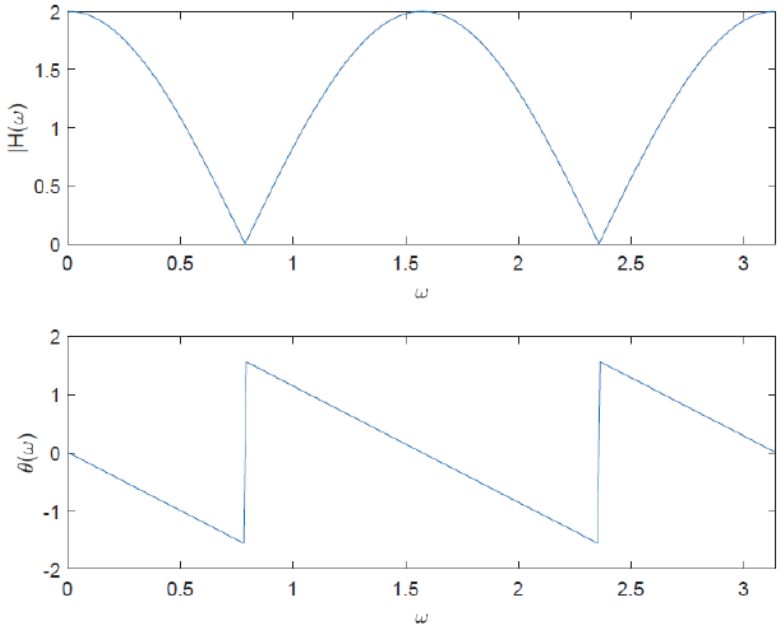
T4 Q4 (tutorial).

(a) Time-shift property:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = 1 + e^{-j4\omega} = e^{-j2\omega}(e^{j2\omega} + e^{-j2\omega}) = 2e^{-j2\omega}\cos(2\omega).$$

$$|H(\omega)| = 2|\cos(2\omega)|, \qquad \theta(\omega) = \begin{cases} -2\omega, & \cos(2\omega) \geq 0 \\ -2\omega + \pi, & \text{otherwise} \end{cases}$$

(π is added to absorb the multiplication by -1 into the exponent.)



T4 Q4(a): the official magnitude and phase sketches (from the tutorial solutions)

(b)(i) $H(\pi/4) = 0$ removes the first tone. $H(\pi/3) = 2e^{-j2\pi/3}\cos(2\pi/3) = e^{j\pi/3}$, so

$$y(n) = \mathbf{3}\sin\left(\tfrac{\pi}{3}\mathbf{n} + \tfrac{\mathbf{13\pi}}{\mathbf{30}}\right)$$

(ii) $H(0) = 2$. $|H(2\pi/5)| = 1.6180$ and $\theta(2\pi/5) = -4\pi/5 + \pi = \pi/5$, so

$$y(n) = 20 + 5(1.6180)\cos\left(\tfrac{2\pi}{5}n + \tfrac{\pi}{2} + \tfrac{\pi}{5}\right) = \mathbf{20 + 8.0902}\cos\left(\tfrac{\mathbf{2\pi}}{\mathbf{5}}\mathbf{n} + \tfrac{\mathbf{7\pi}}{\mathbf{10}}\right)$$

The full method is in D4 (Example 1).

New questions

N1 [3 marks] (easy).

a. $X(\omega) = e^{j\omega} + 1 + e^{-j\omega} = \mathbf{1 + 2}\cos\omega$. It is real because $x(n)$ is real and even.

b. $X(0) = 3 = \sum x$. $X(\pi) = 1 - 2 = \mathbf{-1} = \sum(-1)^n x(n) = -1 + 1 - 1$. (checks)

N2 [4 marks].

a. The sum starts at $n = -1$. Let $m = n + 1$:

$$X(\omega) = \sum_{m=0}^{\infty}(-0.5)^{m-1}e^{-j\omega(m-1)} = (-0.5)^{-1}e^{j\omega}\sum_{m=0}^{\infty}(-0.5e^{-j\omega})^m = \boxed{\frac{-2e^{j\omega}}{1 + 0.5e^{-j\omega}}}$$

b. $X(0) = \sum_{n\geq -1}(-0.5)^n = -2 + \frac{1}{1.5} = \mathbf{-4/3}$. From (a): $-2/1.5 = -4/3$ (checks). $X(\pi) = \sum(-0.5)^n(-1)^n = \sum_{n\geq -1} 0.5^n = 2 + 2 = \mathbf{4}$. From (a): $-2(-1)/(1 - 0.5) = 4$ (checks).

c. $|X(\pi)| = 4 > |X(0)| = 1.33$, so the energy is concentrated near $\omega = \pi$. It is **highpass**. The alternating sign $(-1)^n$ moves a lowpass 0.5^n spectrum by π (modulation property).

N3 [5 marks].

Indices: $x(-2) = 2,\ x(-1) = -1,\ x(0) = 3,\ x(1) = 0,\ x(2) = 1$.

a. $X(0) = 2 - 1 + 3 + 0 + 1 = \mathbf{5}$

b. $X(\pi) = \sum(-1)^n x(n) = 2(+1) + (-1)(-1) + 3(+1) + 0 + 1(+1) = \mathbf{7}$

c. $2\pi x(0) = \mathbf{6\pi}$

d. $2\pi(4 + 1 + 9 + 0 + 1) = \mathbf{30\pi}$

e. The inverse DTFT at $n = 2$ gives $\int Xe^{j2\omega}d\omega = 2\pi x(2) = \mathbf{2\pi}$. (Using $x(-2)$ by mistake gives 4π.)

f. $dX/d\omega = \sum(-jn)x(n)e^{-j\omega n}$, so at $\omega = 0$ it equals $-j\sum nx(n) = -j[(-2)(2) + (-1)(-1) + 0 + 0 + 2(1)] = -j(-1) = \mathbf{j}$.

All six were checked by numerical integration and differentiation.

N4 [6 marks] (multi-part, chained).

a. Convolution property: $Y(\omega) = X(\omega)^2 = \boxed{\dfrac{1}{(1-0.5e^{-j\omega})^2}}$

b. From Example 3(a), $n(0.5)^n u(n) \leftrightarrow \dfrac{0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2}$. Add $(0.5)^n u(n) \leftrightarrow \dfrac{1-0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2}$:

$$\frac{0.5e^{-j\omega} + 1 - 0.5e^{-j\omega}}{(1-0.5e^{-j\omega})^2} = \frac{1}{(1-0.5e^{-j\omega})^2} = Y(\omega)$$

So $y(n) = (n+1)(0.5)^n u(n)$. (A direct numerical convolution gives 1, 1, 0.75, 0.5, 0.3125, ... (checks))

c. $Y(0) = 1/0.5^2 = \mathbf{4}$. Check: $\sum (n+1)0.5^n = 4$ (checks). $Y(\pi) = 1/1.5^2 = \mathbf{0.444}$.

d. $V(\omega) = Y(2\omega) = \boxed{\dfrac{1}{(1-0.5e^{-j2\omega})^2}}$. $V(0) = Y(0) = \mathbf{4}$ (inserting zeros does not change the sum). $V(\pi) = Y(2\pi) = Y(0) = \mathbf{4}$: this is the image created by upsampling.

N5 [5 marks] (sampling).

a. 150 Hz: $\omega = 2\pi(150)/500 = \mathbf{0.6}\pi$. 400 Hz: $\omega = 1.6\pi$. Subtract 2π to get $\mathbf{-0.4}\pi$, so its lines are at $\pm 0.4\pi$. Apparent frequency $= 0.4\pi \cdot 500/(2\pi) = 100$ Hz. **It aliases.**

b. Impulses of area π at $\omega = \pm 0.6\pi$ and of area 0.5π at $\pm 0.4\pi$, repeated every 2π.

c. $\cos(\mathbf{2\pi\,150t}) + \mathbf{0.5}\cos(\mathbf{2\pi\,100t})$

d. $F_s > 2(400) = \mathbf{800\ Hz}$

N6 [5 marks] (hard: modulation and windowing).

a. $\cos(\pi n) = \frac{1}{2}(e^{j\pi n} + e^{-j\pi n})$, so $Y(\omega) = \frac{1}{2}[X(\omega-\pi) + X(\omega+\pi)]$. Since X has period 2π, $X(\omega+\pi) = X(\omega-\pi)$, which gives $\boxed{Y(\omega) = X(\omega-\pi) = \dfrac{1-e^{-j6\omega}}{1+e^{-j\omega}}}$ (because $e^{-j6(\omega-\pi)} = e^{-j6\omega}$). Equivalently, $\cos(\pi n) = (-1)^n$.

b. $Y(0) = X(-\pi) = \sum (-1)^n x(n) = 1-1+1-1+1-1 = \mathbf{0}$.
$Y(\pi) = X(0) = \mathbf{6}$.

c. $|y(n)|^2 = 1$ for 6 samples, so the integral is $2\pi \cdot 6 = \mathbf{12}\pi$. Modulation does not change the energy.

d. It is the figure's curve moved by π: the main lobe of height 6 is now centred at $\omega = \pm\pi$ and there is a zero at $\omega = 0$. The lowpass pulse has become a highpass signal.

6. Quick recall check

Question

1. $4 - 1 + 2 = 5$.
2. $x(-n-3) = x(-(n+3))$: reverse, then advance by 3, which gives $e^{j3\omega}X(-\omega)$.
3. $\omega = 2\pi(1200)/2000 = 1.2\pi$, which reduces to -0.8π. The apparent frequency is $0.8\pi \cdot 2000/(2\pi) = 800$ Hz.
4. The analysis equation: $c_k = \frac{1}{N}\sum x(n)e^{-jk\omega_0 n}$. c_0 is the mean (dc) value over one period.
5. $X(\omega)$ is purely imaginary and odd, e.g. T4 Q1(d), $-2j(2\sin 2\omega + \sin\omega)$.

Frequency response

Answers and worked solutions · Day 07 · Fri 30 Oct · pack D4

5. Practice

From your course files

T4 Q3 (tutorial).

Worked as Example 4.

(a) $H(\omega) = \dfrac{Y(\omega)}{X(\omega)} = \dfrac{\mathbf{1 - \frac{1}{2}e^{-j\omega}}}{\mathbf{1 - \frac{1}{8}e^{-j\omega}}}$

(b) $e^{j\omega n}$ is an eigenfunction of LTI systems, so $\mathbf{H(\pi/5) = 3e^{-j\pi/3}}$ only. $H(\omega)$ at other frequencies cannot be found from this pair.

(c) Spot that the input is $\delta(n)$, so $h(n) = \left(\frac{1}{5}\right)^{n+2}u(n+2)$ and $\mathbf{H(\omega)} = \dfrac{\mathbf{e^{j2\omega}}}{\mathbf{1 - \frac{1}{5}e^{-j\omega}}}$.

T4 Q5 (tutorial).

Input frequency in every case: $\pi/4$ rad. Hint: reduce $5\pi/4$ to $[0, \pi]$.

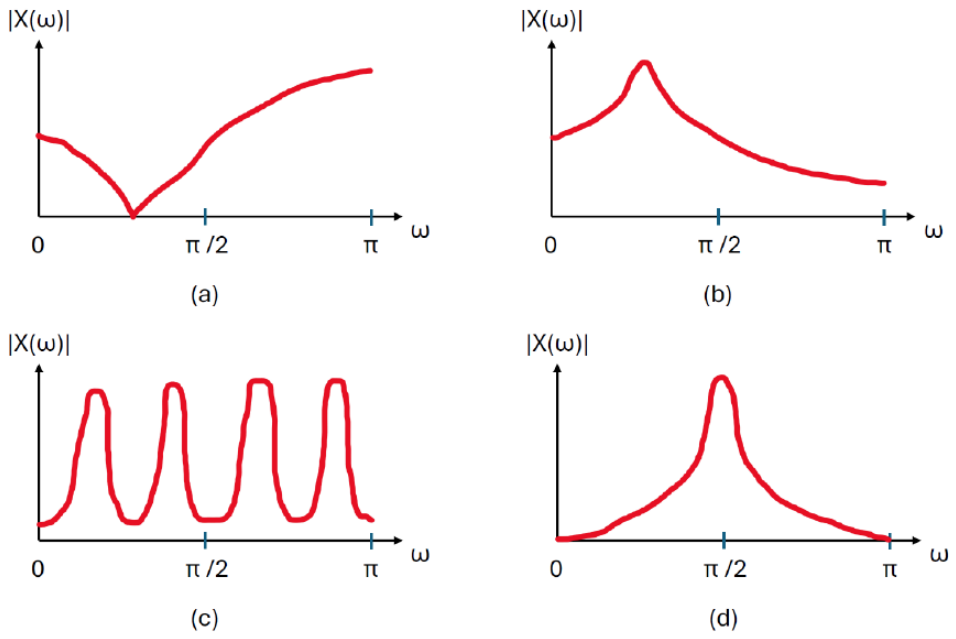
(a) $y(n) = A\cos(\frac{\pi}{2}n)$. Output frequency $\pi/2$ rad, so the system is **time-varying**.

(b) $y(n) = A^2\cos^2(\frac{\pi}{4}n) = \frac{1}{2}A^2 + \frac{1}{2}A^2\cos(\frac{\pi}{2}n)$. Output frequencies 0 and $\pi/2$ rad, so the system is **nonlinear**.

(c) $y(n) = A\cos(\frac{\pi}{4}n)\cos(\pi n) = A\cos(\frac{3\pi}{4}n)$. Output frequency $3\pi/4$ rad, so the system is **time-varying**.

None of the three is LTI.

T4 Q6 (tutorial).



T4 Q6: the official sketches (from the tutorial solutions)

Pole and zero positions below are estimated from the plots. Numbers in brackets come from evaluating those estimated patterns in Python. They are a guide to the shape, not official values.

The course solution says: walk round the unit circle. $|X|$ goes small near a zero and large near a pole. Poles or zeros at the origin have no effect on magnitude. Sketch $0 \le \omega \le \pi$.

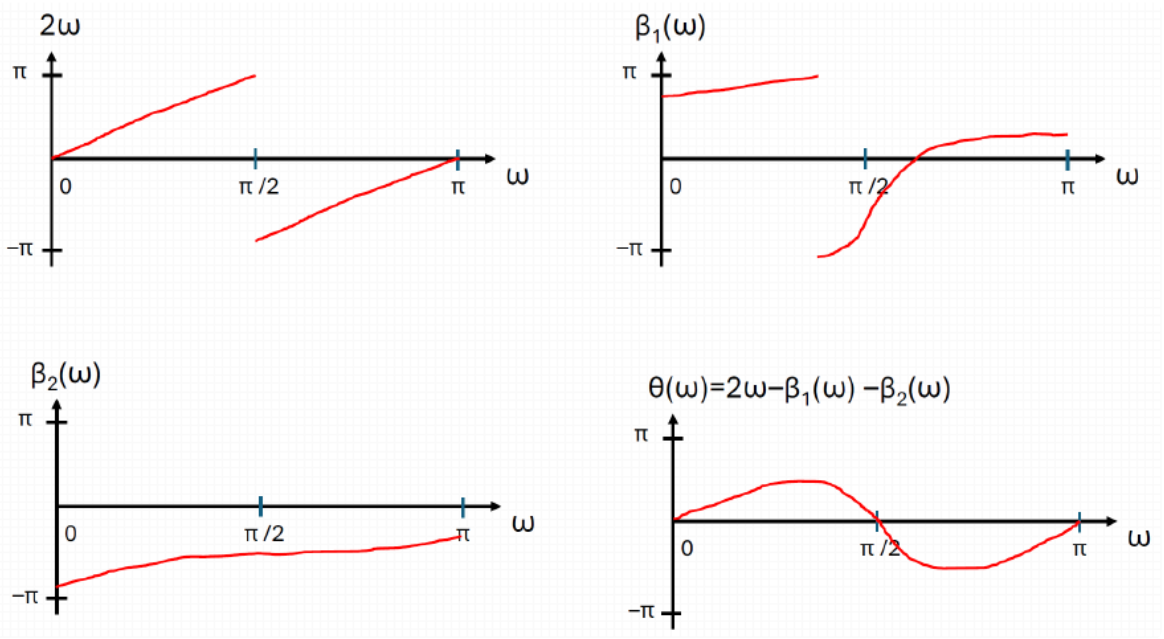
(a) Double pole at the origin. Zeros on (or very near) the unit circle at $e^{\pm j\pi/4}$. This is a second-order FIR notch, like $1 - \sqrt{2}e^{-j\omega} + e^{-j2\omega}$. Shape: it starts at a moderate value at $\omega = 0$ (about 0.59) and falls to **zero at** $\omega \approx \pi/4$. It then rises steadily and smoothly to its **maximum at** $\omega = \pi$ (about 3.41), where $e^{j\omega}$ is furthest from both zeros. It looks like a "V" notch near $\pi/4$ followed by a long rising curve.

(b) Double zero at the origin. Complex-conjugate poles inside the circle at about $0.9e^{\pm j\pi/4}$. Shape: a moderate value at $\omega = 0$, rising to a **single rounded peak at** $\omega \approx \pi/4$, then decaying smoothly to a small value at $\omega = \pi$. A resonance or band-pass shape. It does not reach zero anywhere, because there are no zeros on the circle.

(c) Eight zeros equally spaced on the unit circle at angles $k\pi/4$ ($k = 0, \ldots, 7$, including $z = 1$ and $z = -1$). An 8th-order pole at the origin. This is $1 - e^{-j8\omega}$, so $|X| = 2|\sin 4\omega|$. Shape: **zero at** $\omega = 0, \pi/4, \pi/2, 3\pi/4, \pi$, with **four equal humps** in between. The humps peak at $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$, each with height 2. It is a comb filter.

(d) Zeros on the circle at $z = \pm 1$. Poles at about $\pm j0.9$. This is like $(1 - e^{-j2\omega})/(1 + 0.81e^{-j2\omega})$. Shape: **zero at** $\omega = 0$, rising slowly and then steeply to a **tall sharp peak at** $\omega = \pi/2$ (about 10.5 for poles at radius 0.9), then falling back to **zero at** $\omega = \pi$. It is symmetric about $\pi/2$. A narrow band-pass shape, the same kind of curve as Figure 1 but centred at $\pi/2$.

T4 Q7 (tutorial).



T4 Q7: the official sketches of 2ω, $\beta_1(\omega)$, $\beta_2(\omega)$ and $\theta(\omega)$ (from the tutorial solutions)

Pattern: a double zero at the origin and two poles at about $\pm j0.9$, with $b_0 = 1$. So $X(\omega) = \dfrac{e^{j2\omega}}{(e^{j\omega} - p_1)(e^{j\omega} - p_2)}$, which is $1/(1 + 0.81e^{-j2\omega})$.

Course method: $\theta(\omega) = 2\omega - \beta_1(\omega) - \beta_2(\omega)$, with $\beta_i = \angle(e^{j\omega} - p_i)$. The solution sketches four small plots: 2ω, β_1, β_2 and the sum. The three pieces are hand-drawn roughly and in wrapped form (angles folded into $(-\pi, \pi]$), so your sub-sketches may look different. What matters is the final $\theta(\omega)$. Here is what the pieces actually do, computed for poles at $\pm j0.9$:

- 2ω: a straight line from 0 up to π at $\omega = \pi/2$. In wrapped form it then jumps to $-\pi$ and rises to 0 at $\omega = \pi$.
- Angle from the pole at $+j0.9$: -42° at $\omega = 0$, rising slowly (-15° at $\pi/4$). It then swings very fast through $+90^\circ$ at $\omega = \pi/2$ (about $29^\circ \to 151^\circ$ between 0.45π and 0.55π), because $e^{j\omega}$ passes right next to this pole. It reaches 180°, wraps to -180°, and ends at -138° at $\omega = \pi$.
- Angle from the pole at $-j0.9$: a slow, smooth rise, $42^\circ \to 90^\circ$ (at $\pi/2$) $\to 138^\circ$ (at π).

Resulting $\theta(\omega)$ (what you should draw): $\theta(0) = 0$. It rises to a positive maximum of about $+0.94$ rad ($\approx 54^\circ$) near $\omega \approx 0.4\pi$. It then falls steeply through **0 at** $\omega = \pi/2$ (where the pole is closest). It reaches a minimum of about -0.94 rad near 0.6π and returns to **0 at** $\omega = \pi$. The curve is odd-symmetric about $\omega = \pi/2$. It looks like one smooth "up-then-down" wiggle, with a steep negative slope at the resonance.

Check: $X(0) = 1/1.81$ and $X(\pi) = 1/1.81$ are both real and positive, so the phase is 0 at both ends. $X(\pi/2) = 1/0.19$ is real and positive, so the phase is 0 there too.

New questions

N1.

a. $H(\omega) = 1 + 2e^{-j\omega} + e^{-j2\omega} = e^{-j\omega}(e^{j\omega} + 2 + e^{-j\omega}) = \mathbf{e^{-j\omega}(2 + 2\cos\omega)}$.
$R(\omega) = 2 + 2\cos\omega \geq 0$, so $|H| = 2 + 2\cos\omega$ and $\theta = -\omega$. This is linear phase.

b. At $\omega = \pi/3$: $|H| = 2 + 1 = 3$ and $\theta = -\pi/3$.

$$y(n) = 12\cos\left(\tfrac{\pi}{3}n + \tfrac{\pi}{4} - \tfrac{\pi}{3}\right) = \mathbf{12\cos\left(\tfrac{\pi}{3}n - \tfrac{\pi}{12}\right)}$$

This was checked by direct convolution.

c. $2 + 2\cos\omega = 0$ gives $\boldsymbol{\omega = \pi}$. That is a double zero at $z = -1$, so $(-1)^n$ is blocked.

N2.

a. $H(\omega) = b_0 + b_1 e^{-j\omega}$. DC ($\omega = 0$): $H(0) = b_0 + b_1 = 3$. $(-1)^n$ ($\omega = \pi$): $H(\pi) = b_0 - b_1 = -1$. Solving gives $\mathbf{b_0 = 1,\ b_1 = 2}$. Without the known filter form, two pairs would give only $H(0)$ and $H(\pi)$ (compare T4 Q3b).

b. $H(\pi/2) = 1 + 2(-j) = 1 - 2j$. So $|H| = \sqrt{5} = 2.236$ and $\theta = -\arctan 2 = -1.107$ rad.

$$y(n) = \mathbf{2.236\cos\left(\tfrac{\pi}{2}n - 1.107\right)}$$

This was checked numerically against $\cos(\frac{\pi}{2}n) + 2\cos(\frac{\pi}{2}(n-1))$.

N3.

a. $y = A^2\cos(\frac{\pi}{3}n)\cos(\frac{\pi}{3}n - \frac{\pi}{3}) = \frac{A^2}{2}\left[\cos\frac{\pi}{3} + \cos(\frac{2\pi}{3}n - \frac{\pi}{3})\right]$. Frequencies **0 and** $2\pi/3$ are new, so the system is **nonlinear** and not LTI.

b. $H(\omega) = 1 + e^{-j2\omega} = 2e^{-j\omega}\cos\omega$. At $\pi/3$: $|H| = 1$ and $\theta = -\pi/3$.
$y = A\cos(\frac{\pi}{3}n - \frac{\pi}{3})$. The only frequency is $\pi/3$. This is consistent with LTI, and the system is in fact LTI (an FIR LCCDE).

c. $y = A\cos(-\frac{\pi}{3}n) = A\cos(\frac{\pi}{3}n)$. **No new frequency**, but the system is **not** LTI. Shifting the input by k gives $x(-n-k)$, while shifting the output gives $x(-(n-k)) = x(-n+k)$, so it is time-varying. The frequency test can't prove LTI.

d. $y = 2A\cos(\frac{\pi}{3}n) + 1$. A **new DC component** ($\omega = 0$) appears, so the system is not LTI. It is not linear, because a zero input gives a non-zero output.

N4.

a. $Y(1+0.81e^{-j2\omega}) = X(1+e^{-j2\omega})$, so

$$H(\omega) = \frac{1+e^{-j2\omega}}{1+0.81e^{-j2\omega}} = \frac{e^{j2\omega}+1}{e^{j2\omega}+0.81}$$

b. Zeros: $z^2 = -1$, so $z = \pm j$ (on the unit circle at $\omega = \pm\pi/2$). Poles: $z^2 = -0.81$, so $p = \pm j0.9$ (inside the circle, so the filter is stable). $b_0 = 1$ and $N = M = 2$, so there is no linear phase term.

c. $e^{j\pi/4} = 0.7071 + j0.7071$.

From	Length	Angle
zero $+j$	$V_1 = 0.7654$	-22.5°
zero $-j$	$V_2 = 1.8478$	67.5°
pole $+j0.9$	$U_1 = 0.7329$	-15.26°
pole $-j0.9$	$U_2 = 1.7558$	66.25°

$$|H(\pi/4)| = \frac{0.7654 \times 1.8478}{0.7329 \times 1.7558} = \mathbf{1.0989}$$

$$\theta(\pi/4) = (-22.5 + 67.5) - (-15.26 + 66.25) = \mathbf{-5.99}^\circ = -0.1046 \text{ rad}$$

d.

- $H(0) = 2/1.81 = 1.1050$, so the DC output is 5.525.
- $H(\pi/2) = 0$ (zero on the circle), so the $\cos(\frac{\pi}{2}n)$ term is **removed**.
- The $\pi/4$ term has amplitude $3 \times 1.0989 = 3.297$ and phase $\pi/3 - 0.1046 = 0.9426$ rad ($\approx 0.300\pi$).

$$y_{ss}(n) = \mathbf{5.525 + 3.297\cos\left(\frac{\pi}{4}n + 0.943\right)}$$

This was checked by simulating the LCCDE.

e. $|H|$ is flat at about 1.1 over most of the band (1.105 at 0 and π, 1.099 at $\pi/4$ and $3\pi/4$). It has a **narrow notch dropping to 0 at** $\omega = \pi/2$. The 3 dB points are at about 0.472π and 0.528π. The poles just behind the zeros are what keep the notch narrow.

N5.

a. $H(0) = b_0 \cdot 2/0.2 = 10b_0 = 1$, so $\mathbf{b_0 = 0.1}$.

b. $H = 0.1\left[\frac{1}{1-0.8e^{-j\omega}} + \frac{e^{-j\omega}}{1-0.8e^{-j\omega}}\right]$, so

$$h(n) = \mathbf{0.1(0.8)^n u(n) + 0.1(0.8)^{n-1}u(n-1)}$$

$\sum h = 0.1\left(\frac{1}{0.2} + \frac{1}{0.2}\right) = 1 = H(0)$, as required.

c. Write $H = 0.1\dfrac{e^{j\omega}+1}{e^{j\omega}-0.8}$, with zero $z_1 = -1$ and pole $p_1 = 0.8$, $N = M = 1$. At $e^{j\pi/2} = j$:

- zero vector: $j - (-1) = 1 + j$, length $\sqrt{2} = 1.4142$, angle 45°;
- pole vector: $j - 0.8 = -0.8 + j$, length 1.2806, angle 128.66°.

$$|H(\pi/2)| = 0.1 \times \frac{1.4142}{1.2806} = \mathbf{0.1104}, \qquad \theta(\pi/2) = 45^\circ - 128.66^\circ = \mathbf{-83.66}^\circ = -1.460 \text{ rad}$$

d. $H(\pi) = 0$ (the zero at $z = -1$), so $(-1)^n$ is removed.

$$y_{ss}(n) = \mathbf{0.1104}\cos\left(\tfrac{\pi}{\mathbf{2}}\mathbf{n} - \mathbf{1.460}\right)$$

This was checked by simulation.

e. $|H|^2 = 0.01\dfrac{2(1+\cos\omega)}{1.64 - 1.6\cos\omega} = \dfrac{1}{2}$. This gives $0.02 + 0.02\cos\omega = 0.82 - 0.8\cos\omega$, so $\cos\omega_c = 0.8/0.82 = 0.9756$ and $\boldsymbol{\omega_c = 0.2213}$ rad $(\approx 0.0704\pi)$. It is a narrow low-pass filter: the pole near $z = 1$ boosts DC and the zero at $z = -1$ kills $\omega = \pi$.

6. Quick recall check

Question

1. $7H(0)$, where $H(0) = \sum_n h(n)$ is real.
2. $2\cos(\omega_0 n + 0.3 + \pi)$, or equivalently $-2\cos(\omega_0 n + 0.3)$. The magnitude is 2 and the phase is π.
3. Zero. That frequency is completely blocked (a notch).
4. $|H|$: no, because its distance to every point on the circle is 1. θ: yes, it adds the linear term $+\omega$ via $\omega(N - M)$.
5. No. An LTI system can't create a new frequency, since $Y = HX$. It must be nonlinear and/or time-varying.

DSP · L06

DFT and circular convolution

Answers and worked solutions · Day 09 · Sun 1 Nov · pack D5

5. Practice

From your course files

T5 Q1 (tutorial).

> *Hint: Playbook D. Row 0 of the circulant is* $\{x_2(0), x_2(3), x_2(2), x_2(1)\} = \{4, 2, 2, 3\}$.
>
> $y(n) = x_1(n) \circledast_4 x_2(n) = \sum_{m=0}^{3} x_1(m) x_2((n-m))_4$, $n = 0, 1, 2, 3$:
>
> - $y(0) = x_1(0)x_2(0) + x_1(1)x_2(3) + x_1(2)x_2(2) + x_1(3)x_2(1) = 4 + 4 + 6 + 3 =$
> - $y(1) = x_1(0)x_2(1) + x_1(1)x_2(0) + x_1(2)x_2(3) + x_1(3)x_2(2) = 3 + 8 + 6 + 2 =$
> - $y(2) = x_1(0)x_2(2) + x_1(1)x_2(1) + x_1(2)x_2(0) + x_1(3)x_2(3) = 2 + 6 + 12 + 2 =$
> - $y(3) = x_1(0)x_2(3) + x_1(1)x_2(2) + x_1(2)x_2(1) + x_1(3)x_2(0) = 2 + 4 + 9 + 4 =$
>
> $$\mathbf{y(n) = \{\underset{\uparrow}{17}, 19, 22, 19\}}$$

T5 Q2 (tutorial).

> *Hint: Playbook B then D step 4. Do* $X(0), X(1), X(2)$*; get* $X(3) = X^*(1)$.
>
> Recall $x_1(n) \circledast_4 x_2(n) \overset{\text{DFT}_4}{\longleftrightarrow} X_1(k)X_2(k)$.
>
> - $X_1(k) = \sum_{n=0}^{3} x_1(n) e^{-j2\pi kn/4} = \{7,\ -2 - j,\ 1,\ -2 + j\}$
> - $X_2(k) = \{11,\ 2 - j,\ 1,\ 2 + j\}$
> - $X_1(k)X_2(k) = \{77, -5, 1, -5\}$ (e.g. $(-2 - j)(2 - j) = -4 + 2j - 2j + j^2 = -5$)
> - $y(n) = \frac{1}{4}\sum_{k=0}^{3} Y(k) e^{j2\pi kn/4}$: $y(0) = \frac{1}{4}(77 - 5 + 1 - 5) = 17$, $y(1) = \frac{1}{4}(77 - 5j - 1 + 5j) = 19$, $y(2) = \frac{1}{4}(77 + 5 + 1 + 5) = 22$, $y(3) = 19$.
>
> $$\mathbf{y(n) = \{\underset{\uparrow}{17}, 19, 22, 19\}}$$
>
> the same as Q1.

T5 Q3 (tutorial).

> $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1, 0, 0, 0, 0\}$, so
>
> $$X(k) = \sum_{n=0}^{7} x(n) e^{-j2\pi kn/8} = 1 + e^{-j\pi k/4} + e^{-j\pi k/2} + e^{-j3\pi k/4}$$

$$\mathbf{X(k)} = \{\mathbf{4},\ \mathbf{1-j(1+\sqrt{2})},\ \mathbf{0},\ \mathbf{1+j(1-\sqrt{2})},\ \mathbf{0},\ \mathbf{1-j(1-\sqrt{2})},\ \mathbf{0},\ \mathbf{1+j(1+\sqrt{2})}\}$$

$x_1(n) = x((n+3))_8 = x((n-5))_8$, so the circular time-shift property $x((n-l))_N \leftrightarrow X(k)e^{-j2\pi kl/N}$ gives $\mathbf{X_1(k) = X(k)e^{j3\pi k/4}}$. Likewise $x_2(n) = x((n-2))_8$, so $\mathbf{X_2(k) = X(k)e^{-j\pi k/2}}$.

T5 Q6 (tutorial).

(a) $X(\omega) = \sum_{n=-2}^{3} x(n)e^{-j\omega n} = \mathbf{3 + 4\cos\omega + 2\cos(2\omega)}$ (the pairs $2e^{\pm j\omega}$ and $1 \cdot e^{\pm j2\omega}$ combine by Euler; $x(3) = 0$).

(b) $V(k) = \sum_{n=0}^{5} v(n)e^{-j2\pi kn/6} = 3 + 4\cos\dfrac{\pi k}{3} + 2\cos\dfrac{2\pi k}{3} = \{\mathbf{9, 4, 0, 1, 0, 4}\}$.

(c) $\mathbf{V(k) = X(\omega)|_{\omega=2\pi k/6}}$: $V(k)$ samples $X(\omega)$ at $N = 6$ equally spaced frequencies. In the time domain, sampling $X(\omega)$ corresponds to the periodic extension $x_p(n) = \sum_l x(n - lN)$ with $N = 6$, i.e. $\{\ldots, 1, 2, \underset{\uparrow}{3}, 2, 1, 0, 1, 2, 3, \ldots\}$, and one period of it starting at $n = 0$ is $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

T5 Q4 (tutorial).

The inverse system is $H_I(z) = 1 - 0.5z^{-1}$, with $h_I(n) = \{\underset{\uparrow}{1}, -0.5\}$, so $x(n) = h_I(n) * y(n)$.

1. Zero-pad h_I to length 64: $\bar{h}_I(n) = \{h_I(n), 0, \ldots, 0\}$ (62 zeros). $y(n)$ is already length 64.
2. Compute $\bar{H}_I(k)$ and $Y(k)$ (64-point DFTs), then $\bar{h}_I(n) \circledast_{64} y(n) = \mathrm{IDFT}_{64}\{\bar{H}_I(k)Y(k)\}$.
3. For $n = 1, \ldots, 63$: $\bar{h}_I \circledast_{64} y = h_I(0)y(n) + h_I(1)y(n-1)$, which is the linear convolution, so it equals $x(n)$.
4. For $n = 0$ the circular result is $h_I(0)y(0) + h_I(1)y(63) = y(0) - 0.5y(63)$, but the linear one is $h_I(0)y(0)$ (with $y(n) = 0$ for $n < 0$). Correct it: $\mathbf{x(0) = \left[\bar{h}_I \circledast_{64} y\right]_{n=0} + 0.5\,y(63)}$.

New questions

Q1 (easy) [3 marks].

$X(0) = 1 + 0 + 2 + 1 = 4$ $X(1) = 1 + 0(-j) + 2(-1) + 1(j) = -1 + j$
$X(2) = 1 - 0 + 2 - 1 = 2$ $X(3) = 1 + 0(j) + 2(-1) + 1(-j) = -1 - j$

$\mathbf{X(k) = \{4,\ -1+j,\ 2,\ -1-j\}}$. $X(3) = -1-j = (-1+j)^* = X^*(1)$, as expected for real x. Check: $\frac{1}{4}(4-1+j+2-1-j) = 1 = x(0)$.

Q2 [4 marks].

a. Rows of 4: $\{1,2,3,4\}$, $\{5,6,0,0\}$. Column sums: $\mathbf{\tilde{x}(n) = \{6,8,3,4\}}$.

b. $\tilde{X}(0) = 6+8+3+4 = 21$ and $X(0) = 1+2+\cdots+6 = 21$. Equal, because the samples of $X(\omega)$ are exact even with aliasing; only the time sequence is aliased.

c. $N \geq L = 6$, so $\mathbf{N = 6}$.

Q3 [5 marks].

a. Pad x_2 to $\{1,1,1,0\}$. Circulant (column 0 = x_2):

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+0+1 \\ 1+2+0+1 \\ 1+2+0+0 \\ 0+2+0+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 3 \\ 3 \end{bmatrix}$$

$\mathbf{y_4 = \{2,4,3,3\}}$.

b. Linear: $y = \{1,3,3,3,1,1\}$ (length $4+3-1=6$). Fold $y(4), y(5)$ onto $n = 0, 1$: $\{1+1, 3+1, 3, 3\} = \{2,4,3,3\}$. Matches (a). Check: $\sum y = 12 = 4 \times 3$.

Q4 [6 marks, chained].

a. $L + M - 1 = 3 + 2 - 1 = \mathbf{4}$.

b. $x = \{1,2,1,0\}$: $X(0) = 4$, $X(1) = 1-2j-1 = -2j$, $X(2) = 1-2+1 = 0$, $X(3) = 2j$. So $X = \{4, -2j, 0, 2j\}$. $h = \{1,-1,0,0\}$: $H(k) = 1-(-j)^k$: $H = \{0,\ 1+j,\ 2,\ 1-j\}$.

c. $Y = \{0,\ -2j(1+j),\ 0,\ 2j(1-j)\} = \{0,\ 2-2j,\ 0,\ 2+2j\}$.
$y(n) = \frac{1}{4}\sum_k Y(k) j^{kn}$: $y(0) = \frac{1}{4}(4) = 1$;
$y(1) = \frac{1}{4}[(2-2j)j + (2+2j)(-j)] = \frac{1}{4}(2j+2-2j+2) = 1$;
$y(2) = \frac{1}{4}[-(2-2j) - (2+2j)] = -1$;
$y(3) = \frac{1}{4}[(2-2j)(-j) + (2+2j)j] = \frac{1}{4}(-2j-2+2j-2) = -1$.
$\mathbf{y(n) = \{1,1,-1,-1\}}$. Direct: $\{1,2,1\} * \{1,-1\} = \{1,1,-1,-1\}$. Agrees.

d. $y(n)$ has length $4 \leq N$, so its 4-point DFT is exactly its DTFT sampled at $\omega_k = 2\pi k/4 = \pi k/2$ (no time aliasing). With $N = 3$, $y(3) = -1$ wraps onto $n = 0$: $\mathbf{\{0,1,-1\}}$.

Q5 [6 marks].

a. $X_a(k) = X(k)W_4^k = X(k)(-j)^k$: $k=0$: 10; $k=1$: $(-2+2j)(-j) = 2+2j$; $k=2$: $(-2)(-1) = 2$; $k=3$: $(-2-2j)(j) = 2-2j$.
$\mathbf{X_a = \{10,\ 2+2j,\ 2,\ 2-2j\}}$. (Check: $x_a = \{4,1,2,3\}$, $X_a(2) = 4-1+2-3 = 2$.)

b. $x_b = \{1,4,3,2\}$, $X_b(k) = X((-k))_4 = X(4-k)$:
$\mathbf{X_b = \{10,\ -2-2j,\ -2,\ -2+2j\}}$. For real x this is also $X^*(k)$.

c. $(-1)^n = e^{j2\pi(2)n/4}$, so $X_c(k) = X((k-2))_4$:
$\mathbf{X_c = \{-2,\ -2-2j,\ 10,\ -2+2j\}}$.

d. $\frac{1}{4}(100+8+4+8) = \mathbf{30} = 1+4+9+16$.

Q6 (hard) [6 marks].

a. $N = 4 \geq$ length of y. $Y(k)$: $Y(0) = 15$; $Y(1) = 2-5j-6+2j = -4-3j$; $Y(2) = 2-5+6-2 = 1$; $Y(3) = -4+3j$. $H(k) = 2+(-j)^k$:
$H = \{3,\ 2-j,\ 1,\ 2+j\}$. All nonzero, so division is allowed. $X(k) = Y/H$:
$X(0) = 5$; $X(1) = \frac{-4-3j}{2-j} = \frac{(-4-3j)(2+j)}{5} = \frac{-5-10j}{5} = -1-2j$; $X(2) = 1$;
$X(3) = -1+2j$. IDFT: $x(0) = \frac{1}{4}(5-1-2j+1-1+2j) = 1$;
$x(1) = \frac{1}{4}[5+(-1-2j)j-1+(-1+2j)(-j)] = \frac{1}{4}(5+2-j-1+2+j) = 2$
; $x(2) = \frac{1}{4}[5+1+2j+1+1-2j] = 2$;
$x(3) = \frac{1}{4}[5+(-1-2j)(-j)-1+(-1+2j)j] = \frac{1}{4}(5-2+j-1-2-j) = 0$
. $\mathbf{x(n) = \{1,2,2\}}$. Check: $\{1,2,2\} * \{2,1\} = \{2,5,6,2\}$.

b. $h = \{1,1,0,0\}$ gives $H(k) = 1+(-j)^k = \{2, 1-j, 0, 1+j\}$. $H(2) = 0$, so $X(2) = Y(2)/0$ is undefined. The filter removes the $\omega = \pi$ component, so that part of x cannot be recovered at this N.

6. Quick recall check

Question

1. 2.
2. $[1,\ -j,\ -1,\ j]$.
3. $10+5-1 = 14$ (16 if a radix-2 FFT is required).
4. $X(5) = X^*(3) = 1-2j$.
5. $N^2 = 65\,536$ vs $\frac{N}{2}\log_2 N = 1024$ (64 times fewer).

Spectral analysis and block convolution

Answers and worked solutions · Day 11 · Tue 3 Nov · pack D6

5. Practice

From your course files

T5 Q5 (tutorial).

> *Hint: Playbook A.* $N \geq 6000/50$, *then round up to a power of 2.*
>
> (a) $F_s \geq 2B =$ **6 kHz**.
>
> (b) Resolution $= F_s/N \leq 50$ Hz, so $N \geq \dfrac{6000}{50} = 120$. $m = \log_2 N = 6.9$; rounding up to $m = 7$ gives $\mathbf{N = 2^7 = 128}$.
>
> (c) $T_0 = N/F_s = 128/6000 =$ **0.0213 s**. (The actual resolution is $6000/128 = 46.9$ Hz.)

T5 Q4 (tutorial).

> The inverse system is $H_I(z) = 1 - 0.5z^{-1}$, with $h_I(n) = \{\underset{\uparrow}{1}, -0.5\}$, so $x(n) = h_I(n) * y(n)$.
>
> 1. Zero-pad h_I to length 64: $\bar{h}_I(n) = \{h_I(n), 0, \ldots, 0\}$ (62 zeros). $y(n)$ is already length 64.
> 2. Compute $\bar{H}_I(k)$ and $Y(k)$ (64-point DFTs), then $\bar{h}_I(n) \circledast_{64} y(n) = \mathrm{IDFT}_{64}\{\bar{H}_I(k)Y(k)\}$.
> 3. For $n = 1, \ldots, 63$: $\bar{h}_I \circledast_{64} y = h_I(0)y(n) + h_I(1)y(n-1)$, which is the linear convolution, so it equals $x(n)$.
> 4. For $n = 0$ the circular result is $h_I(0)y(0) + h_I(1)y(63) = y(0) - 0.5y(63)$, but the linear one is $h_I(0)y(0)$ (with $y(n) = 0$ for $n < 0$). Correct it: $\mathbf{x(0) = \left[\bar{h}_I \circledast_{64} y\right]_{n=0} + 0.5\, y(63)}$.
>
> This is overlap-save with $M = 2$: the circular convolution of a length-N block with a length-M filter corrupts exactly the first $M - 1 = 1$ output.

T5 Q6 (tutorial).

> (a) $X(\omega) = \sum_{n=-2}^{3} x(n)e^{-j\omega n} = \mathbf{3 + 4\cos\omega + 2\cos(2\omega)}$ (the pairs $2e^{\pm j\omega}$ and $1 \cdot e^{\pm j2\omega}$ combine by Euler; $x(3) = 0$).
>
> (b) $V(k) = \sum_{n=0}^{5} v(n)e^{-j2\pi kn/6} = 3 + 4\cos\dfrac{\pi k}{3} + 2\cos\dfrac{2\pi k}{3} = \mathbf{\{9, 4, 0, 1, 0, 4\}}$.

(c) $\mathbf{V(k) = X(\omega)|_{\omega=2\pi k/6}}$: $V(k)$ samples $X(\omega)$ at $N = 6$ equally spaced frequencies. In the time domain, sampling $X(\omega)$ corresponds to the periodic extension $x_p(n) = \sum_l x(n - lN)$ with $N = 6$, i.e. $\{\ldots, 1, 2, \underset{\uparrow}{3}, 2, 1, 0, 1, 2, 3, \ldots\}$, and one period of it starting at $n = 0$ is $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

New questions

N1.

a. $F_s/N = 1000/200 = 5$ Hz. $T_0 = N/F_s = 200/1000 = 0.2$ s.

b. $k_0 = 120/5 = 24$, and its mirror image is $N - 24 = 176$. **Bins 24 and 176.** The tone is bin-aligned, so a rectangular window gives no leakage.

c. $k = 37$: $F = 37 \times 5 =$ **185 Hz**. $k = 150 > N/2 = 100$: it represents $-(200 - 150) \times 5 = -250$ **Hz**. For a real signal, it is the mirror image of bin 50 (250 Hz), not "750 Hz". 750 Hz is above $F_s/2$, so it cannot be present after anti-aliasing.

N2.

a. Bin spacing $F_s/N = 1000/512 = 1.95$ Hz. Resolution $1/T_0$ with $T_0 = L/F_s = 100/1000 = 0.1$ s, so the resolution is **10 Hz**.

b. The separation is 5 Hz, which is less than 10 Hz, so **no**. Zero-padding to 4096 only makes the bin spacing finer (0.24 Hz). It interpolates the same smeared DTFT, so the two tones stay as one lobe. **No.**

c. Need $1/T_0 \leq 5$ Hz, so $T_0 \geq 0.2$ s and $L \geq 0.2 \times 1000 =$ **200 samples**. The next power of 2 is $N = L = 256$ (resolution 3.9 Hz).

N3.

a. $M = 2$, so $N = L + M - 1 = 4$, and $\bar{h}(n) = \{1, -1, 0, 0\}$.

b. Blocks $x_0 = \{1, 0, 2\}$ and $x_1 = \{1, 3, 1\}$, each padded with one zero. A length-3 block convolved with length-2 h is 4 long, so the 4-point circular convolution equals the linear convolution:

- $y_0 = \{1, 0, 2\} * \{1, -1\} = \{1, -1, 2, -2\}$
- $y_1 = \{1, 3, 1\} * \{1, -1\} = \{1, 2, -2, -1\}$

c. y_0 is at $n = 0..3$ and y_1 at $n = 3..6$. They overlap by $M - 1 = 1$ sample at $n = 3$: $-2 + 1 = -1$.

$$y(n) = \{\underset{\uparrow}{1}, -1, 2, -1, 2, -2, -1\}$$

Check at $n = 3$: $y(3) = x(3) - x(2) = 1 - 2 = -1$. This matches. Length $6 + 2 - 1 = 7$, also correct.

N4.

a. $M = 3$, so $L = N - M + 1 = 5 - 3 + 1 = \mathbf{3}$.

b. Each block is $M - 1 = 2$ old samples followed by 3 new ones. The first block starts with 2 zeros. The full output has length $8 + 3 - 1 = 10$, so the number of blocks is $\lceil 10/3 \rceil = 4$:

- $x_0 = \{0, 0, 1, 3, 2\}$
- $x_1 = \{3, 2, 1, 0, 2\}$
- $x_2 = \{0, 2, 4, 1, 0\}$ (the last new sample is a padding zero)
- $x_3 = \{1, 0, 0, 0, 0\}$

4 blocks.

c. Linear convolution with h gives length 7. Fold the last 2 samples onto the first 2:

- $x_0 * h = \{0, 0, 2, 5, 2, \mathbf{1}, \mathbf{2}\}$, so $x_0 \circledast_5 h = \{0 + 1, 0 + 2, 2, 5, 2\} = \{1, 2, 2, 5, 2\}$
- $x_1 * h = \{6, 1, 3, 1, 5, \mathbf{-2}, \mathbf{2}\}$, so $x_1 \circledast_5 h = \{6 - 2, 1 + 2, 3, 1, 5\} = \{4, 3, 3, 1, 5\}$

Discard the first $M - 1 = 2$ outputs of each block ($\{1, 2\}$ and $\{4, 3\}$). They contain wrapped-around tail terms (time aliasing). Keep $\{2, 5, 2\}$ and $\{3, 1, 5\}$.

d. The other blocks: $x_2 \circledast_5 h = \{1, 4, 6, 0, 3\}$, so keep $\{6, 0, 3\}$.
$x_3 \circledast_5 h = \{2, -1, 1, 0, 0\}$, so keep $\{1, 0, 0\}$ and use only the first value.

$$y(n) = \{\underset{\uparrow}{2}, 5, 2, 3, 1, 5, 6, 0, 3, 1\}$$

Check: $y(6) = 2x(6) - x(5) + x(4) = 8 - 2 + 0 = 6$. This matches.

Overlap-add needs $\lceil 8/3 \rceil = \mathbf{3\ blocks}$ (the last block's tail supplies $y(9)$). Overlap-save needs one extra block here because each block delivers only L outputs.

N5.

a. **Hanning.** The weak tone is 40 dB down. The rectangular window's leakage never gets below about -26 dB within 6 bins, so it would bury the weak tone. The strong tone is not bin-aligned in general, so you cannot rely on zero leakage. Hanning's side lobes drop below -50 dB by 5 bins. The price is a wider main lobe, which is why (b) asks for 5 bins of separation.

b. Separation 40 Hz $\geq 5F_s/L$, so $L \geq 5 \times 8000/40 = 1000$. Round up: $L = 1024$, $T_0 = 1024/8000 =$ **0.128 s** (bin spacing 7.81 Hz, so the tones are 5.12 bins apart).

c. Rectangular, $L = 1024$: the leakage from 1010 Hz at the weak tone's bins is about -25 dB, which is above the weak tone (-40 dB). **No separate peak is visible.**

Numerical check (Python, $L = 1024$): with Hanning, the weak tone's bin is at -38.4 dB and its neighbours are at -49.5 and -43.1 dB, so there is a clear local peak. With rectangular, the same bin is at -25.9 dB with neighbours at -22.8 and -25.4 dB, so there is no peak. With Hanning and $L = 512$ (2.56 bins apart), there is no peak either.

N6.

a. $2B = 10$ kHz ≤ 12 kHz. **Yes.**

b. $\Delta F = 30$ Hz, so $N \geq 12000/30 = 400$. Round up to $N = 512$. $T_0 = 512/12000 =$ **42.7 ms**. Resolution $12000/512 =$ **23.4 Hz** ≤ 30 Hz.

c. Bin spacing 23.4375 Hz. $k_1 = 1500/23.4375 =$ **64** (bin-aligned, so no leakage on its own). $k_2 = 1530/23.4375 =$ **65.28** (not aligned, so it leaks into all bins, with its peak near $k = 65$). The mirror images are at $512 - 64 = 448$ and $512 - 65.28 = 446.72$. Caveat: $1/T_0 = 23.4$ Hz is the course's rule, and by that rule the tones count as resolved. On a real plot, though, the two tones sit only 1.28 bins apart, so you may not see a clear dip between two peaks. In practice you want a margin above $1/T_0$.

d. Bin spacing $12000/2048 =$ **5.86 Hz**. Resolution $1/T_0 = F_s/L = 12000/128 =$ **93.75 Hz**, which is more than 30 Hz. **No.** The tones merge into one lobe. (Python: the zero-padded spectrum has one main peak, at about 1516 Hz.) Padding only interpolates.

e. $L \geq 2F_s/\Delta F = 2 \times 12000/30 = 800$. Round up to $L = 1024$.
$T_0 = 1024/12000 =$ **85.3 ms**. (Python check: with Hanning and $L = 1024$, there are two peaks near 1500 and 1530 Hz.)

6. Quick recall check

Question

1. $2000/256 = 7.81$ Hz; $T_0 = 256/2000 = 0.128$ s.
2. No. It samples the same DTFT more finely (smaller bin spacing), which is interpolation only. The resolution is still $1/T_0$.
3. $w(n) = \frac{1}{2}(1 - \cos(2\pi n/4))$, so $w = \{0, 0.5, 1, 0.5, 0\}$.
4. $L = N - M + 1 = 13$ new samples. Discard the first $M - 1 = 3$ outputs of each block.

5. Hanning has much lower side lobes (-31 dB vs -13 dB, so less leakage) but a main lobe about twice as wide (worse resolution).

z-transform and ROC

Answers and worked solutions · Day 13 · Thu 5 Nov · pack D7

5. Practice

From your course files

T6 Q1 (tutorial).

Hint: $2^n u(-n-1)$ has a minus sign in its transform, and (a) and (b) have the same algebraic $X(z)$. Only the ROC tells them apart.

(a) $\left(\frac{1}{3}\right)^n u(n) \leftrightarrow \dfrac{1}{1-\frac{1}{3}z^{-1}}$, $|z| > \frac{1}{3}$, and $2^n u(-n-1) \leftrightarrow -\dfrac{1}{1-2z^{-1}}$, $|z| < 2$. So

$$X_1(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-2z^{-1}} = -\frac{\frac{5}{3}z^{-1}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}, \qquad \text{ROC: } |z| > \tfrac{1}{3} \cap |z| < 2 = \tfrac{1}{3} < |z| < 2.$$

Sketch: a ring between the pole circles at $\frac{1}{3}$ and 2 (it contains the unit circle).

(b) Both pieces are causal: $\left(\frac{1}{3}\right)^n u(n) - 2^n u(n) \leftrightarrow \dfrac{1}{1-\frac{1}{3}z^{-1}} - \dfrac{1}{1-2z^{-1}}$, so

$X_2(z) = -\dfrac{\frac{5}{3}z^{-1}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}$ (the same as X_1), with ROC $|z| > \frac{1}{3} \cap |z| > 2 = |z| > 2$.

Sketch: everything outside the circle of radius 2.

(c) Time shift: $x(n+4) \leftrightarrow z^4 X(z)$.

$$X_3(z) = z^4 X_1(z) = -\frac{\frac{5}{3}z^{3}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}, \qquad \text{ROC: } \tfrac{1}{3} < |z| < 2.$$

The ring already excludes $z = \infty$, so the shift does not change the ROC.

(d) Time reversal: $x(-n) \leftrightarrow X(z^{-1})$, and the ROC radii are inverted.

$$X_4(z) = X_1(z^{-1}) = -\frac{\frac{5}{3}z}{1-\frac{7}{3}z+\frac{2}{3}z^{2}}, \qquad \text{ROC: } \tfrac{1}{2} < |z| < 3.$$

The poles move from $\frac{1}{3}, 2$ to $3, \frac{1}{2}$.

T6 Q2 (tutorial).

Hint: the ROC of Y is the ring $\cap\, |z| > \frac{1}{2}$. The partial-fraction step is D8 material.

$y(n) = x_1(n) * x_2(n)$, so $Y(z) = X_1(z)X_2(z)$ with ROC $R_{x_1} \cap R_{x_2}$. From Q1(a), $X_1(z) = -\dfrac{\frac{5}{3}z^{-1}}{1 - \frac{7}{3}z^{-1} + \frac{2}{3}z^{-2}}$, $\frac{1}{3} < |z| < 2$, and $X_2(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$.

$$Y(z) = -\frac{\frac{5}{3}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - 2z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)}, \qquad \text{ROC: } \frac{1}{2} < |z| < 2.$$

Partial fractions (cover-up):

$$Y(z) = -\frac{2}{1 - \frac{1}{3}z^{-1}} - \frac{\frac{4}{3}}{1 - 2z^{-1}} + \frac{\frac{10}{3}}{1 - \frac{1}{2}z^{-1}}.$$

In the ring $\frac{1}{2} < |z| < 2$, the poles at $\frac{1}{3}$ and $\frac{1}{2}$ are inside (causal terms) and the pole at 2 is outside (anti-causal term, $-\frac{1}{1-2z^{-1}} \leftrightarrow 2^n u(-n-1)$):

$$y(n) = -2\left(\frac{1}{3}\right)^n u(n) + \frac{4}{3}\,2^n u(-n-1) + \frac{10}{3}\left(\frac{1}{2}\right)^n u(n).$$

T6 Q3 (tutorial).

Hint: the figure is h_1 in cascade with (h_2 minus the $h_3 \to h_4$ branch). $H_2 = H_3$ lets you factor $H_2(1 - H_4)$.

Transfer functions of the blocks:

$$H_1(z) = \tfrac{1}{2} + \tfrac{1}{4}z^{-1} + \tfrac{1}{2}z^{-2}, \qquad H_2(z) = H_3(z) = 1 - z^{-1}, \qquad H_4(z) = z + 2z^{-2}.$$

Overall system:

$$H(z) = H_1(z)\big[H_2(z) - H_3(z)H_4(z)\big] = H_1(z)H_2(z)\big[1 - H_4(z)\big] = \left(\tfrac{1}{2} + \tfrac{1}{4}z^{-1} + \tfrac{1}{2}z^{-2}\right)\left(1 - z^{-1}\right)\left(-z + 1 - 2z^{-2}\right)$$

$$H(z) = -\tfrac{1}{2}z + \tfrac{3}{4} - \tfrac{1}{2}z^{-1} - \tfrac{1}{4}z^{-2} - \tfrac{1}{2}z^{-4} + z^{-5}.$$

Read the coefficients off as the impulse response (the z^{+1} term is at $n = -1$):

$$h(n) = \Big\{ -\tfrac{1}{2}, \underset{\uparrow}{\tfrac{3}{4}}, -\tfrac{1}{2}, -\tfrac{1}{4}, 0, -\tfrac{1}{2}, 1 \Big\}.$$

The system is **not causal**, because $h(-1) = -\frac{1}{2} \neq 0$ (that is, $h(n) \neq 0$ for $n < 0$).

All answers above were re-derived in Python (sympy) and agree with the official solutions.

New questions

N1.

a. $X(z) = 1 - 2z^{-1} + z^{-2} = (1 - z^{-1})^2$. Only positive n appear, so **ROC:** $|z| > 0$ (all z except 0).

b. $X(z) = \dfrac{(z-1)^2}{z^2}$. A double zero at $z = 1$ and a double pole at $z = 0$. $X(\infty) = 1$, so nothing at ∞. Count 2 = 2.

c. $x(n+2) \leftrightarrow z^2 X(z) = z^2 - 2z + 1$. Samples are now at $n = -2, -1, 0$, so **ROC:** $|z| < \infty$ (all z except ∞). The shift removed the exclusion at 0 and added one at ∞.

N2.

Euler: $\cos(\pi n/2) = \frac{1}{2}\left(e^{j\pi n/2} + e^{-j\pi n/2}\right)$, so $x(n) = \frac{1}{2}\left(\frac{j}{2}\right)^n u(n) + \frac{1}{2}\left(-\frac{j}{2}\right)^n u(n)$.

$$X(z) = \frac{1}{2}\left[\frac{1}{1 - \frac{j}{2}z^{-1}} + \frac{1}{1 + \frac{j}{2}z^{-1}}\right] = \frac{1}{1 + \frac{1}{4}z^{-2}} = \frac{z^2}{z^2 + \frac{1}{4}}$$

ROC: $|z| > \frac{1}{2}$ (both pieces causal, $|\pm j/2| = \frac{1}{2}$).

Poles at $z = \pm j\frac{1}{2}$. A double zero at $z = 0$.

Check: $x(n) = \{1, 0, -\frac{1}{4}, 0, \frac{1}{16}, \ldots\}$, and the series of $1/(1 + \frac{1}{4}z^{-2})$ is $1 - \frac{1}{4}z^{-2} + \frac{1}{16}z^{-4} - \ldots$. This agrees. (Same as the table row with $a = \frac{1}{2}$, $\cos\omega_0 = 0$.)

N3.

a. $\dfrac{z^{-3}}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$. Still causal, so ∞ stays in the ROC.

b. $\dfrac{1}{1 - \frac{1}{2}z}$, $|z| < 2$.

c. $X(z/2) = \dfrac{1}{1 - z^{-1}}$, $|z| > 1$. This is $u(n)$, as it should be: $2^n(\frac{1}{2})^n = 1$.

d. $\dfrac{\frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}$, $|z| > \frac{1}{2}$.

e. $X(z^2) = \dfrac{1}{1 - \frac{1}{2}z^{-2}}$, $|z| > \left(\frac{1}{2}\right)^{1/2} = 0.707$. The poles are now at ± 0.707.

f. $x(0) = \lim_{z\to\infty} X(z) = \mathbf{1}$. This agrees with $(\frac{1}{2})^0 = 1$.

N4.

a. $X(z) = \dfrac{1}{1-\frac{1}{4}z^{-1}}$, $|z| > \frac{1}{4}$. $H(z) = z - \frac{1}{4} = z\left(1 - \frac{1}{4}z^{-1}\right)$, $|z| < \infty$.

b. $Y(z) = z\left(1 - \frac{1}{4}z^{-1}\right) \cdot \dfrac{1}{1-\frac{1}{4}z^{-1}} = z$. ROC: $|z| < \infty$ (the whole plane except ∞). So $\mathbf{y(n) = \delta(n+1)}$.

c. **Not causal**: $h(-1) = 1 \neq 0$ (the z^{+1} term). $R_x \cap R_h = \frac{1}{4} < |z| < \infty$, but the zero of H at $z = \frac{1}{4}$ cancels the pole of X. With no pole left at $\frac{1}{4}$, the ROC grows to include $|z| \leq \frac{1}{4}$. This is the "at least" in the convolution property.

N5.

a. Loop: $H_L = \dfrac{G}{1+GF} = \dfrac{z^{-1}}{1-0.8z^{-1}}$. Cascade with $H_1(z) = 1 + z^{-1}$:

$$H(z) = \frac{z^{-1}(1+z^{-1})}{1-0.8z^{-1}} = \frac{z+1}{z(z-0.8)}$$

b. Zero at $z = -1$. Poles at $z = 0$ and $z = 0.8$. $H(\infty) = 0$, so there is a zero at $z = \infty$. Count: 2 zeros and 2 poles.

c. All blocks are causal and the loop has a delay, so the system is **causal** with **ROC** $|z| > 0.8$.

d. The ROC contains $|z| = 1$, so $H(\omega)$ exists. $H(1) = \dfrac{2}{0.2} = \mathbf{10}$. $H(-1) = \dfrac{0}{\dots} = \mathbf{0}$ (the zero on the unit circle). Low-pass.

N6.

a. Split: $a^n u(n)$ for $n \geq 0$ gives $\dfrac{1}{1-az^{-1}}$, $|z| > |a|$. For $n \leq -1$, $a^{-n}u(-n-1) = \sum_{m\geq 1} a^m z^m = \dfrac{az}{1-az}$, $|z| < \frac{1}{|a|}$.

$$X(z) = \frac{1}{1-az^{-1}} + \frac{az}{1-az} = \frac{(1-a^2)z}{(z-a)(1-az)}, \qquad |a| < |z| < \frac{1}{|a|}$$

The ring is non-empty only if $|a| < 1/|a|$, which means $|a| < 1$.

b. $a = \frac{1}{2}$: $X(z) = \dfrac{\frac{3}{4}z}{(z-\frac{1}{2})(1-\frac{1}{2}z)} = \dfrac{-3z}{(2z-1)(z-2)}$. Poles at $\frac{1}{2}$ and 2. A zero at 0, and a zero at ∞ ($X(\infty) = 0$). ROC: $\frac{1}{2} < |z| < 2$. This is the same annulus as the figure in Example 2.

c. $X(0) = X(1) = \dfrac{-3}{(1)(-1)} = \mathbf{3}$. Check: $\sum(\frac{1}{2})^{|n|} = 2 + 1 = 3$.
$X(\pi) = X(-1) = \dfrac{3}{(-3)(-3)} = \mathbf{\frac{1}{3}}$.

d. $a = 2$: the pieces need $|z| > 2$ and $|z| < \frac{1}{2}$. These do not overlap, so the ROC is empty and **no z-transform exists**.

6. Quick recall check

Question

1. The exterior of a circle, $|z| > r$, through the outermost pole (it includes ∞).
2. $z^3 X(z)$.
3. $0.25 < |z| < 5$.
4. On the unit circle $z = e^{j\omega}$. It exists only if the ROC contains $|z| = 1$.
5. No. The z^{+1} term means $h(-1) = 1 \neq 0$.

Inverse z-transform

Answers and worked solutions · Day 15 · Sat 7 Nov · pack D8

5. Practice

From your course files

T7 Q1 (tutorial).

Hint: factor the 3 out of $(3 - z^{-1})$ first, so that both factors read $(1 - pz^{-1})$.

$$X(z) = \frac{\frac{5}{3}z^{-1}}{(1 - 2z^{-1})\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{1}{1 - 2z^{-1}} - \frac{1}{1 - \frac{1}{3}z^{-1}} \qquad \text{[partial fraction expansion]}$$

Two poles at radii $\frac{1}{3}$ and 2 give three possible ROCs, each with its own signal:

$$x(n) = \begin{cases} 2^n u(n) - (\frac{1}{3})^n u(n), & |z| > 2 \text{ (causal, unstable)} \\ -2^n u(-n-1) - (\frac{1}{3})^n u(n), & \frac{1}{3} < |z| < 2 \text{ (two-sided, the absolutely summable one)} \\ -2^n u(-n-1) + (\frac{1}{3})^n u(-n-1), & |z| < \frac{1}{3} \text{ (anti-causal)} \end{cases}$$

The official solution also lists $2^n u(n) + (\frac{1}{3})^n u(-n-1)$ with ROC $\varnothing$. That one is not a valid signal for this $X(z)$ (its two ROCs $|z| > 2$ and $|z| < \frac{1}{3}$ do not overlap), so there are 3 real answers.

T7 Q2 (tutorial).

Hint: $1 - \frac{10}{3}z^{-1} + z^{-2} = (1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})$.

$$X(z) = \frac{3}{(1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})} = -\frac{\frac{3}{8}}{1 - \frac{1}{3}z^{-1}} + \frac{\frac{27}{8}}{1 - 3z^{-1}} \qquad \text{[partial fraction expansion]}$$

Absolutely summable $x(n)$ means the ROC of $X(z)$ must include the unit circle $|z| = 1$, so the ROC is $\frac{1}{3} < |z| < 3$. The pole at $\frac{1}{3}$ gives a causal term and the pole at 3 an anti-causal term:

$$x(n) = -\frac{3}{8}\left(\frac{1}{3}\right)^n u(n) - \frac{27}{8}\,3^n u(-n-1), \qquad \text{ROC: } \frac{1}{3} < |z| < 3.$$

T7 Q3 (tutorial).

Hint: combine $X(z)$ into one fraction first; then $H = Y/X$.

(a) Using the delay property on the second term of $x(n)$:

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}}, \qquad Y(z) = \frac{1}{1-\frac{1}{3}z^{-1}}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-\frac{1}{2}z^{-1}}{\left(1-\frac{1}{4}z^{-1}\right)\left(1-\frac{1}{3}z^{-1}\right)} = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{7}{12}z^{-1}+\frac{1}{12}z^{-2}} = \frac{3}{1-\frac{1}{4}z^{-1}} - \frac{2}{1-\frac{1}{3}z^{-1}}.$$

As the system is causal, the ROC is $|z| > \frac{1}{3}$ and

$$h(n) = 3\left(\frac{1}{4}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n).$$

(b) Cross-multiply $Y(z)\left(1-\frac{7}{12}z^{-1}+\frac{1}{12}z^{-2}\right) = X(z)\left(1-\frac{1}{2}z^{-1}\right)$ and invert term by term:

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) - \frac{1}{2}x(n-1).$$

(c) Stable, because the ROC of $H(z)$, $|z| > \frac{1}{3}$, includes the unit circle $|z| = 1$.

All three answers were reproduced in Python (sympy partial fractions) and agree with the official solutions.

T8 Q5(c) (tutorial, optional redo).

The full worked solution is Example 2 in section 3 above (and parts (a)–(b) are in pack D10). Key steps:

The zeros of H are 0.8 and $-\frac{3}{2}$ (one inside, one outside the unit circle), so the channel is nonminimum-phase and its stable inverse is noncausal:

$$H_I(z) = H^{-1}(z) = \frac{1-0.8z^{-1}+0.64z^{-2}}{1+0.7z^{-1}-1.2z^{-2}}, \qquad \text{ROC: } 0.8 < |z| < \tfrac{3}{2}.$$

Partial fractions (equal degrees, so a constant $0.64/(-1.2) = -0.5333$ appears):

$$H_I(z) = \frac{0.3478}{1-0.8z^{-1}} + \frac{1.1855}{1+\frac{3}{2}z^{-1}} - 0.5333$$

$$h_I(n) = 0.3478\,(0.8)^n u(n) - 1.1855\left(-\tfrac{3}{2}\right)^n u(-n-1) - 0.5333\,\delta(n),$$

which is absolutely summable, as expected. **Noncausal** (two-sided).

New questions

Q1. Power series [4 marks]

a. Coefficient of z^{-n} is $x(n)$: $z^2 \to x(-2) = 2$, $z^1 \to x(-1) = -1$, $z^0 \to x(0) = 3$, $z^{-2} \to x(2) = -4$.

$$x(n) = \{2,\ -1,\ \underset{\uparrow}{3},\ 0,\ -4\}$$

b. Divide $1 + 2z^{-1}$ by $1 - z^{-1} + \frac{1}{2}z^{-2}$ in ascending powers of z^{-1}:

- 1: remainder $(1 + 2z^{-1}) - (1 - z^{-1} + \frac{1}{2}z^{-2}) = 3z^{-1} - \frac{1}{2}z^{-2}$.
- $3z^{-1}$: remainder $(3z^{-1} - \frac{1}{2}z^{-2}) - (3z^{-1} - 3z^{-2} + \frac{3}{2}z^{-3}) = \frac{5}{2}z^{-2} - \frac{3}{2}z^{-3}$.
- $\frac{5}{2}z^{-2}$: remainder $(\frac{5}{2}z^{-2} - \frac{3}{2}z^{-3}) - (\frac{5}{2}z^{-2} - \frac{5}{2}z^{-3} + \frac{5}{4}z^{-4}) = z^{-3} - \frac{5}{4}z^{-4}$.
- z^{-3}.

$$x(0) = 1,\ x(1) = 3,\ x(2) = 2.5,\ x(3) = 1$$

c. The poles are $\frac{1}{2} \pm j\frac{1}{2}$, radius $1/\sqrt{2}$, so this ROC is the left-sided one. Divide in ascending powers of z: write $\dfrac{2z^{-1} + 1}{\frac{1}{2}z^{-2} - z^{-1} + 1}$. First term $2z^{-1}/(\frac{1}{2}z^{-2}) = 4z$.

Remainder
$(2z^{-1} + 1) - 4z(\frac{1}{2}z^{-2} - z^{-1} + 1) = (2z^{-1} + 1) - (2z^{-1} - 4 + 4z) = 5 - 4z$.
Next term $5/(\frac{1}{2}z^{-2}) = 10z^2$.

Series $4z + 10z^2 + \cdots$, so $x(-1) = 4$, $x(-2) = 10$ (and $x(n) = 0$ for $n \geq 0$). Checked in Python: the next terms are $x(-3) = 12$, $x(-4) = 4$.

Q2. LCCDE, improper $H(z)$ and system types [7 marks]

a. $Y(z)(1 - 0.5z^{-1}) = X(z)(1 + z^{-1})$, so

$$H_1(z) = \frac{1 + z^{-1}}{1 - 0.5z^{-1}} = \frac{z + 1}{z - 0.5}$$

Zero at $z = -1$, pole at $z = 0.5$. ROC $|z| > 0.5$ (causal).

b. $M = N = 1$, so divide first. Quotient $= 1/(-0.5) = -2$. Remainder:
$(1 + z^{-1}) - (-2)(1 - 0.5z^{-1}) = 3$.

$$H_1(z) = -2 + \frac{3}{1 - 0.5z^{-1}} \quad \Rightarrow \quad \mathbf{h_1(n) = -2\delta(n) + 3(0.5)^{nu}(n)}$$

Check: $h_1(0) = 1$, $h_1(1) = 1.5$, $h_1(2) = 0.75$, the same as running the recursion.

c.

System	$H(z)$	Type	Poles	Stable?
S1	$\dfrac{1+z^{-1}}{1-0.5z^{-1}}$	pole-zero (ARMA)	0.5	yes, $0.5<1$
S2	$1-\frac{1}{8}z^{-3}=\dfrac{z^3-\frac{1}{8}}{z^3}$	FIR (MA)	triple pole at $z=0$; zeros at $0.5e^{j2\pi k/3}$, $k=0,1,2$	yes, always (FIR)
S3	$\dfrac{1}{1-0.9z^{-2}}=\dfrac{z^2}{z^2-0.9}$	all-pole (AR)	$\pm\sqrt{0.9}=\pm0.9487$; double zero at $z=0$	yes, $0.9487<1$

Q3. Find all possible sequences [7 marks]

a. Numerator degree 2 < denominator degree 3: proper. Cover-up with $w=z^{-1}$:

$$A_{1/2}=\left[\frac{1+w^2}{(1-\frac{1}{3}w)(1-3w)}\right]_{w=2}=\frac{5}{(\frac{1}{3})(-5)}=-3$$

$$A_{1/3}=\left[\frac{1+w^2}{(1-\frac{1}{2}w)(1-3w)}\right]_{w=3}=\frac{10}{(-\frac{1}{2})(-8)}=\frac{5}{2}$$

$$A_{3}=\left[\frac{1+w^2}{(1-\frac{1}{2}w)(1-\frac{1}{3}w)}\right]_{w=1/3}=\frac{10/9}{(\frac{5}{6})(\frac{8}{9})}=\frac{3}{2}$$

$$X(z)=\frac{-3}{1-\frac{1}{2}z^{-1}}+\frac{5/2}{1-\frac{1}{3}z^{-1}}+\frac{3/2}{1-3z^{-1}}$$

Check at $w=0$: $-3+2.5+1.5=1=X|_{w=0}$.

b. Pole radii $\frac{1}{3}<\frac{1}{2}<3$ give 4 ROCs:

ROC	$x(n)$	Type
$\lvert z\rvert > 3$	$\left[-3(\frac{1}{2})^n + \frac{5}{2}(\frac{1}{3})^n + \frac{3}{2}\,3^n\right]u(n)$	causal (unstable)
$\frac{1}{2} < \lvert z\rvert < 3$	$\left[-3(\frac{1}{2})^n + \frac{5}{2}(\frac{1}{3})^n\right]u(n) - \frac{3}{2}\,3^n u(-n-1)$	two-sided (stable)
$\frac{1}{3} < \lvert z\rvert < \frac{1}{2}$	$\frac{5}{2}(\frac{1}{3})^n u(n) + \left[3(\frac{1}{2})^n - \frac{3}{2}\,3^n\right]u(-n-1)$	two-sided (unstable)
$\lvert z\rvert < \frac{1}{3}$	$\left[3(\frac{1}{2})^n - \frac{5}{2}(\frac{1}{3})^n - \frac{3}{2}\,3^n\right]u(-n-1)$	anti-causal (unstable)

The other $2^3 - 4 = 4$ sign combinations have an empty ROC.

c. Only $\frac{1}{2} < \lvert z\rvert < 3$ contains $\lvert z\rvert = 1$, so only that $x(n)$ has a DTFT.

Check: $X(1) = \dfrac{2}{(\frac{1}{2})(\frac{2}{3})(-2)} = -3$. From the sequence:
$\sum_n x(n) = -3 \cdot 2 + \frac{5}{2} \cdot \frac{3}{2} - \frac{3}{2}\sum_{n\le -1} 3^n = -6 + 3.75 - \frac{3}{2} \cdot \frac{1}{2} = -3$. They agree.

Q4. System ID → LCCDE → stability [10 marks]

a. $X(z) = \dfrac{1}{1 - z^{-1}}$, $\lvert z\rvert > 1$.

$$Y(z) = \frac{2/3}{1 - z^{-1}} - \frac{1/5}{1 - \frac{1}{2}z^{-1}} + \frac{8/15}{1 + 2z^{-1}} = \frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})(1 + 2z^{-1})}, \quad \lvert z\rvert > 2$$

(Combine over the common denominator. Numerator check: at $w = 0$, $\frac{2}{3} - \frac{1}{5} + \frac{8}{15} = 1$; the w and w^2 coefficients cancel, which Python confirms.)

b.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + 2z^{-1})} = \frac{1}{1 + \frac{3}{2}z^{-1} - z^{-2}}$$

Cross-multiply: $Y(z)(1 + \frac{3}{2}z^{-1} - z^{-2}) = X(z)$, so

$$\mathbf{y(n) + \tfrac{3}{2}y(n-1) - y(n-2) = x(n)}$$

It is an all-pole system, with poles 0.5 and -2 and a double zero at $z = 0$.

c. $H(z) = \dfrac{1/5}{1 - \frac{1}{2}z^{-1}} + \dfrac{4/5}{1 + 2z^{-1}}$ (cover-up: $w = 2 \Rightarrow 1/(1+4) = \frac{1}{5}$;
$w = -\frac{1}{2} \Rightarrow 1/(1 + \frac{1}{4}) = \frac{4}{5}$). Causal → ROC $\lvert z\rvert > 2$:

$$\mathbf{h(n) = \left[\frac{1}{5}\left(\frac{1}{2}\right)^{n} + \frac{4}{5}(-2)^{n}\right]u(n)}$$

Unstable. The ROC $|z| > 2$ excludes $|z| = 1$ (the pole at -2 is outside the unit circle). Consistency check: $R_h \cap R_x = \{|z| > 2\}$, which matches the ROC of Y.

d. Stable → the ROC must contain $|z| = 1$ → $\frac{1}{2} < |z| < 2$. Pole $\frac{1}{2}$ is causal, pole -2 is anti-causal:

$$\mathbf{h(n) = \frac{1}{5}\left(\frac{1}{2}\right)^{nu}(n) - \frac{4}{5}(-2)^{nu}(-n-1)}$$

Not causal (two-sided): $h(n) \neq 0$ for $n < 0$.

e. No. Now $R_y = R_h \cap R_x = \{1 < |z| < 2\}$, so the $(-2)^n$ term in the output becomes anti-causal: $y(n) = \left[\frac{2}{3} - \frac{1}{5}\left(\frac{1}{2}\right)^n\right]u(n) - \frac{8}{15}(-2)^n u(-n-1)$. Same LCCDE, different ROC, different system.

All values were checked in Python (scipy `lfilter` reproduces $y(0..5) = 1, -0.5, 2.75, -3.625, 9.1875, -16.406$ from the causal h).

Q5. Improper $H(z)$ with complex poles [8 marks]

a. Zeros: $1 + z^{-2} = 0 \Rightarrow z = \pm j$ (on the unit circle). Poles: $p^2 - p + \frac{1}{2} = 0 \Rightarrow p = \frac{1}{2} \pm j\frac{1}{2} = 0.7071e^{\pm j\pi/4}$. Causal, and both poles are inside the unit circle, so it is **stable**.

b. Divide in ascending powers of z^{-1}:

- 1: remainder $(1 + z^{-2}) - (1 - z^{-1} + \frac{1}{2}z^{-2}) = z^{-1} + \frac{1}{2}z^{-2}$
- z^{-1}: remainder $(z^{-1} + \frac{1}{2}z^{-2}) - (z^{-1} - z^{-2} + \frac{1}{2}z^{-3}) = \frac{3}{2}z^{-2} - \frac{1}{2}z^{-3}$
- $\frac{3}{2}z^{-2}$: remainder $(\frac{3}{2}z^{-2} - \frac{1}{2}z^{-3}) - (\frac{3}{2}z^{-2} - \frac{3}{2}z^{-3} + \frac{3}{4}z^{-4}) = z^{-3} - \frac{3}{4}z^{-4}$
- z^{-3}

$$h(0) = 1,\ h(1) = 1,\ h(2) = 1.5,\ h(3) = 1$$

c. $M = N = 2$: quotient $= 1/\frac{1}{2} = 2$. Remainder $(1 + z^{-2}) - 2(1 - z^{-1} + \frac{1}{2}z^{-2}) = -1 + 2z^{-1}$:

$$H(z) = 2 + \frac{-1 + 2z^{-1}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

Cover-up for $p = \frac{1}{2} + j\frac{1}{2}$: $1/p = 1 - j$. Numerator: $-1 + 2(1 - j) = 1 - 2j$. Other factor: $1 - p^*(1/p) = 1 + j$ (as in Example 3b).

$$A = \frac{1 - 2j}{1 + j} = \frac{(1 - 2j)(1 - j)}{2} = -\frac{1}{2} - j\frac{3}{2}, \qquad |A| = \frac{\sqrt{10}}{2},\ \angle A = -108.43^\circ$$

$2r^n[\mathrm{Re}\, A \cos\theta n - \mathrm{Im}\, A \sin\theta n] = r^n[-\cos\frac{\pi n}{4} + 3\sin\frac{\pi n}{4}]$, so

$$\boxed{h(n) = 2\delta(n) + \left(\frac{1}{\sqrt{2}}\right)^n \left[3\sin\frac{\pi n}{4} - \cos\frac{\pi n}{4}\right] u(n)}$$

Equivalently $2\delta(n) + \sqrt{10}\,(0.7071)^n \cos(\frac{\pi n}{4} - 108.43^\circ)u(n)$.

Check: $n = 0$: $2 + (0 - 1) = 1$. $n = 1$: $0.7071(2.1213 - 0.7071) = 1$. $n = 2$: $0.5(3 - 0) = 1.5$. $n = 3$: $0.3536(2.1213 + 0.7071) = 1$. Matches (b). Python's `lfilter` agrees to $n = 9$.

d. A zero at $z = e^{j\pi/2} = j$ means $H(\omega)|_{\omega=\pi/2} = 0$. The steady-state output is **0**.

6. Quick recall check

Question

1. $-4(0.8)^n u(-n-1)$.
2. No. Causal means ROC $|z| > 1.1$, which excludes the unit circle.
3. Four (ROCs: $|z| < 0.5$, $0.5 < |z| < 2$, $2 < |z| < 5$, $|z| > 5$). The stable one is $0.5 < |z| < 2$.
4. $y(n) - 0.25y(n-2) = x(n) + 2x(n-1)$.
5. FIR (all-zero, MA). Triple pole at $z = 0$. Always stable.

Unilateral z-transform and mixed practice

Answers and worked solutions · Day 16 · Sun 8 Nov · pack D9

5. Practice

From your course files

T3 Q3 (tutorial; do (b), redone with $Y = HX$).

Hint: at rest, so $Y = HX$. This is the zero-state part of Example 2.

(a) $\lambda^2 - \frac{3}{4}\lambda + \frac{1}{8} = 0 \Rightarrow \lambda = \frac{1}{2}, \frac{1}{4}$. The modes are 2^{-n} and 4^{-n}.

(b) Transform the LCCDE with zero initial conditions: $H(z) = \dfrac{1 - z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)}$, and $X(z) = \dfrac{1}{1 - \frac{1}{3}z^{-1}}$, $|z| > \frac{1}{3}$. So

$$Y(z) = \frac{1 - z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{-6}{1 - \frac{1}{2}z^{-1}} + \frac{-9}{1 - \frac{1}{4}z^{-1}} + \frac{16}{1 - \frac{1}{3}z^{-1}}, \qquad |z| > \tfrac{1}{2}.$$

(Cover-up: at $z^{-1} = 2$: $\frac{1-2}{(1-\frac{1}{2})(1-\frac{2}{3})} = -6$; at $z^{-1} = 4$: $\frac{1-4}{(1-2)(1-\frac{4}{3})} = -9$; at $z^{-1} = 3$: $\frac{1-3}{(1-\frac{3}{2})(1-\frac{3}{4})} = 16$.) All three terms are causal:

$$y[n] = \left(-6 \cdot 2^{-n} - 9 \cdot 4^{-n} + 16 \cdot 3^{-n}\right)u[n],$$

which is exactly the official (time-domain) answer.

T3 Q5 (tutorial; do (b) and (c) via $H(z)$).

Hint: take the causal ROC (outside the largest pole). For (c), $H = \frac{1}{(1-z^{-1})^2}$, then use the $(n+1)p^n$ pair with $p = 1$.

(b)

$$H(z) = \frac{1}{1 + 0.2z^{-1} - 0.08z^{-2}} = \frac{1}{(1 - 0.2z^{-1})(1 + 0.4z^{-1})} = \frac{\frac{1}{3}}{1 - 0.2z^{-1}} + \frac{\frac{2}{3}}{1 + 0.4}$$

, ROC $|z| > 0.4$, so

$$h[n] = \left(\tfrac{1}{3}(0.2)^n + \tfrac{2}{3}(-0.4)^n\right)u[n].$$

(c) $H(z) = \dfrac{1}{1 - 2z^{-1} + z^{-2}} = \dfrac{1}{(1 - z^{-1})^2}$, a double pole at $z = 1$, ROC $|z| > 1$. With $(n+1)p^n u[n] \leftrightarrow \dfrac{1}{(1 - pz^{-1})^2}$ and $p = 1$:

$$h[n] = (1+n)u[n].$$

Both agree with the official solutions, which find the same $h[n]$ from the characteristic modes and $h[0]$, $h[1]$.

T7 Q1 (tutorial).

Hint: factor the 3 out of $(3 - z^{-1})$ first, so that both factors read $(1 - pz^{-1})$.

$$X(z) = \frac{\frac{5}{3}z^{-1}}{(1-2z^{-1})\left(1-\frac{1}{3}z^{-1}\right)} = \frac{1}{1-2z^{-1}} - \frac{1}{1-\frac{1}{3}z^{-1}} \qquad [\text{partial fraction expansion}]$$

Two poles at radii $\frac{1}{3}$ and 2 give three possible ROCs, each with its own signal:

$$x(n) = \begin{cases} 2^n u(n) - (\frac{1}{3})^n u(n), & |z| > 2 \text{ (causal, unstable)} \\ -2^n u(-n-1) - (\frac{1}{3})^n u(n), & \frac{1}{3} < |z| < 2 \text{ (two-sided, the absolutely summable one)} \\ -2^n u(-n-1) + (\frac{1}{3})^n u(-n-1), & |z| < \frac{1}{3} \text{ (anti-causal)} \end{cases}$$

The official solution also lists $2^n u(n) + (\frac{1}{3})^n u(-n-1)$ with ROC $\varnothing$. That one is not a valid signal for this $X(z)$ (its two ROCs $|z| > 2$ and $|z| < \frac{1}{3}$ do not overlap), so there are 3 real answers.

T7 Q2 (tutorial).

Hint: $1 - \frac{10}{3}z^{-1} + z^{-2} = (1 - \frac{1}{3}z^{-1})(1 - 3z^{-1})$.

$$X(z) = \frac{3}{\left(1-\frac{1}{3}z^{-1}\right)(1-3z^{-1})} = -\frac{\frac{3}{8}}{1-\frac{1}{3}z^{-1}} + \frac{\frac{27}{8}}{1-3z^{-1}} \qquad [\text{partial fraction expansion}]$$

Absolutely summable $x(n)$ means the ROC of $X(z)$ must include the unit circle $|z| = 1$, so the ROC is $\frac{1}{3} < |z| < 3$. The pole at $\frac{1}{3}$ gives a causal term and the pole at 3 an anti-causal term:

$$x(n) = -\frac{3}{8}\left(\frac{1}{3}\right)^n u(n) - \frac{27}{8}\,3^n u(-n-1), \qquad \text{ROC: } \tfrac{1}{3} < |z| < 3.$$

T7 Q3 (tutorial).

Hint: combine $X(z)$ into one fraction first; then $H = Y/X$.

(a) Using the delay property on the second term of $x(n)$:

$$X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1-\frac{1}{4}z^{-1}}{1-\frac{1}{2}z^{-1}}, \qquad Y(z) = \frac{1}{1-\frac{1}{3}z^{-1}}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{1}{2}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{7}{12}z^{-1} + \frac{1}{12}z^{-2}} = \frac{3}{1 - \frac{1}{4}z^{-1}} - \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

As the system is causal, the ROC is $|z| > \frac{1}{3}$ and

$$h(n) = 3\left(\frac{1}{4}\right)^n u(n) - 2\left(\frac{1}{3}\right)^n u(n).$$

(b) Cross-multiply $Y(z)\left(1 - \frac{7}{12}z^{-1} + \frac{1}{12}z^{-2}\right) = X(z)\left(1 - \frac{1}{2}z^{-1}\right)$ and invert term by term:

$$y(n) - \frac{7}{12}y(n-1) + \frac{1}{12}y(n-2) = x(n) - \frac{1}{2}x(n-1).$$

(c) Stable, because the ROC of $H(z)$, $|z| > \frac{1}{3}$, includes the unit circle $|z| = 1$.

All three answers were reproduced in Python (sympy partial fractions) and agree with the official solutions.

New questions

Part A: unilateral z-transform (graded easy to hard)

a. $Y - 0.5\left[z^{-1}Y + 4\right] = \frac{1}{1-z^{-1}}$, so $Y(1 - 0.5z^{-1}) = 2 + \frac{1}{1-z^{-1}}$:

$$Y(z) = \frac{2}{1 - 0.5z^{-1}} + \frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})}$$

c. The first term is the zero-input part: $y_{zi}(n) = 2(0.5)^n u(n)$. For the second term, the residue at $z^{-1} = 1$ is $\frac{1}{0.5} = 2$ and at $z^{-1} = 2$ it is $\frac{1}{1-2} = -1$. So $y_{zs}(n) = \left(2 - (0.5)^n\right)u(n)$.

d. $\boxed{y(n) = \left(2 + (0.5)^n\right)u(n)}$. Check: $y(0) = 1 + 0.5(4) = 3$ (checks), $y(1) = 1 + 1.5 = 2.5$ (checks).

a. $a_1 = -0.7$, $a_2 = 0.1$. $I(z) = -0.7(1) + 0.1\left[z^{-1}(1) + 5\right] = -0.2 + 0.1z^{-1}$. With $A(z) = (1 - 0.5z^{-1})(1 - 0.2z^{-1})$, we get $A(z)Y = 1 + 0.2 - 0.1z^{-1}$.

b. $Y_{zi} = \frac{0.2 - 0.1z^{-1}}{A(z)} = \frac{0.2(1 - 0.5z^{-1})}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})} = \frac{0.2}{1 - 0.2z^{-1}}$. So $y_{zi}(n) = 0.2(0.2)^n u(n)$. The ICs excite only the 0.2^n mode.

c. $Y_{zs} = \frac{1}{A(z)} = H(z)$, because $X(z) = 1$. The residue at $z^{-1} = 2$ is $\frac{1}{1-0.4} = \frac{5}{3}$, and at $z^{-1} = 5$ it is $\frac{1}{1-2.5} = -\frac{2}{3}$. So $y_{zs}(n) = h(n) = \left(\frac{5}{3}(0.5)^n - \frac{2}{3}(0.2)^n\right)u(n)$. The impulse response is defined with zero ICs (L04 slide 13), so it is the zero-state response to $\delta(n)$.

d. $\boxed{y(n) = \left(\frac{5}{3}(0.5)^n - \frac{7}{15}(0.2)^n\right)u(n)}$. The recursion gives $y(0) = 1 + 0.7(1) - 0.1(5) = 1.2$ and $y(1) = 0.7(1.2) - 0.1(1) = 0.74$. The formula gives $\frac{5}{3} - \frac{7}{15} = 1.2$ (checks) and $\frac{5}{6} - \frac{7}{75} = 0.74$ (checks).

Here $a_1 = 0$ and $a_2 = -0.25$. The input term $x(n-1)$ gives $z^{-1}X$ with no extra term, because $x(-1) = 0$.

$I(z) = -0.25\left[z^{-1}(2) + (-4)\right] = 1 - 0.5z^{-1}$. Also
$A(z) = 1 - 0.25z^{-2} = (1 - 0.5z^{-1})(1 + 0.5z^{-1})$.

Zero-input: $Y_{zi} = \frac{-(1-0.5z^{-1})}{(1-0.5z^{-1})(1+0.5z^{-1})} = \frac{-1}{1+0.5z^{-1}}$, so $y_{zi}(n) = -(-0.5)^n u(n)$.

Zero-state: $Y_{zs} = \frac{1+z^{-1}}{(1-0.5z^{-1})(1+0.5z^{-1})(1-\frac{1}{3}z^{-1})}$. The residues are:

- at $z^{-1} = 2$: $\frac{3}{(2)(\frac{1}{3})} = 4.5$
- at $z^{-1} = -2$: $\frac{-1}{(2)(\frac{5}{3})} = -0.3$
- at $z^{-1} = 3$: $\frac{4}{(-0.5)(2.5)} = -3.2$

So $y_{zs}(n) = \left(4.5(0.5)^n - 0.3(-0.5)^n - 3.2(\frac{1}{3})^n\right)u(n)$.

TOTAL:

$$\boxed{y(n) = \left(4.5(0.5)^n - 1.3(-0.5)^n - 3.2(\tfrac{1}{3})^n\right)u(n)}$$

Check: the recursion gives $y(0) = x(0) + x(-1) + 0.25y(-2) = 1 + 0 - 1 = 0$, and the formula gives $4.5 - 1.3 - 3.2 = 0$ (checks). Also $y(1) = \frac{1}{3} + 1 + 0.25(2) = \frac{11}{6}$, and the formula gives $2.25 + 0.65 - 1.0667 = 1.8333$ (checks).

$A(z) = (1 - 0.5z^{-1})^2$, a double pole at 0.5.
$I(z) = -1(2) + 0.25\left[2z^{-1} + 0\right] = -2 + 0.5z^{-1}$.

a. $Y_{zi} = \frac{2-0.5z^{-1}}{(1-0.5z^{-1})^2} = \frac{B}{1-0.5z^{-1}} + \frac{C}{(1-0.5z^{-1})^2}$. Then $C =$ the numerator at $z^{-1} = 2$, which is $2 - 1 = 1$. Setting $z^{-1} = 0$ gives $2 = B + C$, so $B = 1$.

$$y_{zi}(n) = \left[(0.5)^n + (n+1)(0.5)^n\right]u(n) = (n+2)(0.5)^n u(n)$$

b. $Y_{zs} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})^2}$. The residue at $z^{-1} = 1$ is $\frac{1}{0.25} = 4$. C at $z^{-1} = 2$ is $\frac{1}{1-2} = -1$. Setting $z^{-1} = 0$ gives $1 = 4 + B - 1$, so $B = -2$.

$$y_{zs}(n) = \left[4 - 2(0.5)^n - (n+1)(0.5)^n\right]u(n) = \left[4 - (n+3)(0.5)^n\right]u(n)$$

c. $\boxed{y(n) = \left(4 - (0.5)^n\right)u(n)}$. The $n(0.5)^n$ terms cancel exactly. You can see why in the combined form: the numerator $1 + (2 - 0.5z^{-1})(1 - z^{-1})$ factors as $(1 - 0.5z^{-1})(3 - z^{-1})$, which cancels one of the two pole factors. So $Y = \frac{3-z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$.

d. $y(0) = 1 + 2 - 0 = 3$ and $y(1) = 1 + 3 - 0.25(2) = 3.5$. The formula gives $4 - 1 = 3$ (checks) and $4 - 0.5 = 3.5$ (checks).

a. $H(z) = \frac{1}{1-0.9z^{-1}+0.2z^{-2}} = \frac{1}{(1-0.5z^{-1})(1-0.4z^{-1})}$. The poles are 0.5 and 0.4, so the causal system is stable. The steady state is $H(1) = \frac{1}{0.5 \times 0.6} = \frac{10}{3}$.

b. $I(z) = -0.9c + 0.2cz^{-1}$, so

$$Y(z) = \frac{1 + (0.9c - 0.2cz^{-1})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.4z^{-1})}$$

c. Set the numerator to zero at $z^{-1} = 2$: $1 + (0.9c - 0.4c)(1 - 2) = 1 - 0.5c = 0$, so $\boxed{c = y(-1) = 2}$.

d. With $c = 2$, the numerator is $2.8 - 2.2z^{-1} + 0.4z^{-2} = (1 - 0.5z^{-1})(2.8 - 0.8z^{-1})$. So

$$Y = \frac{2.8 - 0.8z^{-1}}{(1 - z^{-1})(1 - 0.4z^{-1})}$$

The residue at $z^{-1} = 1$ is $\frac{2}{0.6} = \frac{10}{3}$, and at $z^{-1} = 2.5$ it is $\frac{0.8}{-1.5} = -\frac{8}{15}$.

$$\boxed{y(n) = \left(\tfrac{10}{3} - \tfrac{8}{15}(0.4)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 1 + 0.9(2) - 0 = 2.8$, and the formula gives $\frac{10}{3} - \frac{8}{15} = 2.8$ (checks).

a. Factor the denominator: $1 - 0.25z^{-1} - 0.125z^{-2} = (1 - 0.5z^{-1})(1 + 0.25z^{-1})$. So

$$H(z) = \frac{1 - z^{-1}}{(1 - 0.5z^{-1})(1 + 0.25z^{-1})} = \frac{z(z - 1)}{(z - 0.5)(z + 0.25)}$$

The zeros are at $z = 0$ and $z = 1$. The poles are at 0.5 and -0.25. For the causal system the ROC is $|z| > 0.5$, which contains the unit circle, so the system is **stable**.

b. $I(z) = -0.25(4) - 0.125\left[4z^{-1} - 8\right] = -1 - 0.5z^{-1} + 1 = -0.5z^{-1}$. So $Y_{zi} = \frac{0.5z^{-1}}{(1-0.5z^{-1})(1+0.25z^{-1})}$.

The residue at $z^{-1} = 2$ is $\frac{1}{1.5} = \frac{2}{3}$, and at $z^{-1} = -4$ it is $\frac{-2}{3} = -\frac{2}{3}$.

$$y_{zi}(n) = \tfrac{2}{3}\left[(0.5)^n - (-0.25)^n\right]u(n)$$

Note $y_{zi}(0) = 0$. This is consistent, because $0.25(4) + 0.125(-8) = 0$.

c. $Y_{zs} = \frac{1-z^{-1}}{(1-0.5z^{-1})^2(1+0.25z^{-1})}$. The input pole at 0.5 lands on a system pole, so there is a double pole and therefore an $(n + 1)(0.5)^n$ term.

- C ($z^{-1} = 2$): $\frac{1-2}{1+0.5} = -\frac{2}{3}$

- A for the (-0.25) term ($z^{-1} = -4$): $\frac{1+4}{(1+2)^2} = \frac{5}{9}$
- B: setting $z^{-1} = 0$ gives $1 = B + C + A$, so $B = \frac{10}{9}$

$$y_{zs}(n) = \left[\tfrac{10}{9}(0.5)^n - \tfrac{2}{3}(n+1)(0.5)^n + \tfrac{5}{9}(-0.25)^n\right]u(n) = \left[\left(\tfrac{4}{9} - \tfrac{2}{3}n\right)(0.5)^n + \tfrac{5}{9}(-0.25)^n\right]u(n)$$

d. $$\boxed{y(n) = \left[\left(\tfrac{10}{9} - \tfrac{2}{3}n\right)(0.5)^n - \tfrac{1}{9}(-0.25)^n\right]u(n)}$$

The recursion gives $y(0) = 1 - 0 + 0.25(4) + 0.125(-8) = 1$, and the formula gives $\frac{10}{9} - \frac{1}{9} = 1$ (checks). Next, $y(1) = 0.5 - 1 + 0.25(1) + 0.125(4) = 0.25$, and the formula gives $\frac{5}{9} - \frac{1}{3} + \frac{1}{36} = \frac{1}{4}$ (checks).

Part B: mixed z-domain set (exam-style, about 12 min each)

a. In powers of z, $X(z) = \frac{z(2z-2.5)}{(z-0.5)(z+2)}$. The zeros are at $z = 0$ and $z = 1.25$. The poles are at $z = 0.5$ and $z = -2$.

b. The residue at $z^{-1} = 2$ is $\frac{2-5}{1+4} = -0.6$, and at $z^{-1} = -0.5$ it is $\frac{2+1.25}{1+0.25} = 2.6$.

$$X(z) = \frac{-0.6}{1 - 0.5z^{-1}} + \frac{2.6}{1 + 2z^{-1}}$$

c.

ROC	$x(n)$	Type
$\lvert z\rvert > 2$	$-0.6(0.5)^n u(n) + 2.6(-2)^n u(n)$	causal
$0.5 < \lvert z\rvert < 2$	$-0.6(0.5)^n u(n) - 2.6(-2)^n u(-n-1)$	two-sided
$\lvert z\rvert < 0.5$	$0.6(0.5)^n u(-n-1) - 2.6(-2)^n u(-n-1)$	anti-causal

d. Only the ring $0.5 < |z| < 2$ contains $|z| = 1$, so only the two-sided signal has a DTFT. $X(\omega)\big|_{\omega=0} = X(1) = \frac{2-2.5}{(0.5)(3)} = -\frac{1}{3}$.

e. $X^+(z)$ keeps only the $n \geq 0$ samples, which here are $-0.6(0.5)^n u(n)$. So $X^+(z) = \frac{-0.6}{1-0.5z^{-1}}$. It differs from $X(z)$ because the signal is not causal. Its left-sided part, $-2.6(-2)^n u(-n-1)$, is thrown away.

Numerical check: summing each sequence against z^{-n} at a test point inside each ROC reproduced $X(z)$ to at least 10 significant figures.

a. Factor the denominator: $1 + 0.1z^{-1} - 0.3z^{-2} = (1 - 0.5z^{-1})(1 + 0.6z^{-1})$. So $H(z) = \frac{1}{(1-0.5z^{-1})(1+0.6z^{-1})}$ with ROC $|z| > 0.6$. The poles are 0.5 and -0.6.

The residue at $z^{-1} = 2$ is $\frac{1}{2.2} = \frac{5}{11}$. At $z^{-1} = -\frac{5}{3}$ it is $\frac{1}{1+\frac{5}{6}} = \frac{6}{11}$.

$$h(n) = \left(\tfrac{5}{11}(0.5)^n + \tfrac{6}{11}(-0.6)^n\right)u(n)$$

b. Yes. The system is causal and both poles are inside $|z| = 1$, so the ROC $|z| > 0.6$ contains the unit circle.

c. $I(z) = 0.1(1) - 0.3\left[z^{-1}(1) + 0\right] = 0.1 - 0.3z^{-1}$.

Zero-input: $Y_{zi} = \frac{-0.1+0.3z^{-1}}{(1-0.5z^{-1})(1+0.6z^{-1})}$. The residue at $z^{-1} = 2$ is $\frac{0.5}{2.2} = \frac{5}{22}$, and at $z^{-1} = -\frac{5}{3}$ it is $\frac{-0.6}{11/6} = -\frac{18}{55}$.

$$y_{zi}(n) = \left(\tfrac{5}{22}(0.5)^n - \tfrac{18}{55}(-0.6)^n\right)u(n)$$

Zero-state: $Y_{zs} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})(1+0.6z^{-1})}$. The residues are:

- at $z^{-1} = 1$: $\frac{1}{0.5\times 1.6} = 1.25$
- at $z^{-1} = 2$: $\frac{1}{(-1)(2.2)} = -\frac{5}{11}$
- at $z^{-1} = -\frac{5}{3}$: $\frac{1}{(\frac{8}{3})(\frac{11}{6})} = \frac{9}{44}$

$$y_{zs}(n) = \left(1.25 - \tfrac{5}{11}(0.5)^n + \tfrac{9}{44}(-0.6)^n\right)u(n)$$

TOTAL:

$$\boxed{y(n) = \left(1.25 - \tfrac{5}{22}(0.5)^n - \tfrac{27}{220}(-0.6)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 1 - 0.1(1) + 0.3(0) = 0.9$, and the formula gives $1.25 - 0.2273 - 0.1227 = 0.9$ (checks).

d. $y(n)$ settles to $1.25 = H(1)$. The zero-input response contains only the system modes $(0.5)^n$ and $(-0.6)^n$, and both decay. For a stable system the ICs therefore affect only the transient.

a. $Y(z) = \frac{4}{1-z^{-1}} - \frac{4}{1-0.5z^{-1}} + \frac{1}{1-0.25z^{-1}}$. Over the common denominator, the numerator is

$$4(1 - 0.75z^{-1} + 0.125z^{-2}) - 4(1 - 1.25z^{-1} + 0.25z^{-2}) + (1 - 1.5z^{-1} + 0.5z^{-2}) = 1$$

Then $H = Y/X = Y(1 - z^{-1})$:

$$H(z) = \frac{1 + 0.5z^{-1}}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{1 + 0.5z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

b. $y(n) - 0.75y(n-1) + 0.125y(n-2) = x(n) + 0.5x(n-1)$.

c. The residue at $z^{-1} = 2$ is $\frac{2}{0.5} = 4$, and at $z^{-1} = 4$ it is $\frac{3}{-1} = -3$. So $h(n) = \left(4(0.5)^n - 3(0.25)^n\right)u(n)$. The system is stable because both poles are inside the unit circle and the ROC is $|z| > 0.5$. Also $H(1) = 4$, which agrees with the step steady state.

d. $I(z) = -0.75(4) + 0.125\left[4z^{-1} + 4\right] = -2.5 + 0.5z^{-1}$. So $Y_{zi} = \frac{2.5 - 0.5z^{-1}}{(1-0.5z^{-1})(1-0.25z^{-1})}$. The residue at $z^{-1} = 2$ is $\frac{1.5}{0.5} = 3$, and at $z^{-1} = 4$ it is $\frac{0.5}{-1} = -0.5$. The zero-state part is the given step response.

$$\boxed{y(n) = \left[4 - (0.5)^n + 0.5(0.25)^n\right]u(n)}, \quad y(0) = 3.5$$

Why it is not constant: a step that starts at $n = 0$ has $x(-1) = 0$. At $n = 0$, the term $0.5x(n-1)$ therefore contributes 0, not 0.5. A true steady state needs the input to have been on for ever, so that $x(-1) = 1$. In that case the recursion gives $y(0) = 1 + 0.5 + 3 - 0.5 = 4$. The missing 0.5 sets off a transient that decays.

a. In powers of z, $H = \frac{1.5z}{(z-0.5)(z-2)}$. The zeros are at $z = 0$ and $z = \infty$ (because $H(\infty) = 0$). The poles are at 0.5 and 2. That is two poles and two zeros, as L07 slide 17 requires.

b. The residue at $z^{-1} = 0.5$ is $\frac{0.75}{0.75} = 1$, and at $z^{-1} = 2$ it is $\frac{3}{-3} = -1$. So $H = \frac{1}{1-2z^{-1}} - \frac{1}{1-0.5z^{-1}}$.

ROC	$h(n)$	Causal?	Stable?
$\lvert z\rvert > 2$	$(2^n - 0.5^n)u(n)$	yes	no ($\lvert z\rvert = 1$ is not in the ROC)
$0.5 < \lvert z\rvert < 2$	$-2^n u(-n-1) - 0.5^n u(n)$	no (two-sided)	yes
$\lvert z\rvert < 0.5$	$-2^n u(-n-1) + 0.5^n u(-n-1)$	no (anti-causal)	no

c. $y(n) - 2.5y(n-1) + y(n-2) = 1.5x(n-1)$.

d. $Y = \frac{1.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})(1-2z^{-1})}$. The residues are 2 at the pole 2, -3 at the pole 1 and 1 at the pole 0.5. So $y(n) = \left(2 \cdot 2^n - 3 + 0.5^n\right)u(n)$, which is unbounded.

e. Let $y(-1) = c$. Then $I(z) = -2.5c + cz^{-1}$, and

$$Y = \frac{1.5z^{-1} + (2.5c - cz^{-1})(1 - z^{-1})}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 2z^{-1})}$$

To remove the 2^n mode, the numerator must be zero at $z^{-1} = 0.5$: $0.75 + (2c)(0.5) = 0$, so $\boxed{c = y(-1) = -0.75}$. The numerator then factors as $(1 - 2z^{-1})(-1.875 + 0.375z^{-1})$. That leaves $Y = \frac{-1.875 + 0.375z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$, and

$$\boxed{y(n) = \left(-3 + 1.125(0.5)^n\right)u(n)}$$

Check: the recursion gives $y(0) = 0 + 2.5(-0.75) - 0 = -1.875$, and the formula gives $-3 + 1.125 = -1.875$ (checks).

f. The cancellation needs $y(-1)$ to be exactly -0.75. In simulation, $y(-1) = -0.7501$ gives $y(20) \approx -283$ and $y(39) \approx -1.5 \times 10^8$. Any noise or rounding re-excites the 2^n mode. Stability is a property of $H(z)$ and its ROC (L08 slide 8), not of the initial conditions.

6. Quick recall check

Question

1. $z^{-2}Y(z) + z^{-1}y(-1) + y(-2)$.
2. $\frac{z^{-1}}{1-z^{-1}}$. There is no extra term because $x(-1) = 0$.
3. The zero-state part. The zero-input part contains only the system poles (the characteristic modes), never the input poles.
4. $(n+1)p^n u(n)$.
5. Set $z^{-1} = 0$: $y(0) = 3$ (initial value theorem; this is L08's slide example).

Digital filters

Answers and worked solutions · Day 17 · Mon 9 Nov · pack D10

5. Practice

From your course files

T8 Q1 (tutorial).

Hint: three equations: $H(2\pi/3) = 0$, $H(0) = 1$, $b_0 = b_2$.

$h(n) = \{\underset{\uparrow}{b_0}, b_1, b_2\}$, so $H(\omega) = b_0 + b_1 e^{-j\omega} + b_2 e^{-j2\omega}$. The requirements give

$$0 = b_0 + b_1 e^{-j2\pi/3} + b_2 e^{-j4\pi/3}\ [\text{i}], \qquad 1 = b_0 + b_1 + b_2\ [\text{ii}], \qquad 0 = b_0 - b_2\ [\text{iii}].$$

With $b_0 = b_2$, [i] becomes $e^{-j2\pi/3}\left(2b_0 \cos\frac{2\pi}{3} + b_1\right) = 0$, so $b_1 = b_0$. Then [ii] gives

$$b_0 = b_1 = b_2 = \tfrac{1}{3}.$$

$$H(\omega) = \tfrac{1}{3}e^{-j\omega}\left(e^{j\omega} + 1 + e^{-j\omega}\right) = \tfrac{1}{3}(1 + 2\cos\omega)e^{-j\omega}$$

$$|H(\omega)| = \tfrac{1}{3}|1 + 2\cos\omega|, \qquad \theta(\omega) = \begin{cases} -\omega, & 1 + 2\cos\omega \geq 0 \\ -\omega \pm \pi, & \text{otherwise} \end{cases}$$

Check the sketch: $|H(0)| = 1$, a null at $\omega = 2\pi/3$, $|H(\pi)| = \frac{1}{3}$; the phase is linear with slope -1 and jumps by π at the null.

T8 Q2 (tutorial).

Hint: 5 taps, so the nulls are at $2\pi k/5$.

Geometric series: $H(z) = \dfrac{1}{5}\dfrac{1 - z^{-5}}{1 - z^{-1}}$, so

$$H(\omega) = \frac{1}{5}\frac{e^{-j5\omega/2}\left(e^{j5\omega/2} - e^{-j5\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right)} = \frac{1}{5}e^{-j2\omega}\frac{\sin\frac{5\omega}{2}}{\sin\frac{\omega}{2}}$$

$$|H(\omega)| = \frac{1}{5}\left|\frac{\sin(5\omega/2)}{\sin(\omega/2)}\right|, \qquad \theta(\omega) = \begin{cases} -2\omega, & \dfrac{\sin(5\omega/2)}{\sin(\omega/2)} \geq 0 \\ -2\omega \pm \pi, & \text{otherwise.} \end{cases}$$

$H(\omega) = 0$ at $\omega = 2\pi k/5$, $k = 1, 2$ on $0 \leq \omega \leq \pi$. In continuous time,

$$\Omega = \frac{2\pi k}{5} F_s = 400\pi k \text{ rad/s}, \quad \text{i.e. } F = 200k \text{ Hz}, \; k = 1, 2: \quad F = 200 \text{ Hz and } 400 \text{ Hz}.$$

T8 Q3 (tutorial).

Hint: substitute $z^{-1} = -1$ (that is, $\omega = \pi$) for the gain condition.

Null frequency: $\omega_0 = 2\pi \cdot 50/500 = \pi/5$ rad. Zeros at $e^{\pm j\omega_0}$ give

$$H(\omega) = b_0 \left(1 - 2\cos\omega_0 \, e^{-j\omega} + e^{-j2\omega}\right).$$

At $\omega = \pi$: $1 = b_0 \left(1 + 2\cos\frac{\pi}{5} + 1\right)$, so

$$b_0 = \frac{1/2}{1 + \cos\frac{\pi}{5}} = 0.2764, \qquad H(z) = \frac{1/2}{1 + \cos\frac{\pi}{5}} \left(1 - 2z^{-1}\cos\frac{\pi}{5} + z^{-2}\right) = 0.2764 \left(1 - 1.6180 z^{-1} + z^{-2}\right).$$

T8 Q4 (tutorial).

Hint: minimum phase, so the causal inverse is stable.

(a) $H(z) = 1 - 0.9z^{-4} = \dfrac{z^4 - 0.9}{z^4}$: zeros at $z_k = 0.9^{1/4} e^{j\pi k/2} = 0.9740 \, e^{j\pi k/2}$, $k = 0, 1, 2, 3$ (on the axes, just inside the unit circle), and a 4th-order pole at 0.

(b) $H(\omega) = 1 - 0.9e^{-j4\omega}$, so

$$|H(\omega)|^2 = 1 - 1.8\cos 4\omega + 0.81 = 1.81 - 1.8\cos 4\omega.$$

On $0 \le \omega \le \pi$, $e^{j\omega}$ passes close to zeros at $\omega = 0, \pi/2, \pi$ and is furthest from them at $\pi/4, 3\pi/4$. The 4th-order pole at 0 does not affect $|H|$.

ω	**0**	$\pi/4$	$\pi/2$	$3\pi/4$	π
$\lvert H(\omega)\rvert$	0.1	1.9	0.1	1.9	0.1

Sketch: a smooth periodic ripple with period $\pi/2$, minima of 0.1 (not 0, because the zeros are just inside the circle) at $0, \pi/2, \pi$, and maxima of 1.9 at $\pi/4, 3\pi/4$. It looks like two "humps" on $[0, \pi]$, a 4-tooth comb.

(c)

$$H_I(z) = H^{-1}(z) = \frac{1}{1 - 0.9z^{-4}}, \qquad \text{ROC: } |z| > 0.9^{1/4}.$$

Stable, because this ROC includes the unit circle ($H(z)$ is minimum-phase, so its causal inverse is stable).

(d) The poles of H become zeros and the zeros become poles: $H_I(z) = \dfrac{z^4}{z^4 - 0.9}$ has poles at $0.974e^{j\pi k/2}$ and a 4th-order zero at 0. $|H_I(\omega)| = 1/|H(\omega)|$:

ω	**0**	$\pi/4$	$\pi/2$	$3\pi/4$	π
$\lvert H_I(\omega)\rvert$	10	0.526	10	0.526	10

Sketch: sharp peaks of height 10 at $0, \pi/2, \pi$ (the poles are very close to the circle there), and broad valleys of 0.526 at $\pi/4, 3\pi/4$. Peaks are narrow, valleys are wide: the opposite shape to (b).

T8 Q5 (tutorial).

Hint: section 1.7. Equal degrees, so the constant is $K = 0.64/(-1.2)$.

(a) Factorise:

$$H(z) = \frac{\left(1 - 0.8z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)}{\left(1 - 0.8e^{j\pi/3}z^{-1}\right)\left(1 - 0.8e^{-j\pi/3}z^{-1}\right)}$$

Zeros at 0.8 and $-\frac{3}{2}$; poles at $0.8e^{\pm j\pi/3}$.

(b) Nonminimum-phase: one zero is inside and one is outside the unit circle.

(c) Because it is nonminimum-phase, its stable inverse is noncausal:

$$H_I(z) = H^{-1}(z) = \frac{1 - 0.8z^{-1} + 0.64z^{-2}}{1 + 0.7z^{-1} - 1.2z^{-2}}, \qquad \text{ROC: } 0.8 < |z| < \tfrac{3}{2}.$$

Partial fractions:

$$H_I(z) = \frac{0.3478}{1 - 0.8z^{-1}} + \frac{1.1855}{1 + \frac{3}{2}z^{-1}} - 0.5333$$

$$h_I(n) = 0.3478\,(0.8)^n u(n) - 1.1855\left(-\tfrac{3}{2}\right)^n u(-n-1) - 0.5333\,\delta(n),$$

which is absolutely summable, as expected. It is **noncausal** (two-sided).

New questions

N1.

a. Flip the sign of the odd sample: $h_{HP}(n) = \{0.25, \underset{\uparrow}{-0.5}, 0.25\}$, so $H_{HP}(z) = 0.25 - 0.5z^{-1} + 0.25z^{-2} = 0.25(1 - z^{-1})^2$. This is $H_{LP}(-z)$.

b. $H_{LP}(z) = 0.25(1 + z^{-1})^2$ has a double zero at $z = -1$. H_{HP} has a double zero at $z = +1$ (reflected through the origin).

$|H_{LP}(\omega)| = \cos^2(\omega/2)$ and $|H_{HP}(\omega)| = \sin^2(\omega/2)$.

ω	0	$\pi/2$	π
LP	1	0.5	0
HP	0	0.5	1

Both are symmetric, so both have linear phase with delay 1 sample.

N2.

a. $\omega_0 = 2\pi(50)/1000 = \pi/10 = 0.3142$ rad. $2\cos\omega_0 = 1.9021$.
$H(z) = b_0(1 - 1.9021z^{-1} + z^{-2})$. At DC ($z = 1$):
$b_0(2 - 1.9021) = b_0(0.09789) = 1$, so $b_0 = 10.216$.

$$H(z) = 10.216 - 19.432z^{-1} + 10.216z^{-2}$$

b. With $|H(\omega)| = b_0|2\cos\omega - 2\cos\omega_0|$ (from $H = b_0e^{-j\omega}(2\cos\omega - 2\cos\omega_0)$):

- 100 Hz ($\omega = \pi/5$): $|H| = 2.902$.
- 500 Hz ($\omega = \pi$): $|H| = 39.86$.

The FIR notch is so wide that, once you force DC gain 1, everything above the notch is amplified hugely. Not usable.

c. Denominator
$1 - 2(0.95)\cos(\pi/10)z^{-1} + 0.9025z^{-2} = 1 - 1.8070z^{-1} + 0.9025z^{-2}$. At $z = 1$:
$b_0\dfrac{2 - 1.9021}{1 - 1.8070 + 0.9025} = b_0\dfrac{0.09789}{0.09550} = 1$, so $b_0 = 0.9755$.

$$H(z) = 0.9755\,\frac{1 - 1.9021z^{-1} + z^{-2}}{1 - 1.8070z^{-1} + 0.9025z^{-2}}$$

d. Python: $|H| = 1.012$ at 100 Hz and 1.026 at 500 Hz. Nearly flat away from the notch. The -3 dB band is about 42.2–57.8 Hz, close to the $2(1 - r)F_s/(2\pi) = 15.9$ Hz estimate.

N3.

a. Nulls at $F_k = kF_s/(M + 1)$. Need $F_s/(M + 1) = 1$ kHz, so $M + 1 = 8$, $M = 7$. Nulls at 1, 2, 3, 4 kHz.

b. $\omega = 2\pi(500)/8000 = \pi/8$. $|H| = \dfrac{1}{8}\left|\dfrac{\sin(8\cdot\pi/16)}{\sin(\pi/16)}\right| = \dfrac{1}{8}\cdot\dfrac{1}{0.19509} = 0.6407$.
Symmetric, so linear phase with delay $M/2 = 3.5$ samples $= 3.5/8000$ s $= 0.4375$ ms.

N4.

a. $\omega_0 = 2\pi(1000)/8000 = \pi/4$. $2r\cos\omega_0 = 2(0.95)(0.7071) = 1.3435$, $r^2 = 0.9025$.

$$H(z) = \frac{b_0(1 - z^{-2})}{1 - 1.3435z^{-1} + 0.9025z^{-2}}$$

$$y(n) = 1.3435\,y(n-1) - 0.9025\,y(n-2) + b_0[x(n) - x(n-2)]$$

b. Use distances: $|1 - e^{-j2\omega_0}| = 2\sin\omega_0$, the near pole contributes $1 - r$, the far pole $|1 - re^{-j2\omega_0}| = \sqrt{1 - 2r\cos 2\omega_0 + r^2}$. At $\omega_0 = \pi/4$, $\cos 2\omega_0 = 0$:

$$|H(\omega_0)| = \frac{b_0 \cdot 2\sin(\pi/4)}{(0.05)\sqrt{1.9025}} = b_0(20.506)$$

So $b_0 = 0.04877$. $|H(0)| = |H(\pi)| = 0$ (the zeros at ± 1). Python confirms the peak is at $\omega \approx 0.787 \approx \pi/4$ with height 1.0003.

c. $H(z) = \dfrac{\sin(\pi/4)}{1 - 1.4142z^{-1} + z^{-2}}$, so
$y(n) = 1.4142\,y(n-1) - y(n-2) + 0.7071\,x(n)$. $h(n) = \sin(\pi(n+1)/4)u(n)$:
$h(0) = 0.7071$, $h(1) = 1$, $h(2) = 0.7071$, $h(3) = 0$. Period 8 samples, i.e. a 1 kHz tone at 8 kHz. Marginally stable (poles on the circle).

N5.

a. $1 - 1.5z^{-1} - z^{-2} = (1 - 2z^{-1})(1 + 0.5z^{-1})$. Zeros at 2 and -0.5. Pole at 0.4. Causal and stable (pole inside). One zero inside, one outside: **nonminimum-phase (mixed)**.

b. H_I has poles at 2 and -0.5. Causal ROC $|z| > 2$ excludes $|z| = 1$: **unstable**.

c. $H_I(z) = \dfrac{1 - 0.4z^{-1}}{(1 - 2z^{-1})(1 + 0.5z^{-1})}$, ROC $0.5 < |z| < 2$. Proper (degree 1 over degree 2), so no constant.

- A_1 (pole 2): $\dfrac{1 - 0.4/2}{1 + 0.5/2} = \dfrac{0.8}{1.25} = 0.64$
- A_2 (pole -0.5): $\dfrac{1 - 0.4/(-0.5)}{1 - 2/(-0.5)} = \dfrac{1.8}{5} = 0.36$

- Check: $0.64 + 0.36 = 1 = H_I(\infty)$.

Pole -0.5 inside the circle: causal. Pole 2 outside: anti-causal.

$$h_I(n) = 0.36(-0.5)^n u(n) - 0.64(2)^n u(-n-1)$$

Two-sided, so **non-causal**. The pole of H at 0.4 has become a zero of H_I.

d. $h_I(-2) = -0.64/4 = -0.16$, $h_I(-1) = -0.32$, $h_I(0) = 0.36$, $h_I(1) = -0.18$, $h_I(2) = 0.09$. Python: $h * h_I = \delta(n)$.

N6.

a. Pole at -0.5, zero at -2. They are reciprocals, the allpass signature (L09 slide 23 with $N = 1$, $A(z) = 1 + 0.5z^{-1}$). $H(\omega) = \dfrac{0.5 + e^{-j\omega}}{1 + 0.5e^{-j\omega}} = e^{-j\omega}\dfrac{1 + 0.5e^{j\omega}}{1 + 0.5e^{-j\omega}}$. The fraction is a number over its own conjugate, so its magnitude is 1. Hence $|H(\omega)| = 1$.

b. The zero at -2 is outside the unit circle, so it is **not** minimum phase (it is maximum phase). The causal inverse has a pole at -2: unstable. Its stable inverse is anti-causal.

c. Let $\phi(\omega) = \angle(1 + ae^{-j\omega}) = -\tan^{-1}\dfrac{a\sin\omega}{1 + a\cos\omega}$. Then $\theta(\omega) = -\omega - 2\phi(\omega)$ (numerator angle is $-\phi$). Differentiating, $\dfrac{d\phi}{d\omega} = -\dfrac{a\cos\omega + a^2}{1 + 2a\cos\omega + a^2}$, so

$$\tau = 1 + 2\frac{d\phi}{d\omega} = \frac{1 + 2a\cos\omega + a^2 - 2a\cos\omega - 2a^2}{1 + 2a\cos\omega + a^2} = \frac{1 - a^2}{1 + 2a\cos\omega + a^2}.$$

With $a = 0.5$: $\tau(0) = 0.75/2.25 = 0.333$, $\tau(\pi/2) = 0.75/1.25 = 0.6$, $\tau(\pi) = 0.75/0.25 = 3$ samples (matches `scipy.signal.group_delay`). The group delay varies with ω, which is exactly why an allpass works as a phase equaliser: it changes delay without changing magnitude.

6. Quick recall check

Question

1. Yes. It is symmetric ($h(n) = h(4-n)$). Group delay $= (5-1)/2 = 2$ samples.
2. $F_s/(M+1) = 2000/10 = 200$ Hz.
3. $H_{HP}(z) = \dfrac{1 - 2z^{-1} + z^{-2}}{1 + 0.5z^{-1} + 0.2z^{-2}}$ (odd powers flip sign).
4. Non-causal (two-sided), ROC $0.5 < |z| < 1.2$. H is nonminimum-phase.
5. On the unit circle at $e^{\pm j\omega_0}$. $h(n)$ is a sinusoid that never decays, so it is not absolutely summable.

IIR design: bilinear transform and Butterworth

Answers and worked solutions · Day 19 · Wed 11 Nov · pack D11

5. Practice

From your course files

T9 Q4 (tutorial).

Hint: only the frequencies are warped. The magnitude limits stay the same.

Bilinear transformation requires frequency prewarping, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$0.8 \le |H(\Omega)| \le 1, \quad 0 \le \Omega \le \frac{2}{T}\tan 0.15\pi \;\; (= 1.0191 \text{ with } T = 1)$$

$$|H(\Omega)| \le 0.1, \quad \Omega \ge \frac{2}{T}\tan 0.3\pi \;\; (= 2.7528 \text{ with } T = 1).$$

T9 Q2 (tutorial).

Hint: the official solution uses $2/T = 20$ (that is, $T = 0.1$). Map the quadratic with $b = 0.2$, $c = 9.01$.

Substitute $s = 20\,\dfrac{1 - z^{-1}}{1 + z^{-1}}$ and multiply top and bottom by $(1 + z^{-1})^2$:

- numerator: $\left[20(1 - z^{-1}) + 0.1(1 + z^{-1})\right](1 + z^{-1}) = 20.1 + 0.2z^{-1} - 19.9z^{-2}$
- denominator: $400(1 - z^{-1})^2 + 0.2 \cdot 20(1 - z^{-2}) + 9.01(1 + z^{-1})^2 = 413.01 - 781.98z^{-1} + 40$

$$H(z) = \frac{20.1 + 0.2z^{-1} - 19.9z^{-2}}{413.01 - 781.98z^{-1} + 405.01z^{-2}} = \frac{0.0487 + 0.0005z^{-1} - 0.0482z^{-2}}{1 - 1.8934z^{-1} + 0.9806z^{-2}} \quad (\text{after normalising } a_0 = 1)$$

Zeros $z_1 = -1$, $z_2 = 0.99$; poles $p_{1,2} = 0.9467 \pm j0.2905$.

T9 Q3 (tutorial).

Hint: warping makes the digital integrator fall to zero at $\omega = \pi$. Avoid inputs with DC (pole at $z = 1$).

(a)

$$H(z) = H_a(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{T}{2}\,\frac{1 + z^{-1}}{1 - z^{-1}}, \qquad |z| > 1 \text{ (causal system)}$$

$$y(n) - y(n-1) = \frac{T}{2}x(n) + \frac{T}{2}x(n-1).$$

(b) $H_a(\Omega) = \dfrac{1}{j\Omega} = -\dfrac{j}{\Omega}$, so $|H_a(\Omega)| = \dfrac{1}{|\Omega|}$ (infinite at $\Omega = 0$, a hyperbola either side) and $\theta(\Omega) = -\frac{\pi}{2}$ for $\Omega > 0$, $+\frac{\pi}{2}$ for $\Omega < 0$.

(c)

$$H(\omega) = \frac{T}{2}\,\frac{1+e^{-j\omega}}{1-e^{-j\omega}} = \frac{T}{2}\,\frac{e^{-j\omega/2}\left(e^{j\omega/2}+e^{-j\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2}-e^{-j\omega/2}\right)} = -j\,\frac{T}{2}\,\frac{\cos(\omega/2)}{\sin(\omega/2)} = \frac{T}{2}\cot(\omega/2)\,e^{-j\pi/2}$$

$$|H(\omega)| = \frac{T}{2}|\cot(\omega/2)|, \quad -\pi \le \omega \le \pi; \qquad \theta(\omega) = \begin{cases} -\frac{\pi}{2}, & 0 < \omega \le \pi \\ \frac{\pi}{2}, & -\pi \le \omega < 0 \end{cases}$$

Strictly, $H(\omega)$ does not exist because the ROC $|z| > 1$ excludes the unit circle, but it can still be computed if the pole at $z = 1$ is avoided (L04 slides 12–13).

(d) For $T = 1$ ($\omega = \Omega$), $1/|\Omega|$ and $\frac{T}{2}|\cot(\omega/2)|$ match very well at all frequencies except near $\omega = \pi$, where the frequency warping of the bilinear transformation makes the digital integrator decay to zero. The phase responses are identical.

(e) Input signals with a non-zero DC component should be avoided, because they would make the digital integrator unstable (unbounded output).

T9 Q6 (tutorial).

Hint: radius $= \Omega_c$. 100 Hz $= 200\pi$ rad/s.

$\Omega_c = 50$ rad/s. 80 dB at 100 Hz means $10\log_{10}|H(200\pi)|^2 = -80$ dB:

$$\frac{1}{1+\left(\frac{200\pi}{50}\right)^{2N}} = 10^{-8} \quad \Rightarrow \quad N = \frac{\log(10^8-1)}{2\log(4\pi)} = 3.6390.$$

Rounding up, $N = 4$.

T9 Q5 (tutorial).

Hint: factor method: $(s + 1.2799)$ and $(s^2 + 1.2799s + 1.6380)$ with $K = 2$. Denominator constant 26.8876.

Edges. $\omega_p = \dfrac{2\pi\cdot 3}{30} = \dfrac{\pi}{5}$, $\omega_s = \dfrac{2\pi\cdot 9}{30} = \dfrac{3\pi}{5}$. Let $T = 1$ and prewarp:
$\Omega_p = 2\tan\frac{\pi}{10} = 0.6498$, $\Omega_s = 2\tan\frac{3\pi}{10} = 2.7528$.

Linear scale. $\delta_p = 10^{-1/20} = 0.8913$, $\delta_s = 10^{-20/20} = 0.1$.

Order.

$$N = \frac{\log\left[\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}\right]}{2\log\dfrac{\Omega_s}{\Omega_p}} = 2.0596 \;\Rightarrow\; N = 3.$$

Cutoff. For the bilinear transformation it is fine to exceed the passband spec (no aliasing), so meet the stopband exactly:

$$\Omega_c = \Omega_s\left(\frac{\delta_s^2}{1-\delta_s^2}\right)^{1/(2N)} = 1.2799.$$

Analogue filter. Poles at -1.2799 and $1.2799e^{\pm j2\pi/3}$:

$$H(s) = \frac{1.2799^3}{(s+1.2799)(s^2+1.2799s+1.6380)} = \frac{2.0966}{s^3+2.5598s^2+3.2762s+2.0966}.$$

Bilinear ($s = 2\frac{1-z^{-1}}{1+z^{-1}}$, multiply through by $(1+z^{-1})^3$):

$$H(z) = \frac{2.0965+6.2895z^{-1}+6.2895z^{-2}+2.0965z^{-3}}{26.8876-21.3974z^{-1}+13.4983z^{-2}-2.2167z^{-3}} = \frac{0.0780+0.2339z^{-1}+0.2339z^{-2}+0.0780z^{-3}}{1-0.7958z^{-1}+0.5020z^{-2}-0.0824z^{-3}}$$

Difference equation.

$$y(n) - 0.7958y(n-1) + 0.5020y(n-2) - 0.0824y(n-3) = 0.0780x(n) + 0.2339x(n-1) + 0.2339x(n-2) + 0.0780x(n-3)$$

New questions

Question 1

a. Prewarp: $\Omega_p = 2\tan(0.1\pi) = 0.64984$ and $\Omega_s = 2\tan(0.25\pi) = 2.00000$.

$$0.85 \le |H(\Omega)| \le 1,\;\; 0 \le \Omega \le 0.6498; \qquad |H(\Omega)| \le 0.15,\;\; \Omega \ge 2.0000$$

(In dB: passband ripple 1.41 dB, stopband attenuation 16.48 dB.)

b. $\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)} = \dfrac{0.7225\times 0.9775}{0.0225\times 0.2775} = 113.11$, and $\Omega_s/\Omega_p = 3.07768$.

$$N = \frac{\log 113.11}{2\log 3.07768} = 2.1030 \Rightarrow \mathbf{N = 3}$$

Passband exact: $\Omega_c = 0.64984\left(\frac{0.7225}{0.2775}\right)^{1/6} = 0.7622$. Stopband exact: $\Omega_c = 2\left(\frac{0.0225}{0.9775}\right)^{1/6} = 1.0667$.

Any $0.7622 \le \Omega_c \le 1.0667$ meets both specs. (scipy `buttord` gives $N = 3$.)

Question 2

a. $\omega_c = 2\pi(1)/8 = \pi/4$. So $\Omega_c = \frac{2}{T}\tan(\omega_c/2) = 16000\tan(\pi/8) =$ **6627.4 rad/s**. (This is 1054.8 Hz, not 1000 Hz. That is warping at work.)

b. $K = 2/T = 16000$. The first-order factor maps to $(K + \Omega_c) + (\Omega_c - K)z^{-1} = 22627.4 - 9372.6z^{-1}$. The numerator becomes $\Omega_c(1 + z^{-1}) = 6627.4(1 + z^{-1})$.

$$H(z) = \frac{0.29289(1 + z^{-1})}{1 - 0.41421z^{-1}}, \qquad y(n) = 0.41421\,y(n-1) + 0.29289\,x(n) + 0.29289\,x(n-1)$$

Checks: $H(1) = 0.58579/0.58579 = 1$ (DC gain 1). $H(-1) = 0$ (zero at $\omega = \pi$). $|H(e^{j\pi/4})| = -3.01$ dB. The pole at $z = 0.414$ is stable.

c. The digital cutoff is $\omega = 2\arctan(\Omega_c T/2) = 2\arctan(\pi/8) = 0.74839$ rad, so $f = 0.74839 \times 8000/(2\pi) =$ **952.9 Hz**. That is 47 Hz low.

Question 3

a. $\Omega_c = 3141.59$ rad/s. The angles are $\frac{\pi}{2} + \frac{(2k+1)\pi}{8}$, that is $112.5^\circ, 157.5^\circ, 202.5^\circ, 247.5^\circ$:

$$s_{0,3} = -1202.2 \pm j2902.5, \qquad s_{1,2} = -2902.5 \pm j1202.2 \text{ rad/s}$$

b. Each pair gives $s^2 - 2\Omega_c\cos\theta\, s + \Omega_c^2$, with $\Omega_c^2 = 9.8696 \times 10^6$:

$$H(s) = \frac{9.7409 \times 10^{13}}{(s^2 + 2404.5s + 9.8696 \times 10^6)(s^2 + 5804.9s + 9.8696 \times 10^6)}$$

(The numerator is Ω_c^4.)

c. $\Omega/\Omega_c = 1500/500 = 3$, so the attenuation is $10\log_{10}(1 + 3^8) =$ **38.17 dB**.

d. $N = \log(10^6 - 1)/(2\log 3) = 6.2877 \Rightarrow N = 7$. (With $N = 4$, 60 dB is only reached at 2812 Hz.)

Question 4

a. $\omega_p = 2\pi/8 = \pi/4$ and $\omega_s = 6\pi/8 = 3\pi/4$. $\Omega_p = 2\tan(\pi/8) = 0.82843$ and $\Omega_s = 2\tan(3\pi/8) = 4.82843$. The ratio is 5.82843.

$$N = \frac{\log(99/0.25893)}{2\log 5.82843} = 1.6867 \Rightarrow \mathbf{N = 2}$$

b. Stopband exact: $\Omega_c = 4.82843/99^{1/4} = \mathbf{1.53072}$. So $\Omega_c^2 = 2.34312$ and $\sqrt{2}\Omega_c = 2.16477$. The poles are $-1.08239 \pm j1.08239$.

$$H(s) = \frac{2.34312}{s^2 + 2.16477s + 2.34312}$$

c. With $K = 2$: $4 + 4.32954 + 2.34312 = 10.67266$; $2(2.34312 - 4) = -3.31377$; $4 - 4.32954 + 2.34312 = 2.01358$.

$$H(z) = \frac{2.34312(1 + z^{-1})^2}{10.67266 - 3.31377z^{-1} + 2.01358z^{-2}} = \frac{0.21954 + 0.43909z^{-1} + 0.21954z^{-2}}{1 - 0.31049z^{-1} + 0.18867z^{-2}}$$

$$\boxed{y(n) = 0.31049\,y(n-1) - 0.18867\,y(n-2) + 0.21954\,x(n) + 0.43909\,x(n-1) + 0.21954\,x(n-2)}$$

d. At Ω_p: $-10\log_{10}[1 + (0.82843/1.53072)^4] = -0.357$ dB (better than the 1 dB allowed). At Ω_s: $-10\log_{10}[1 + (4.82843/1.53072)^4] = -20.00$ dB (met exactly). The poles are $0.1552 \pm j0.4057$, with $|z| = 0.434 < 1$, so the filter is stable.

VERIFIED with scipy: `butter(2, 1.53072, analog=True)` followed by `bilinear(..., fs=1)` gives the same coefficients, and `freqz` gives −0.357 dB and −20.00 dB at $\pi/4$ and $3\pi/4$.

Question 5

a. $\omega_p = 2\pi(2)/12 = \pi/3$ and $\omega_s = 2\pi(4)/12 = 2\pi/3$. $\Omega_p = 2\tan(\pi/6) = 1.15470$ and $\Omega_s = 2\tan(\pi/3) = 3.46410$. The ratio is exactly 3.

$$N = \frac{\log(99/0.25893)}{2\log 3} = 2.7063 \Rightarrow \mathbf{N = 3}$$

b. Passband exact: $\Omega_c = 1.15470/(0.25893)^{1/6} = \mathbf{1.44635}$. The LHP poles are at $120^\circ, 180^\circ, 240^\circ$: $-0.72318 \pm j1.25258$ and -1.44635. With $\Omega_c^2 = 2.09193$ and $\Omega_c^3 = 3.02566$:

$$H(s) = \frac{3.02566}{(s + 1.44635)(s^2 + 1.44635s + 2.09193)} = \frac{3.02566}{s^3 + 2.89270s^2 + 4.18386s + 3.02566}$$

c. Map the factors with $K = 2$:

- linear: $(2 + 1.44635) + (1.44635 - 2)z^{-1} = 3.44635 - 0.55365z^{-1}$
- quadratic: $4 + 2.89270 + 2.09193 = 8.98463$; $2(2.09193 - 4) = -3.81614$; $4 - 2.89270 + 2.09193 = 3.19923$

Multiply the two:
$(3.44635 - 0.55365z^{-1})(8.98463 - 3.81614z^{-1} + 3.19923z^{-2}) = 30.96419 - 18.1260$
.

The numerator is $3.02566(1 + 3z^{-1} + 3z^{-2} + z^{-3})$. Divide by 30.96419:

$$H(z) = \frac{0.097715 + 0.293145z^{-1} + 0.293145z^{-2} + 0.097715z^{-3}}{1 - 0.585389z^{-1} + 0.424312z^{-2} - 0.057203z^{-3}}$$

$$\begin{aligned} y(n) = {} & 0.585389\,y(n-1) - 0.424312\,y(n-2) + 0.057203\,y(n-3) \\ & + 0.097715\,x(n) + 0.293145\,x(n-1) + 0.293145\,x(n-2) + 0.097715\,x(n-3) \end{aligned}$$

d. At Ω_p: $-10\log_{10}[1 + (1.15470/1.44635)^6] = -1.00$ dB (met exactly). At Ω_s: $-10\log_{10}[1 + (3.46410/1.44635)^6] = -22.78$ dB (meets ≤ -20 dB). The poles in z are 0.16065 (from the linear factor, $0.55365/3.44635$) and $0.21237 \pm j0.55765$ ($|z| = 0.597$). All are inside the unit circle, so the filter is stable. DC gain: $\sum b_k = 0.78172 = \sum a_k$, so the gain is 1.

e. With $T = 1/12000$: $\Omega_p = 24000\tan(\pi/6)$ and $\Omega_s = 24000\tan(\pi/3)$. Both scale by 12000, so the ratio and N are unchanged, and $\Omega_c = 17356$ rad/s (also 12000 times larger). In the map, $K = 24000$ also scales by 12000. Every factor coefficient therefore scales by the same power of 12000, which cancels on normalising. scipy gives identical coefficients. The 3-dB point is at $\omega = 2\arctan(\Omega_c T/2) = 2\arctan(0.72318) = 0.3986\pi$, so $f = 0.3986 \times 6000 =$ **2392 Hz**.

> **VERIFIED:** `buttord(1/3, 2/3, 1, 20)` gives $(3,\ 0.39859)$, and `butter(3, 0.39859)` matches the coefficients above.

Question 6

a. The ratio is $\Omega_s/\Omega_p = 2$, so $N = \log(99/0.25893)/(2\log 2) = 4.289 \Rightarrow N = 5$. That is two orders more than necessary, because without prewarping the transition band looks narrower than it really is.

b. $\Omega_c = (\pi/3)/(0.25893)^{1/6} = 1.31169$. After the bilinear map, the digital gain at ω equals the analogue gain at $2\tan(\omega/2)$, **not** at ω:

- at ω_p: $\Omega = 1.15470$, so the gain is $-10\log_{10}[1 + (1.15470/1.31169)^6] =$ **−1.66 dB**. **This fails the 1 dB ripple spec.**
- at ω_s: $\Omega = 3.46410$, so the gain is $-10\log_{10}[1 + (3.46410/1.31169)^6] =$ **−25.32 dB** (passes).

The digital 3-dB point moved down to $2\arctan(1.31169/2) = 0.3695\pi$, compared with 0.3986π in the correct design. Warping compresses the frequency axis, so a filter designed with unwarped edges cuts off too early. (Checked with `freqz`: −1.660 dB and −25.318 dB.)

6. Quick recall check

Question

1. $s = 0 \to z = 1$ (DC gain kept). $s = \infty \to z = -1$ (zeros at $\omega = \pi$).
2. $\Omega = \frac{2}{T}\tan(\omega/2)$. It changes only the band-edge frequencies. δ_p and δ_s are unchanged.
3. $\delta_p = 10^{-1/20} = 0.8913$, and $(1 - \delta_p^2)/\delta_p^2 = 10^{0.1} - 1 = 0.2589$.
4. A circle of radius Ω_c centred on the origin, in the LHP. For $N = 3$ the angles are 120°, 180° and 240°.
5. $N = 4$. You always round up, because $N = 3$ would miss the spec.

Chebyshev and FIR design

Answers and worked solutions · Day 20 · Thu 12 Nov · pack D12

5. Practice

From your course files

T9 Q1 (tutorial).

Hint: integrate $e^{j\omega n}$ from $-\omega_c$ to ω_c.

Inverse DTFT of $H(\omega)$:

$$h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{j\omega n}\,d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n}\,d\omega = \frac{1}{j2\pi n}\left[e^{j\omega n}\right]_{-\omega_c}^{\omega_c} = \frac{1}{\pi n}\sin(\omega_c n)$$

$$h(n) = 2f_c\,\mathrm{sinc}(\omega_c n), \qquad \mathrm{sinc}\,x = \begin{cases} 1, & x = 0 \\ \dfrac{\sin x}{x}, & \text{otherwise} \end{cases}$$

where $f_c = \omega_c/2\pi$. (The course sinc is $\sin x/x$; MATLAB's is $\sin(\pi x)/(\pi x)$.)

The ideal LPF is not realisable in practice for two reasons:

- $h(n)$ is not causal, because $h(n) \neq 0$ for $n < 0$;
- $h(n)$ is of infinite duration.

T9 Q7 (tutorial).

Hint: Playbook A. Prewarp first, and use $\cosh^{-1}$ (section 1.2).

Edges. $\omega_p = \dfrac{2\pi\cdot 2}{40} = \dfrac{\pi}{10}$, $\omega_s = \dfrac{2\pi\cdot 4}{40} = \dfrac{\pi}{5}$. Let $T = 1$ and prewarp: $\Omega_p = 2\tan\frac{\pi}{20} = 0.3168$, $\Omega_s = 2\tan\frac{\pi}{10} = 0.6498$.

Linear scale. $\delta_p = 10^{-1/20} = 0.8913$, $\delta_s = 10^{-50/20} = 0.0032$.

$$\epsilon = \sqrt{\frac{1}{\delta_p^2} - 1} = 0.5087$$

$$N = \frac{\cosh^{-1}\left(\dfrac{1}{\epsilon}\,\dfrac{\sqrt{1-\delta_s^2}}{\delta_s}\right)}{\cosh^{-1}\left(\dfrac{\Omega_s}{\Omega_p}\right)} = 5.2851 \;\Rightarrow\; N = 6.$$

> **NOTE:** the official solution prints "$\cos^{-1}$" here. Both arguments are greater than 1, so it has to be $\cosh^{-1}$, and 5.2851 only comes out with $\cosh^{-1}$. With unrounded values, $\epsilon = 0.50885$ and $N = 5.2929$. The official 5.2851 comes from using the rounded values inside the formula ($\delta_s = 0.0032$, $\epsilon = 0.5087$ and $\Omega_s/\Omega_p = 0.6498/0.3168$). Both round up to $N = 6$, so the course answer stands.

T9 Q8 (tutorial).

Hint: the ends differ in sign, so antisymmetric. M odd, so the middle sample is 0.

The first and last samples differ, so $h(n)$ cannot be symmetric; it must be antisymmetric. $M = 5$ is odd, so the mid-point sample is $h\left(\frac{M-1}{2}\right) = h(2) = 0$:

$$h(n) = \{\underset{\uparrow}{1}, 2, 0, -2, -1\}$$

$$H(\omega) = 1 + 2e^{-j\omega} - 2e^{-j3\omega} - e^{-j4\omega} = e^{-j2\omega}\left(e^{j2\omega} - e^{-j2\omega}\right) + 2e^{-j2\omega}\left(e^{j\omega} - e^{-j\omega}\right) = 2(\sin 2\omega + 2\sin\omega)\, e^{-j(2\omega - \pi/2)}$$

The phase response is linear:

$$\theta(\omega) = \begin{cases} -2\omega + \pi/2, & \sin 2\omega + 2\sin\omega \geq 0 \\ -2\omega - \pi/2, & \text{otherwise.} \end{cases}$$

T9 Q9 (tutorial).

Hint: same steps as Example 3, with $\omega_c = \pi/6$. Hanning $w = \{0, 0.5, 1, 0.5, 0\}$.

(a)

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} e^{j\omega n}\, d\omega = \frac{1}{6}\,\mathrm{sinc}\left(\frac{\pi n}{6}\right)$$

(b) Shift by $(M-1)/2 = 2$ and window:

$$h(n) = h_d\left(n - \tfrac{M-1}{2}\right)w(n) = \frac{1}{6}\,\mathrm{sinc}\left(\frac{\pi}{6}(n-2)\right)w(n), \qquad w(n) = 1,\ 0 \leq n \leq 4.$$

$$h(n) = \left\{\tfrac{1}{6}\mathrm{sinc}\left(-\tfrac{\pi}{3}\right), \tfrac{1}{6}\mathrm{sinc}\left(-\tfrac{\pi}{6}\right), \tfrac{1}{6}\mathrm{sinc}(0), \tfrac{1}{6}\mathrm{sinc}\left(\tfrac{\pi}{6}\right), \tfrac{1}{6}\mathrm{sinc}\left(\tfrac{\pi}{3}\right)\right\} = \{\underset{\uparrow}{0.1378}, 0.1592, 0.1667, 0.1592, 0.1378\}$$

(c) Hanning window $w(n) = \frac{1}{2}(1 - \cos(\pi n/2))$ for $0 \leq n \leq 4$, that is $\{0, 0.5, 1, 0.5, 0\}$:

$$h(n) = \{0.1378, 0.1592, 0.1667, 0.1592, 0.1378\} \times \{0, 0.5, 1, 0.5, 0\} = \{\underset{\uparrow}{0}, 0.0796, 0.1667, 0.0796, 0\}$$

T9 Q4 (tutorial; only if not done in D11).

Hint: only the frequencies are warped. The magnitude limits stay the same.

Bilinear transformation requires frequency prewarping, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$0.8 \le |H(\Omega)| \le 1, \quad 0 \le \Omega \le \frac{2}{T}\tan 0.15\pi \ \ (= 1.0191 \text{ with } T = 1)$$

$$|H(\Omega)| \le 0.1, \quad \Omega \ge \frac{2}{T}\tan 0.3\pi \ \ (= 2.7528 \text{ with } T = 1).$$

T9 Q2 (tutorial; only if not done in D11).

Hint: the official solution uses $2/T = 20$ (that is, $T = 0.1$). Map the quadratic with $b = 0.2$, $c = 9.01$.

Substitute $s = 20\,\dfrac{1 - z^{-1}}{1 + z^{-1}}$ and multiply top and bottom by $(1 + z^{-1})^2$:

- numerator: $\left[20(1 - z^{-1}) + 0.1(1 + z^{-1})\right](1 + z^{-1}) = 20.1 + 0.2z^{-1} - 19.9z^{-2}$
- denominator: $400(1 - z^{-1})^2 + 0.2 \cdot 20(1 - z^{-2}) + 9.01(1 + z^{-1})^2 = 413.01 - 781.98z^{-1} + 405.01z^{-2}$

$$H(z) = \frac{20.1 + 0.2z^{-1} - 19.9z^{-2}}{413.01 - 781.98z^{-1} + 405.01z^{-2}} = \frac{0.0487 + 0.0005z^{-1} - 0.0482z^{-2}}{1 - 1.8934z^{-1} + 0.9806z^{-2}} \quad (\text{after normalising } a_0 = 1)$$

Zeros $z_1 = -1$, $z_2 = 0.99$; poles $p_{1,2} = 0.9467 \pm j0.2905$.

T9 Q3 (tutorial; only if not done in D11).

Hint: warping makes the digital integrator fall to zero at $\omega = \pi$. Avoid inputs with DC (pole at $z = 1$).

(a)

$$H(z) = H_a(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{T}{2}\,\frac{1 + z^{-1}}{1 - z^{-1}}, \qquad |z| > 1 \text{ (causal system)}$$

$$y(n) - y(n-1) = \frac{T}{2}x(n) + \frac{T}{2}x(n-1).$$

(b) $H_a(\Omega) = \dfrac{1}{j\Omega} = -\dfrac{j}{\Omega}$, so $|H_a(\Omega)| = \dfrac{1}{|\Omega|}$ (infinite at $\Omega = 0$, a hyperbola either side) and $\theta(\Omega) = -\frac{\pi}{2}$ for $\Omega > 0$, $+\frac{\pi}{2}$ for $\Omega < 0$.

(c)

$$H(\omega) = \frac{T}{2}\,\frac{1 + e^{-j\omega}}{1 - e^{-j\omega}} = \frac{T}{2}\,\frac{e^{-j\omega/2}\left(e^{j\omega/2} + e^{-j\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right)} = -j\,\frac{T}{2}\,\frac{\cos(\omega/2)}{\sin(\omega/2)} = \frac{T}{2}\cot(\omega/2)\,e^{-j\pi/2}$$

$$|H(\omega)| = \frac{T}{2}|\cot(\omega/2)|, \quad -\pi \le \omega \le \pi; \qquad \theta(\omega) = \begin{cases} -\frac{\pi}{2}, & 0 < \omega \le \pi \\ \frac{\pi}{2}, & -\pi \le \omega < 0 \end{cases}$$

Strictly, $H(\omega)$ does not exist because the ROC $|z| > 1$ excludes the unit circle, but it can still be computed if the pole at $z = 1$ is avoided (L04 slides 12–13).

(d) For $T = 1$ ($\omega = \Omega$), $1/|\Omega|$ and $\frac{T}{2}|\cot(\omega/2)|$ match very well at all frequencies except near $\omega = \pi$, where the frequency warping of the bilinear transformation makes the digital integrator

decay to zero. The phase responses are identical.

(e) Input signals with a non-zero DC component should be avoided, because they would make the digital integrator unstable (unbounded output).

New questions

New Q1 [3 marks]

a. $\delta_p = 10^{-0.5/20} = 0.9441$ and $\delta_s = 10^{-30/20} = 0.03162$.

$\epsilon = \sqrt{1/0.9441^2 - 1} = \mathbf{0.3493}$ ($\epsilon^2 = 0.1220$).

Numerator: $\frac{\sqrt{1-0.001}}{0.3493 \times 0.03162} = 90.48$, and $\cosh^{-1}(90.48) = 5.1983$.

Denominator: $\cosh^{-1}(2) = 1.3170$.

$N = 5.1983/1.3170 = 3.947 \Rightarrow \mathbf{N = 4}$.

b. Butterworth (L10 slide 17): $N = \dfrac{\log \frac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}}{2\log 2} = 6.4996 \Rightarrow \mathbf{N = 7}$.

The Chebyshev filter saves three orders. scipy check: `cheb1ord(1000,2000,0.5,30,analog=True)` gives 4, and `buttord` gives 7. For reference, `ellipord` gives 3.

New Q2 [4 marks]

$\epsilon = \sqrt{10^{0.1} - 1} = 0.5088$.

$\beta = \left(\frac{\sqrt{1.2589}+1}{0.5088}\right)^{1/3} = 4.1702^{1/3} = 1.6096$.

$r_1 = \frac{1.6096^2+1}{2(1.6096)} = 1.1154$ and $r_2 = \frac{1.6096^2-1}{2(1.6096)} = 0.4942$.

$\phi_k = 90^\circ + (2k+1)30^\circ = 120^\circ, 180^\circ, 240^\circ$. The poles are:

- $p_0 = 0.4942\cos 120^\circ + j1.1154 \sin 120^\circ = \mathbf{-0.2471 + j0.9660}$
- $p_1 = \mathbf{-0.4942}$
- $p_2 = \mathbf{-0.2471 - j0.9660}$

The denominator is
$(s + 0.4942)(s^2 + 0.4942s + 0.9942) = s^3 + 0.9883s^2 + 1.2384s + 0.4913$.

N is odd, so $K = 0.4913$, and

$$H(s) = \frac{0.4913}{s^3 + 0.9883s^2 + 1.2384s + 0.4913}$$

$|H| = 1$ where $T_3(\Omega) = 4\Omega^3 - 3\Omega = 0$, which is at $\Omega = 0$ and $\Omega = \sqrt{3}/2 = 0.866$ rad/s.
$|H| = 0.8913$ (−1 dB) where $T_3^2 = 1$, which is at $\Omega = 0.5$ and $\Omega = 1$ rad/s. That gives 4 extrema across $0 \le \Omega \le 1$, the equiripple passband.

scipy check: `cheby1(3,1,1,analog=True)` and `cheb1ap(3,1)` give the same result.

New Q3 [5 marks]

a. Symmetric: $\{1,3,3,1\}$ (Type II). Antisymmetric: $\{1,3,-3,-1\}$ (Type IV).

b. $\{1,3,3,1\} = (1+z^{-1})^3$ has a triple zero at $z=-1$, so $H(\pi)=0$. It cannot pass high frequencies.

$\{1,3,-3,-1\}$: $H(0) = 1+3-3-1 = 0$, a zero at $z=1$. Factorising, $z^3+3z^2-3z-1 = (z-1)(z^2+4z+1)$, so the zeros are $z=1$, -0.2679 and -3.7321. The last two are a reciprocal pair ($0.2679 = 1/3.7321$). $|H(\pi)| = |1-3-3+1| = 4$. The response is 0 at DC and 4 at $\omega=\pi$, so the **antisymmetric** filter is the highpass candidate.

c. $\alpha = 1.5$.

$$H(\omega) = e^{-j1.5\omega}\left[(e^{j1.5\omega}-e^{-j1.5\omega}) + 3(e^{j0.5\omega}-e^{-j0.5\omega})\right] = 2\left[\sin 1.5\omega + 3\sin 0.5\omega\right]e^{-j(1.5\omega-\pi/2)}$$

The group delay is **1.5 samples**. Checked numerically: the unwrapped phase $+1.5\omega$ equals $\pi/2$ at every test frequency.

New Q4 [8 marks]

a. $h_d(n) = \dfrac{\sin(\pi n/2)}{\pi n} = \frac{1}{2}\mathrm{sinc}\left(\frac{\pi n}{2}\right)$:

- $h_d(0) = 0.5$
- $h_d(\pm 1) = 1/\pi = 0.3183$
- $h_d(\pm 2) = 0$
- $h_d(\pm 3) = -1/(3\pi) = -0.1061$

b. $\alpha = 3$, so $h(n) = h_d(n-3)$:

$$h(n) = \{-0.1061,\ 0,\ 0.3183,\ 0.5,\ 0.3183,\ 0,\ -0.1061\}.$$

c. $w(n) = 0.54 - 0.46\cos(\pi n/3) = \{0.08,\ 0.31,\ 0.77,\ 1,\ 0.77,\ 0.31,\ 0.08\}$, so

$$h(n) = \{-0.0085,\ 0,\ 0.2451,\ 0.5,\ 0.2451,\ 0,\ -0.0085\}.$$

d. $A(\omega) = h(3) + 2h(2)\cos\omega + 2h(1)\cos 2\omega + 2h(0)\cos 3\omega$.

	$A(0)$	$A(\pi/2)$	$\lvert A(\pi)\rvert$
Rectangular	$0.5 + 0.6366 - 0.2122 = 0.9244$	0.5	$0.5 - 0.6366 + 0.2122 = 0.0756$ (−22.4 dB)
Hamming	$0.5 + 0.4902 - 0.0170 = 0.9732$	0.5	$0.5 - 0.4902 + 0.0170 = 0.0268$ (−31.4 dB)

Hamming gives the better stopband: −31.4 dB versus −22.4 dB at $\omega=\pi$. Both are exactly 0.5 (−6 dB) at the cutoff.

e. Symmetric with $M = 7$ (odd), so **Type I**. The group delay is $\alpha = 3$ samples.

scipy check: `firwin(7, 0.5, window='boxcar'/'hamming', scale=False)` reproduces (b) and (c).

New Q5 [5 marks]

Check the prototype first: $|H_{LP}(\omega)| = |\cos(\omega/2)|$, which is 0.7071 at $\pi/2$. Correct.

$a = -\dfrac{\cos((\pi/2+\pi/3)/2)}{\cos((\pi/2-\pi/3)/2)} = -\dfrac{\cos 75^\circ}{\cos 15^\circ} = -0.2679$. (Exactly $a = \sqrt{3} - 2$.)

Substitute $z^{-1} \to -\dfrac{z^{-1}+a}{1+az^{-1}}$:

$$H_{HP}(z) = \frac{1}{2}\left(1 - \frac{z^{-1}+a}{1+az^{-1}}\right) = \frac{(1-a)}{2}\cdot\frac{1-z^{-1}}{1+az^{-1}} = \boxed{\frac{0.6340\,(1-z^{-1})}{1-0.2679z^{-1}}}$$

Checks (freqz):

- $|H(0)| = 0$
- $|H(\pi/3)| = 0.7071$ (−3.01 dB), the same gain the prototype had at its edge
- $|H(\pi/2)| = 0.866$
- $|H(\pi)| = 0.6340 \times 2/1.2679 = 1.000$

So the filter is a highpass with its −3 dB edge at $\pi/3$.

New Q6 [6 marks]

a. $\Omega_p = 2\tan(0.1\pi) = 0.6498$ and $\Omega_s = 2\tan(0.175\pi) = 1.2256$. The ratio is 1.8860.

$\delta_p = 0.9441$ and $\delta_s = 0.01$, so $\epsilon = \mathbf{0.3493}$.

$\cosh^{-1}\left(\frac{\sqrt{1-10^{-4}}}{0.3493\times 0.01}\right) = \cosh^{-1}(286.26) = 6.3501$ and $\cosh^{-1}(1.8860) = 1.2485$.

$N = 5.086 \Rightarrow \mathbf{N = 6}$. It is only just over 5, but you still round up.

b. Butterworth needs $N = 8.92 \to 9$, which is more taps but a smoother phase. Elliptic needs $N = 4$ (scipy `ellipord`), which is the lowest order but the most nonlinear phase. Chebyshev I and II both need 6.

c. $|H| = 0$ when $T_6(\Omega_s/\Omega) = 0$, which is when $\Omega_s/\Omega = \cos\frac{(2k+1)\pi}{12}$ for $k = 0, 1, 2$. That gives $\Omega = 1.2256/\cos 15^\circ$, $1.2256/\cos 45^\circ$ and $1.2256/\cos 75^\circ$, i.e. **1.2688, 1.7333 and 4.7354 rad/s**.

These are the zeros of the equiripple stopband. All of them lie above Ω_s.

scipy check: `cheb2ord(0.2,0.35,0.5,40)` returns $N = 6$, and the `cheby2` design gives −0.50 dB at 0.2π and −40.06 dB at 0.35π.

6. Quick recall check

Question

1. Elliptic is the most efficient. Bessel has the best (linear) passband phase.
2. Even N, so $|H(0)| = \delta_p$, which is −2 dB.
3. The ends are 2 and -2, so antisymmetric. $M = 5$ is odd, so $h(2) = 0$, and $h(1) = -h(3) = 1$. $h = \{2, 1, 0, -1, -2\}$.
4. $w(n) = 0.54 - 0.46\cos(\pi n) = \{0.08,\ 1,\ 0.08\}$.
5. It always has a zero at $z = -1$, so $H(\pi) = \sum(-1)^n h(n) = 0$.

DSP · MOCK

DSP mock A

Answers and worked solutions · Day 22 · Sat 14 Nov · pack DX1

Marking guide

Mark yourself honestly. Give method marks where the method is right and only the arithmetic slips. Give no marks for a correct number with no working on a part worth 2 or more marks.

Question 1 [8 marks]

(a) The highest frequency is $F_{\max} = 1100\pi/(2\pi) = 550$ Hz. Nyquist rate $= 2F_{\max} = \mathbf{1100}$ **Hz**. $F_s = 800$ Hz is below this, so the 550 Hz component will alias. The 150 Hz component is fine.

(b) $T = 1/800$ s. Each component has $f = F/F_s$:

- 150 Hz: $f = 150/800 = 0.1875$, so $\omega = 2\pi(0.1875) = 0.375\pi$. Already in range.
- 550 Hz: $f = 550/800 = 0.6875$, so $\omega = 1.375\pi$. Subtract 2π: $\omega = -0.625\pi$.

So $2\sin(1.375\pi n) = 2\sin(-0.625\pi n) = -2\sin(0.625\pi n)$, and

$$\boxed{x(n) = 3\cos(0.375\pi n) - 2\sin(0.625\pi n)}$$

(c) The ideal D/A puts each principal frequency back at $F = fF_s$: $0.1875 \times 800 = 150$ Hz and $0.3125 \times 800 = 250$ Hz.

$$\boxed{y_a(t) = 3\cos(300\pi t) - 2\sin(500\pi t)}$$

The 550 Hz tone comes back as a 250 Hz tone with its sign flipped.

(d) $y(n) = x(2n) = x_a(2nT)$. This is the same as sampling at 400 Hz.

$$y(n) = 3\cos(0.75\pi n) - 2\sin(1.25\pi n).$$

$1.25\pi - 2\pi = -0.75\pi$, so $-2\sin(1.25\pi n) = -2\sin(-0.75\pi n) = +2\sin(0.75\pi n)$. Both components now sit at $\omega_0 = 0.75\pi$:

$$y(n) = 3\cos(0.75\pi n) + 2\sin(0.75\pi n).$$

Use $a\cos\theta + b\sin\theta = \sqrt{a^2+b^2}\cos(\theta - \tan^{-1}(b/a))$:

$$\boxed{y(n) = 3.606\cos(0.75\pi n - 0.588)}, \quad A = \sqrt{13} = 3.606,\ \omega_0 = 0.75\pi,\ \phi = -0.588 \text{ rad } (-33.7^\circ)$$

(e)

System	Linear	Time invariant	Causal	Static	BIBO stable
(i) $x(n) - 0.5n\,x(n-1)$	Yes	**No**: coefficient n changes with time	Yes	**No**: uses $x(n-1)$	**No**: $\lvert x\rvert \le 1$ gives $\lvert y\rvert$ growing like $0.5n$
(ii) $\lvert x(n+1)\rvert$	**No**: $\lvert -x\rvert \neq -\lvert x\rvert$	Yes	**No**: uses future input $x(n+1)$	**No**: depends on a sample other than $x(n)$	Yes: $\lvert y\rvert = \lvert x\rvert \le B$

- a. 1 mark: 1100 Hz and “not adequate, 550 Hz aliases”.
- b. 2 marks: 1 for 0.375π, 1 for $-2\sin(0.625\pi n)$ with the sign right. Half credit for $+2\sin(0.625\pi n)$.
- c. 1 mark: 250 Hz with the minus sign.
- d. 2 marks: 1 for showing both terms at 0.75π, 1 for A and ϕ.
- e. 2 marks: 1 per system, all five properties right.

COMMON MISTAKES

- Folding a sine without flipping its sign. Cosine is even, sine is odd: $\sin(-\theta) = -\sin\theta$.
- Writing $\omega = 2\pi F$ instead of $\omega = 2\pi F/F_s$.
- In (d), treating $x(2n)$ as halving the frequency. Discarding samples *doubles* every digital frequency, which is why aliasing gets worse.
- Calling (ii) static. Memoryless means $y(n)$ depends on $x(n)$ only. $x(n+1)$ is not $x(n)$.
- Calling (i) stable because it has no feedback. The growing coefficient n makes it unstable.

REREAD: D1 (sampling, aliasing, system properties).

Question 2 [7 marks]

(a) Characteristic equation: $\lambda^2 - \frac{1}{4}\lambda - \frac{1}{8} = 0$, so $(\lambda - \frac{1}{2})(\lambda + \frac{1}{4}) = 0$.

$$\lambda_1 = \tfrac{1}{2},\ \lambda_2 = -\tfrac{1}{4}; \qquad \text{modes } \left(\tfrac{1}{2}\right)^n,\ \left(-\tfrac{1}{4}\right)^n$$

Both $|\lambda| < 1$ and the system is causal, so it is **BIBO stable**.

(b) $y(n) = \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) + x(n) + x(n-1)$, with $x(0..3) = 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}$.

$$\begin{aligned}
y(0) &= 1 \\
y(1) &= \tfrac{1}{4}(1) + \tfrac{1}{3} + 1 = \tfrac{19}{12} = 1.5833 \\
y(2) &= \tfrac{1}{4}\cdot\tfrac{19}{12} + \tfrac{1}{8}(1) + \tfrac{1}{9} + \tfrac{1}{3} = \tfrac{139}{144} = 0.9653 \\
y(3) &= \tfrac{1}{4}\cdot\tfrac{139}{144} + \tfrac{1}{8}\cdot\tfrac{19}{12} + \tfrac{1}{27} + \tfrac{1}{9} = \tfrac{1015}{1728} = 0.5874
\end{aligned}$$

(c) For $n \geq 1$ the input $\delta(n) + \delta(n-1)$ is zero, so $h(n) = c_1(\frac{1}{2})^n + c_2(-\frac{1}{4})^n$ for $n \geq 1$ (and also at $n = 0$ here). Get the initial values by recursion: $h(0) = 1$, $h(1) = \frac{1}{4}(1) + 1 = 1.25$.

$$\begin{aligned} c_1 + c_2 &= 1 \\ \tfrac{1}{2}c_1 - \tfrac{1}{4}c_2 &= 1.25 \end{aligned} \quad \Rightarrow \quad c_1 = 2,\ c_2 = -1$$

Check by partial fractions: $H(z) = \dfrac{1 + z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})} = \dfrac{2}{1 - \frac{1}{2}z^{-1}} - \dfrac{1}{1 + \frac{1}{4}z^{-1}}$. Then

$$\boxed{h(n) = \left[2\left(\tfrac{1}{2}\right)^n - \left(-\tfrac{1}{4}\right)^n\right]u(n)}$$

Check: $h(2) = 0.5 - 0.0625 = 0.4375$; recursion gives $\frac{1}{4}(1.25) + \frac{1}{8}(1) = 0.4375$.

(d) Use $a^n u(n) * b^n u(n) = \dfrac{a^{n+1} - b^{n+1}}{a - b}u(n)$.

$$2\left(\tfrac{1}{2}\right)^n * \left(\tfrac{1}{3}\right)^n = 2\,\frac{(\frac{1}{2})^{n+1} - (\frac{1}{3})^{n+1}}{1/6} = 6\left(\tfrac{1}{2}\right)^n - 4\left(\tfrac{1}{3}\right)^n$$

$$\left(-\tfrac{1}{4}\right)^n * \left(\tfrac{1}{3}\right)^n = \frac{(-\frac{1}{4})^{n+1} - (\frac{1}{3})^{n+1}}{-7/12} = \tfrac{3}{7}\left(-\tfrac{1}{4}\right)^n + \tfrac{4}{7}\left(\tfrac{1}{3}\right)^n$$

Subtract the second from the first:

$$\boxed{y(n) = \left[6\left(\tfrac{1}{2}\right)^n - \tfrac{3}{7}\left(-\tfrac{1}{4}\right)^n - \tfrac{32}{7}\left(\tfrac{1}{3}\right)^n\right]u(n)}$$

Check: $y(0) = 6 - \frac{3}{7} - \frac{32}{7} = 1$. $y(1) = 3 + \frac{3}{28} - \frac{32}{21} = 1.5833$. Both match part (b).

- a. 1 mark: both modes and a stability statement.
- b. 2 marks: 1 for the recursion set up correctly, 1 for all four values.
- c. 2 marks: 1 for the form and initial values (or the partial fractions), 1 for c_1, c_2.
- d. 2 marks: 1 for the geometric-convolution formula used twice, 1 for the final coefficients.

COMMON MISTAKES

- Using $h(0) = 1$ and $h(1) = \frac{1}{4}$, i.e. forgetting the $x(n-1)$ term. The $+\delta(n-1)$ adds 1 to $h(1)$.
- Getting the characteristic equation signs wrong. Move the y terms to the form $\lambda^2 - \frac{1}{4}\lambda - \frac{1}{8}$ exactly as they appear on the left-hand side.
- In (d), dropping the $u(n)$, or forgetting that $(-\frac{1}{4}) - \frac{1}{3} = -\frac{7}{12}$ is negative.
- Not checking (d) against (b). Two lines of checking catch most errors in this question.

REREAD: D2 (LCCDE, modes, $h(n)$, convolution).

Question 3 [7 marks]

(a) $h(n) = \{\underset{\uparrow}{0.5}, 1, 0.5\}$.

$$H(\omega) = 0.5 + e^{-j\omega} + 0.5e^{-j2\omega} = e^{-j\omega}\left(0.5e^{j\omega} + 1 + 0.5e^{-j\omega}\right) = \boxed{(1+\cos\omega)\,e^{-j\omega}}$$

Since $1 + \cos\omega \geq 0$ for all ω:

$$|H(\omega)| = 1 + \cos\omega, \qquad \theta(\omega) = -\omega, \quad -\pi < \omega < \pi$$

There are no $\pm\pi$ jumps (the amplitude never goes negative). $|H(0)| = 2$ and $|H(\pi)| = 0$, so this is a lowpass filter.

(b) Yes. $\theta(\omega) = -\omega n_0$ with $n_0 = 1$, so the group delay is $\tau(\omega) = -\frac{d\theta}{d\omega} = \mathbf{1\ sample}$. This agrees with $h(n)$ being symmetric about $n = 1$.

(c) Parseval: $\displaystyle\int_{-\pi}^{\pi} |H(\omega)|^2 d\omega = 2\pi \sum_n |h(n)|^2 = 2\pi(0.25 + 1 + 0.25) = \boxed{3\pi}$.

(d) Use $A\cos(\omega_0 n + \phi) \to A|H(\omega_0)|\cos(\omega_0 n + \phi + \theta(\omega_0))$ for each term.

Term	ω_0	$\lvert H\rvert$	θ	Output
4	0	2	0	8
$2\cos(\frac{\pi}{2}n + \frac{\pi}{6})$	$\pi/2$	1	$-\pi/2$	$2\cos(\frac{\pi}{2}n - \frac{\pi}{3})$
$3\cos(\frac{2\pi}{3}n)$	$2\pi/3$	0.5	$-2\pi/3$	$1.5\cos(\frac{2\pi}{3}n - \frac{2\pi}{3})$
$5(-1)^n = 5\cos(\pi n)$	π	0	(not needed)	0

$$\boxed{y(n) = 8 + 2\cos\left(\tfrac{\pi}{2}n - \tfrac{\pi}{3}\right) + 1.5\cos\left(\tfrac{2\pi}{3}n - \tfrac{2\pi}{3}\right)}$$

Because the phase is linear, every surviving term is scaled by its gain and delayed by exactly one sample. A simulation of the difference equation matches this exactly for $n \geq 2$.

- a. 2 marks: 1 for $H(\omega)$ factored, 1 for magnitude and phase.
- b. 1 mark: "yes, 1 sample".
- c. 1 mark: 3π with Parseval named.
- d. 3 marks: 1 for the DC term, 1 for the two cosines, 1 for recognising the $(-1)^n$ term is removed.

COMMON MISTAKES

- Leaving $(-1)^n$ out of the frequency analysis. It is a sinusoid at $\omega = \pi$.
- Forgetting to multiply the DC term by $H(0) = 2$.
- Writing Parseval without the 2π: $\int_{-\pi}^{\pi} |H|^2 d\omega = 2\pi \sum |h|^2$, not $\sum |h|^2$.
- Adding the phase to the amplitude, or using degrees and radians in the same expression.

REREAD: D4 (frequency response, sinusoidal steady state). For part (c), D3 (DTFT properties, Parseval).

Question 4 [8 marks]

(a) Lengths 4 and 3, so the result has length $4 + 3 - 1 = 6$.

$$x_1 * x_2 = \{\underset{\uparrow}{2}, 5, 3, 0, -1, -1\}$$

(b) 4-point circular convolution = linear convolution wrapped modulo 4. Add samples 4 and 5 onto samples 0 and 1:

$$x_1 \circledast_4 x_2 = \{2 + (-1),\ 5 + (-1),\ 3,\ 0\} = \boxed{\{\underset{\uparrow}{1}, 4, 3, 0\}}$$

Circular equals linear when $N \geq L_1 + L_2 - 1 = \mathbf{6}$.

(c) For $N = 4$, $W_4 = e^{-j\pi/2} = -j$, so $X_1(k) = \sum_{n=0}^{3} x_1(n)(-j)^{kn}$.

$$\begin{aligned} X_1(0) &= 1 + 2 + 0 - 1 = 2 \\ X_1(1) &= 1 + 2(-j) + 0 - 1(j) = 1 - 3j \\ X_1(2) &= 1 - 2 + 0 - 1(-1) = 0 \\ X_1(3) &= 1 + 2(j) + 0 - 1(-j) = 1 + 3j \end{aligned}$$

$$\boxed{X_1(k) = \{2,\ 1 - 3j,\ 0,\ 1 + 3j\}}$$

Check: $X_1(3) = X_1^*(1)$, as it must be for a real sequence.

(d) Resolution $= F_s/N \leq 30$ Hz, so $N \geq 8000/30 = 266.7$. The next power of 2 is $\boxed{N = 512}$ ($m = 9$).

$$T_0 = N/F_s = 512/8000 = \boxed{0.064\ \text{s} = 64\ \text{ms}}$$

(e) No. The true resolution is set by the record length: $1/T_0 = F_s/L = 8000/200 = 40$ Hz, which is wider than the 30 Hz tone spacing. Zero-padding only samples the same DTFT more finely. It does not add information, so the two peaks stay merged.

- a. 1 mark: correct six samples.
- b. 2 marks: 1 for the wrapped result, 1 for $N \geq 6$.
- c. 2 marks: 1 for $X_1(0), X_1(2)$; 1 for $X_1(1), X_1(3)$.
- d. 2 marks: 1 for $N = 512$, 1 for T_0.
- e. 1 mark: "no" with the 40 Hz reasoning.

COMMON MISTAKES

- Using $e^{+j2\pi kn/N}$ in the forward DFT. The forward DFT has the minus sign.
- Rounding N to 267 or 300. The question asks for $N = 2^m$.
- Believing that zero-padding improves the ability to separate tones. It gives a smoother picture of the same, smeared spectrum.
- In (b), wrapping the wrong end. Samples with index $n \geq N$ are added to index $n - N$.

REREAD: D5 (DFT, circular convolution) for (a)–(c). D6 (spectral analysis, resolution, zero-padding) for (d)–(e).

Question 5 [8 marks]

(a) Denominator: $1 + 1.5z^{-1} - z^{-2} = (1 - 0.5z^{-1})(1 + 2z^{-1})$.

$$H(z) = \frac{3 + z^{-1}}{(1 - 0.5z^{-1})(1 + 2z^{-1})} = \frac{3z\left(z + \frac{1}{3}\right)}{(z - 0.5)(z + 2)}$$

Zero: $z = -\frac{1}{3}$ (and one at $z = 0$ when written in positive powers). **Poles:** $z = 0.5$ and $z = -2$.

(b) Partial fractions in z^{-1}:

$$A = \left.\frac{3 + z^{-1}}{1 + 2z^{-1}}\right|_{z^{-1}=2} = \frac{5}{5} = 1, \qquad B = \left.\frac{3 + z^{-1}}{1 - 0.5z^{-1}}\right|_{z^{-1}=-1/2} = \frac{2.5}{1.25} = 2$$

$$H(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{2}{1 + 2z^{-1}}$$

ROC	$h(n)$	Type
$\lvert z\rvert > 2$	$[(0.5)^n + 2(-2)^n]u(n)$	causal, unstable
$0.5 < \lvert z\rvert < 2$	$(0.5)^n u(n) - 2(-2)^n u(-n-1)$	noncausal, stable
$\lvert z\rvert < 0.5$	$-[(0.5)^n + 2(-2)^n]u(-n-1)$	anticausal, unstable

(c) Stable requires the ROC to include the unit circle $|z| = 1$. Only $\mathbf{0.5 < |z| < 2}$ does. That ROC is an annulus, so $h(n)$ is two-sided: **noncausal**. A causal system would need the ROC outside the outermost pole ($|z| > 2$), which excludes the unit circle. So this system cannot be both causal and stable.

(d) Cross-multiply $Y(z)(1 + 1.5z^{-1} - z^{-2}) = X(z)(3 + z^{-1})$:

$$\boxed{y(n) + 1.5y(n-1) - y(n-2) = 3x(n) + x(n-1)}$$

(e) The ROC includes $|z| = 1$, so $H(\omega) = H(z)|_{z=e^{j\omega}}$ exists. At $\omega = \pi$, $z = -1$:

$$H(\pi) = \frac{3 - 1}{1 - 1.5 - 1} = \frac{2}{-1.5} = -\frac{4}{3}$$

$$\boxed{y(n) = -\tfrac{4}{3}\cos(\pi n) = \tfrac{4}{3}\cos(\pi n + \pi)}$$

A numerical two-sided convolution with the stable $h(n)$ gives $y(0) = -1.3333$, which agrees.

- a. 1 mark: both poles and the zero.
- b. 3 marks: 1 for A and B, 2 for the three ROC/$h(n)$ pairs (lose 1 per wrong pair, minimum 0).
- c. 2 marks: 1 for the ROC with reason, 1 for "noncausal" with reason.
- d. 1 mark: correct LCCDE.
- e. 1 mark: $-4/3$ and the output.

COMMON MISTAKES

- Using $-(a)^n u(-n-1)$ for a pole *inside* the ROC's inner edge. Rule: for each pole, if the ROC is outside it the term is right-sided $u(n)$; if the ROC is inside it the term is left-sided $-u(-n-1)$.
- Missing the minus sign on the left-sided term.
- Saying "stable because both poles are inside the unit circle". The pole at -2 is outside. Stability depends on the ROC, not just the pole positions.
- Evaluating H on the unit circle for the causal system (ROC $|z| > 2$). Its frequency response does not exist.

REREAD: D7 (z-transform and ROC), D8 (inverse z-transform, stability and causality). D9 (mixed practice) for extra drill.

Question 6 [7 marks]

(a) $\omega_0 = 2\pi \times 125/1000 = \boxed{\pi/4}$ rad, so $2\cos\omega_0 = \sqrt{2}$.

$$H(z) = b_0\left(1 - \sqrt{2}z^{-1} + z^{-2}\right)$$

At $\omega = \pi$, $e^{-j\pi} = -1$ and $e^{-j2\pi} = 1$:

$$1 = b_0\left(1 + \sqrt{2} + 1\right) \Rightarrow b_0 = \frac{1}{2+\sqrt{2}} = \boxed{0.2929}$$

$$H(z) = 0.2929\left(1 - 1.4142z^{-1} + z^{-2}\right)$$

(b) $H(\omega) = b_0 e^{-j\omega}(2\cos\omega - \sqrt{2})$, so $|H(\omega)| = b_0|2\cos\omega - \sqrt{2}|$.

- 0 Hz ($\omega = 0$): $|H| = 0.2929(2 - 1.4142) = 3 - 2\sqrt{2} = \mathbf{0.172}$ (-15.3 dB).
- 250 Hz ($\omega = \pi/2$): $|H| = 0.2929 \times 1.4142 = \sqrt{2} - 1 = \mathbf{0.414}$ (-7.7 dB).

The notch is very wide. A second-order FIR notch has no poles to sharpen it, so it also strongly attenuates frequencies well away from 125 Hz.

(c) No. The zeros $e^{\pm j\pi/4}$ lie **on** the unit circle. The inverse $1/H(z)$ would have poles on the unit circle, which is marginally stable at best (it is a sinusoidal oscillator, not BIBO stable). A stable causal inverse needs all zeros strictly inside the unit circle (minimum phase).

(d) The comb has M zeros at $z_k = e^{j2\pi k/(M+1)}$, $k = 1, \ldots, M$, i.e. at $F = kF_s/(M+1)$. For nulls at multiples of 200 Hz:

$$\frac{F_s}{M+1} = 200 \Rightarrow M + 1 = \frac{1200}{200} = 6 \Rightarrow \boxed{M = 5}$$

Nulls in $0 < F \leq 600$ Hz: $\boxed{200,\ 400,\ 600\ \text{Hz}}$ ($k = 1, 2, 3$; 600 Hz is $\omega = \pi$, the zero at $z = -1$).

(e) $h(n) = \frac{1}{6}$ for $0 \leq n \leq 5$ is symmetric, so it has linear phase with delay $M/2 = \mathbf{2.5}$ **samples** $= 2.5/1200 = \mathbf{2.08}$ **ms**. DC gain: $H(0) = \sum h(n) = 6 \times \frac{1}{6} = \mathbf{1}$.

- a. 2 marks: 1 for ω_0, 1 for b_0.
- b. 1 mark: both gains and a sensible comment.
- c. 1 mark: “no”, zeros on the unit circle.
- d. 2 marks: 1 for $M = 5$, 1 for the three frequencies.
- e. 1 mark: delay and DC gain.

COMMON MISTAKES

- Normalising at $\omega = 0$ instead of $\omega = \pi$ as asked. That gives $b_0 = 1/(2 - \sqrt{2}) = 1.707$.
- Using $\omega_0 = 2\pi F$ (forgetting to divide by F_s).
- In (d), choosing $M = 6$ because “6 = 1200/200”. The zero spacing is $F_s/(M + 1)$, not F_s/M.
- Forgetting the null at $F_s/2 = 600$ Hz. It is there whenever $M + 1$ is even.

REREAD: D10 (digital filters: notch, comb, minimum phase). D4 for evaluating $|H(\omega)|$.

Question 7 [9 marks]

(a) $\omega = 2\pi F/F_s$:

$$\omega_p = \frac{2\pi(1000)}{8000} = \frac{\pi}{4}, \qquad \omega_s = \frac{2\pi(2000)}{8000} = \frac{\pi}{2}$$

(b) $T = 1$, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$\Omega_p = 2\tan\frac{\pi}{8} = 0.8284 \text{ rad/s}, \qquad \Omega_s = 2\tan\frac{\pi}{4} = 2.0000 \text{ rad/s}$$

(c) $\delta_p = 10^{-3/20} = 0.7079$, $\delta_s = 10^{-15/20} = 0.1778$.

$$N = \frac{\log\left(\dfrac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}\right)}{2\log\left(\dfrac{\Omega_s}{\Omega_p}\right)} = \frac{\log(30.77)}{2\log(2.4142)} = 1.944$$

Round up: $\boxed{N = 2}$.

(d) Match the stopband exactly (as in T9 Q5; with the bilinear transform there is no aliasing, so exceeding the passband spec is the sensible choice):

$$\Omega_c = \Omega_s\left(\frac{\delta_s^2}{1-\delta_s^2}\right)^{\frac{1}{2N}} = 2\left(\frac{0.03162}{0.96838}\right)^{1/4} = \boxed{0.8502 \text{ rad/s}}$$

For $N = 2$ the stable poles are $s_k = \Omega_c e^{j(\pi/2 + \pi(2k+1)/4)}$, $k = 0, 1$, i.e. $\Omega_c e^{\pm j3\pi/4}$. Their product is Ω_c^2 and their sum is $-\sqrt{2}\Omega_c$:

$$H(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\,\Omega_c s + \Omega_c^2} = \boxed{\frac{0.7228}{s^2 + 1.2024s + 0.7228}}$$

(Matching the passband instead gives $\Omega_c = 0.8294$. That also meets the spec and earns full marks if stated.)

(e) With $\Omega_c = 0.85$: $\Omega_c^2 = 0.7225$ and $\sqrt{2}\Omega_c = 1.2021$. Substitute $s = 2\dfrac{1-z^{-1}}{1+z^{-1}}$ and multiply top and bottom by $(1+z^{-1})^2$:

$$H(z) = \frac{0.7225(1+z^{-1})^2}{4(1-z^{-1})^2 + 2(1.2021)(1-z^{-1})(1+z^{-1}) + 0.7225(1+z^{-1})^2}$$

Collect powers of z^{-1} in the denominator:

$$\begin{aligned} z^0 &: 4 + 2.4042 + 0.7225 = 7.1267 \\ z^{-1} &: -8 + 1.4450 = -6.5550 \\ z^{-2} &: 4 - 2.4042 + 0.7225 = 2.3183 \end{aligned}$$

$$H(z) = \frac{0.7225 + 1.4450z^{-1} + 0.7225z^{-2}}{7.1267 - 6.5550z^{-1} + 2.3183z^{-2}} = \boxed{\frac{0.1014 + 0.2028z^{-1} + 0.1014z^{-2}}{1 - 0.9198z^{-1} + 0.3253z^{-2}}}$$

Difference equation:

$$\boxed{y(n) = 0.9198\,y(n-1) - 0.3253\,y(n-2) + 0.1014\,x(n) + 0.2028\,x(n-1) + 0.1014\,x(n-2)}$$

Checks:

- $\omega = 0$ ($z = 1$): $H = \dfrac{0.1014 + 0.2028 + 0.1014}{1 - 0.9198 + 0.3253} = \dfrac{0.4056}{0.4055} \approx 1$. Correct for a Butterworth lowpass.
- $\omega = \pi$ ($z = -1$): numerator $= 0.1014(1-1)^2 = 0$, so $H(\pi) = 0$. The bilinear transform maps $s = \infty$ to $z = -1$.

The design gives -2.79 dB at ω_p (spec: no worse than -3 dB) and -15.0 dB at ω_s. Its poles are at $0.570\,e^{\pm j0.633}$, inside the unit circle.

- a. 1 mark: both ω_p and ω_s.
- b. 1 mark: both prewarped values.
- c. 2 marks: 1 for δ_p, δ_s in linear scale, 1 for $N = 2$ from 1.944.
- d. 2 marks: 1 for Ω_c, 1 for $H(s)$.
- e. 3 marks: 1 for the substitution set up, 1 for normalised $H(z)$, 1 for the difference equation with correct signs (the check is part of this mark).

COMMON MISTAKES

- Skipping prewarping and using $\Omega = \omega/T$. That gives N and Ω_c for the wrong filter.
- Using $\Omega = \tan(\omega/2)$ without the factor $2/T$. Either is fine only if you are consistent and use the same factor in the bilinear substitution.
- Putting dB values straight into the N formula. Convert to linear: $\delta = 10^{-A/20}$.
- Rounding $N = 1.944$ down to 1.
- Sign errors moving from $H(z)$ to the difference equation. Denominator $1 - 0.9198z^{-1}$ becomes $+0.9198\,y(n-1)$ on the right-hand side.
- Forgetting to normalise so that $a_0 = 1$.

REREAD: D11 (IIR design: bilinear transform and Butterworth).

Question 8 [6 marks]

(a) Inverse DTFT of the ideal lowpass with $\omega_c = \pi/4$:

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} e^{j\omega n}d\omega = \frac{\sin(\pi n/4)}{\pi n} = \boxed{\tfrac{1}{4}\operatorname{sinc}\left(\tfrac{\pi n}{4}\right)}, \qquad h_d(0) = \tfrac{1}{4}$$

(course definition $\operatorname{sinc} x = \sin x/x$).

(b) Shift by $(M-1)/2 = 3$ and window: $h(n) = h_d(n-3)\,w(n)$, $0 \le n \le 6$, with Hamming $w(n) = 0.54 - 0.46\cos(2\pi n/6)$.

n	$n-3$	$h_d(n-3)$	$w(n)$	$h(n)$
0	-3	0.0750	0.08	0.0060
1	-2	0.1592	0.31	0.0493
2	-1	0.2251	0.77	0.1733
3	0	0.2500	1.00	0.2500
4	1	0.2251	0.77	0.1733
5	2	0.1592	0.31	0.0493
6	3	0.0750	0.08	0.0060

$$\boxed{h(n) = \{\underset{\uparrow}{0.0060}, 0.0493, 0.1733, 0.2500, 0.1733, 0.0493, 0.0060\}}$$

(These agree with `scipy.signal.firwin(7, 0.25, window='hamming', scale=False)`.)

(c) Symmetric: $h(n) = h(6-n)$. Linear phase with group delay $(M-1)/2 = \mathbf{3}$ **samples**.

(d) $H(0) = \sum h(n) = \mathbf{0.7073}$ (-3.0 dB). It is not 1 because the window truncates and tapers $h_d(n)$. The ideal filter's DC gain $\sum_{n=-\infty}^{\infty} h_d(n) = 1$ only holds for the full infinite sequence. With $M = 7$ and a Hamming taper, most of that sum is cut off. To fix it, divide every coefficient by $\sum h(n)$, i.e. $h(n)/0.7073$.

- a. 1 mark: $h_d(n)$ including the $n = 0$ value.
- b. 3 marks: 1 for the shift by 3, 1 for the window values, 1 for the products.
- c. 1 mark: symmetric, 3 samples.
- d. 1 mark: 0.7073 with a valid reason and fix.

COMMON MISTAKES

- Forgetting the shift: using $h_d(n)$ for $0 \le n \le 6$ gives a non-symmetric, non-linear-phase filter.
- Using MATLAB's sinc definition ($\sin(\pi x)/(\pi x)$) with the course's argument. Check: $h_d(0)$ must equal $\omega_c/\pi = 0.25$.
- Using the Hanning formula (endpoints 0) instead of Hamming (endpoints 0.08).
- Using $\cos(2\pi n/M)$ instead of $\cos(2\pi n/(M-1))$ in the window.

REREAD: D12 (Chebyshev and FIR design).

Score interpretation

Add up your marks out of 60. Then look at *where* you lost them, because that matters more than the total.

Score	What it means	What to do in the next 3 days
51–60 (85%+)	Exam-ready. Your methods are reliable under time pressure.	Do mock B if available. Spend the rest on your reference sheet and on speed.
42–50 (70–84%)	Solid. A few method gaps or slips.	Reread the packs for the two questions where you lost the most marks. Redo those questions from a blank page without the solution.
30–41 (50–69%)	Pass level, but fragile. Probably one or two whole topics are weak.	Spend one study block on each weak topic (pack sections 2 and 3, then the new practice questions). Put the method steps on your reference sheet.
Below 30	Big gaps. Most likely in L05–L10, which carry 45 of the 60 marks.	Go in this order: D8 (inverse z), D11 (Butterworth), D4 (frequency response), D5 (DFT). These are mechanical and give the most marks per hour.

Diagnose by question. Use this to choose what to reread.

Lost 3 or more marks on	Likely gap	Reread
Q1	Aliasing and folding sines, or property tests	D1
Q2	Initial values of $h(n)$, geometric convolution	D2
Q3	Factoring $H(\omega)$, phase, Parseval	D3, D4
Q4	DFT sums, wrap-around, resolution vs zero-padding	D5, D6
Q5	Choosing left- or right-sided terms from the ROC	D7, D8, D9
Q6	$\omega = 2\pi F/F_s$, normalisation, comb zero spacing	D10
Q7	Prewarping, order formula, bilinear algebra	D11
Q8	Shift by $(M-1)/2$, window formula, sinc definition	D12

Timing check. If you did not finish, note which question you were on at 120 minutes. Running out of time on Q7 usually means too long was spent on Q1–Q2. Those are worth only 15 marks, so aim to finish them in 30 minutes. If you finished early but scored below 42, slow down and use the checks built into the questions (Q2(d) against Q2(b), $H(1)$ and $H(-1)$ in Q7, conjugate symmetry in Q4(c)).

DSP · MOCK

DSP mock B

Answers and worked solutions · Day 24 · Mon 16 Nov · pack DX2

Marking guide

Mark your script against these solutions. Give part marks for correct method even when the arithmetic slips. Every number below has been checked in Python (numpy/scipy/sympy). The “Reread” line names the study pack to go back to if you lost marks.

Question 1: Correlation and delay estimation [5]

(a) [1] x is nonzero for $n = 0, 1, 2$.

- $r_{xx}(0) = 1^2 + 2^2 + (-1)^2 = 6$
- $r_{xx}(1) = x(1)x(0) + x(2)x(1) = 2 - 2 = 0$
- $r_{xx}(2) = x(2)x(0) = -1$

Autocorrelation is even, so

$$r_{xx}(l) = \{-1, 0, \underset{\uparrow}{6}, 0, -1\}, \quad l = -2, \ldots, 2.$$

(b) [2] $r_{yx}(l) = \sum_n y(n)x(n-l) = y(l) + 2y(l+1) - y(l+2)$. Slide x along y. (This equals $y(l) * x(-l)$, so you can also do it as a convolution with the flipped pulse.)

l	-2	-1	0	1	2	3	4	5	6
$r_{yx}(l)$	-1	2	2	-4	-1	**12**	-1	0	1

For example, $r_{yx}(3) = y(3) + 2y(4) - y(5) = 2 + 8 + 2 = 12$.

(c) [1] $\hat{d} = \arg\max_l |r_{yx}(l)| = 3$ samples. From $d = 2F_s r/c$:

$$r = \frac{\hat{d}\,c}{2F_s} = \frac{3 \times 3 \times 10^8}{2 \times 10^7} = \mathbf{45\ m}.$$

(d) [1] Near the peak, $r_{yx}(\hat{d}) \approx a\, r_{xx}(0)$, so $\hat{a} = 12/6 = \mathbf{2}$. Range resolution is $c/(2F_s) = 3 \times 10^8/(2 \times 10^7) = \mathbf{15\ m}$ per sample of delay.

MARKS: (a) 1 for all five values. (b) 2 for the full table, or 1 if one or two entries are wrong. (c) ½ for $\hat{d}$, ½ for r. (d) ½ for a, ½ for resolution.

COMMON MISTAKES

- Mixing up the conventions. T3 Q7 uses $R_{xy}[l] = \sum x[k]y[k+l]$, which is the lecture's $r_{yx}(l)$. Whatever the convention, the peak must sit at a **positive** lag equal to the delay. If you get -3, your lag sign is flipped.
- Picking the largest value rather than the largest magnitude. Here it makes no difference, but the course's rule is $\arg\max |r_{yx}|$.
- Forgetting the factor 2 in $d = 2F_s r/c$. The signal makes a round trip.

REREAD: D1 (L03 correlation and radar case study).

Question 2: LCCDE with complex modes [6]

(a) [1] Characteristic equation: $\lambda^2 - \lambda + 0.5 = 0$, so

$$\lambda_{1,2} = 0.5 \pm j0.5 = \tfrac{1}{\sqrt{2}}\, e^{\pm j\pi/4}.$$

The modes are $\left(\tfrac{1}{\sqrt{2}}\right)^n e^{\pm j\pi n/4}$. Both have $|\lambda| = 0.7071 < 1$, so the causal system is **stable**.

(b) [1] $h(n) = \delta(n) + h(n-1) - 0.5h(n-2)$:

$$h(n) = \{\underset{\uparrow}{1}, 1, 0.5, 0, -0.25, -0.25, -0.125\}, \quad n = 0, \ldots, 6.$$

(c) [2] Complex-conjugate modes combine into a real form:

$$h(n) = \left(\tfrac{1}{\sqrt{2}}\right)^n \left[A\cos\tfrac{\pi n}{4} + B\sin\tfrac{\pi n}{4}\right] u(n).$$

- $h(0) = 1 \Rightarrow A = 1$.
- $h(1) = \tfrac{1}{\sqrt{2}}\left(\tfrac{1}{\sqrt{2}} + \tfrac{B}{\sqrt{2}}\right) = \tfrac{1+B}{2} = 1 \Rightarrow B = 1$.

$$\boxed{h(n) = \left(\tfrac{1}{\sqrt{2}}\right)^n \left[\cos\tfrac{\pi n}{4} + \sin\tfrac{\pi n}{4}\right] u(n) = 2^{(1-n)/2} \cos\left(\tfrac{\pi(n-1)}{4}\right) u(n)}$$

Check: $n = 2$ gives $\tfrac{1}{2}(0+1) = 0.5$, and $n = 4$ gives $\tfrac{1}{4}(-1+0) = -0.25$. Both match (b).

(d) [2] $H(\omega) = \dfrac{1}{1 - e^{-j\omega} + 0.5e^{-j2\omega}}$.

- $H(0) = \dfrac{1}{1 - 1 + 0.5} = 2$.
- $H(\pi/2) = \dfrac{1}{1 - (-j) + 0.5(-1)} = \dfrac{1}{0.5 + j} = 0.4 - j0.8 = 0.8944\angle -1.107$ rad (-63.43°).

$$\boxed{y_{ss}(n) = 8 + 1.789\cos\left(\tfrac{\pi}{2}n - 1.107\right)}$$

MARKS: (a) ½ modes, ½ stability with reason. (b) 1. (c) 1 for the real form, 1 for A and B. (d) 1 for each frequency (gain and phase applied correctly).

COMMON MISTAKES

- Writing the real form with the wrong radius, for example $(0.5)^n$ instead of $|\lambda|^n = (1/\sqrt{2})^n$.

- Fitting A and B with $h(-1) = h(-2) = 0$ and getting confused. Use $h(0)$ and $h(1)$ from the recursion.
- Taking $e^{-j\pi/2} = +j$. It is $-j$.
- Doubling the DC term or forgetting to scale it: the DC part is $4H(0) = 8$.

REREAD: D2 (modes and $h(n)$ from an LCCDE), D4 (sinusoidal steady state).

Question 3: DTFT shortcut evaluations [5]

The samples are $x(-2) = 1$, $x(-1) = -2$, $x(0) = 3$, $x(1) = 0$, $x(2) = -1$ and $x(3) = -4$.

(a) [1] $X(0) = \sum_n x(n) = 1 - 2 + 3 + 0 - 1 - 4 = \mathbf{-3}$. A negative real number has $\angle X(0) = \boldsymbol{\pi}$ (or $-\pi$).

(b) [1] $X(\pi) = \sum_n (-1)^n x(n) = 1 + 2 + 3 + 0 - 1 + 4 = \mathbf{9}$. (Signs: $n = -2$ gives $+$, $n = -1$ gives $-$, and so on.)

(c) [1] From the inverse DTFT, $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$. So the integral is $2\pi x(2) = \mathbf{-2}\pi \approx -6.283$.

(d) [1] Parseval:
$\int_{-\pi}^{\pi} |X(\omega)|^2 d\omega = 2\pi \sum |x(n)|^2 = 2\pi(1 + 4 + 9 + 0 + 1 + 16) = \mathbf{62}\pi \approx 194.8$.

(e) [1] Differentiation in frequency: $n\,x(n) \leftrightarrow j\,dX/d\omega$, so $|dX/d\omega|^2 = |\text{DTFT of } nx(n)|^2$. Apply Parseval to $n\,x(n)$:

$$\int_{-\pi}^{\pi} \left| \frac{dX}{d\omega} \right|^2 d\omega = 2\pi \sum n^2 |x(n)|^2 = 2\pi(4 + 4 + 0 + 0 + 4 + 144) = \mathbf{312}\pi \approx 980.2.$$

MARKS: 1 per part. In (a), give ½ if $X(0)$ is right but the angle is given as 0.

COMMON MISTAKES

- Putting the origin in the wrong place. The arrow is under the 3, so the first sample is at $n = -2$. This flips the signs in $X(\pi)$ and changes (c) and (e).
- Saying $\angle X(0) = 0$ because $X(0)$ is real. Real and negative means $\pm\pi$.
- Forgetting the 2π in Parseval, or using $\frac{1}{2\pi}$ on the wrong side.
- Using $\sum n|x(n)|^2$ instead of $\sum n^2|x(n)|^2$ in (e).

REREAD: D3 (DTFT properties; T4 Q2 is the course version of this question).

Question 4: Deconvolution via the DFT [6]

(a) [1] The linear convolution has length $L + M - 1 = 3 + 2 - 1 = 4$, which is the length of y. With $N \geq 4$, the N-point circular convolution equals the linear one, so $Y(k) = H(k)X(k)$ holds exactly. Then $X(k) = Y(k)/H(k)$ (provided no $H(k) = 0$), and the IDFT returns $x(n)$ with one trailing zero.

(b) [2] With $W_4 = e^{-j\pi/2} = -j$, we have $Y(k) = \sum_n y(n)(-j)^{nk}$.

- $Y(0) = 4 + 0 + 1 + 1 = 6$
- $Y(1) = 4 + 0(-j) + 1(-1) + 1(j) = 3 + j$
- $Y(2) = 4 - 0 + 1 - 1 = 4$
- $Y(3) = \overline{Y(1)} = 3 - j$

$H(k) = 2 + (-j)^k$, which gives $H(k) = \{3,\ 2 - j,\ 1,\ 2 + j\}$.

(c) [2]

- $X(0) = 6/3 = 2$
- $X(1) = \dfrac{3 + j}{2 - j} = \dfrac{(3 + j)(2 + j)}{5} = \dfrac{5 + 5j}{5} = 1 + j$
- $X(2) = 4/1 = 4$
- $X(3) = 1 - j$

IDFT, $x(n) = \frac{1}{4}\sum_k X(k)\, j^{nk}$:

- $x(0) = \frac{1}{4}(2 + 1 + j + 4 + 1 - j) = 2$
- $x(1) = \frac{1}{4}(2 + (1 + j)j - 4 + (1 - j)(-j)) = \frac{1}{4}(2 + j - 1 - 4 - j - 1) = -1$
- $x(2) = \frac{1}{4}(2 - (1 + j) + 4 - (1 - j)) = 1$
- $x(3) = 0$

$$\boxed{x(n) = \{2, \underset{\uparrow}{-1}, 1\}}$$

Check: $\{2, 1\} * \{2, -1, 1\} = \{4, 0, 1, 1\}$. Correct.

(d) [1] For $h = \{1, 1\}$, $H(2) = 1 + (-1) = 0$. The channel has a zero at $z = -1$ ($\omega = \pi$), which is exactly DFT bin $k = 2$. $Y(2) = 0$ whatever the input, so $X(2) = 0/0$ is undefined. The information at $\omega = \pi$ is destroyed, and the inverse $1/(1 + z^{-1})$ has a pole on the unit circle.

MARKS: (a) 1. (b) 1 for $Y(k)$, 1 for $H(k)$. (c) 1 for $X(k)$, 1 for the IDFT. (d) 1.

COMMON MISTAKES

- Using $W_4 = +j$ in the forward DFT. The forward kernel is $e^{-j2\pi nk/N}$.

- Forgetting the $1/N$ in the IDFT.
- Dividing complex numbers without the conjugate. Multiply top and bottom by $\overline{H(k)}$.
- Choosing $N = 3$ "because x has length 3". N must cover the **output** length.

REREAD: D5 (DFT and IDFT by hand), D6 (DFT filtering and T5 Q4 deconvolution).

Question 5: Overlap-save block convolution [6]

(a) [1] $M = 3$, $N = 4$, so $L = N - M + 1 = \mathbf{2}$ new samples per block. Discard the first $M - 1 = \mathbf{2}$ outputs of each block.

(b) [1] Prepend $M - 1 = 2$ zeros. Each block holds the last 2 samples of the previous block followed by 2 new samples. Output length is $6 + 3 - 1 = 8$, so 4 blocks are needed (pad the tail with zeros):

Block	Input $b(n)$, $n = 0..3$
1	$\{0, 0, 1, 2\}$
2	$\{1, 2, 0, -1\}$
3	$\{0, -1, 3, 1\}$
4	$\{3, 1, 0, 0\}$

(c) [3] With $h = \{1, -1, 2, 0\}$, the samples kept involve no wrap-around:

$$y_c(2) = b(2) - b(1) + 2b(0), \qquad y_c(3) = b(3) - b(2) + 2b(1).$$

Block	Full circular output $y_c(0..3)$	Kept $y_c(2), y_c(3)$
1	$\{0, 4, 1, 1\}$	$1, 1$
2	$\{2, -1, 0, 3\}$	$0, 3$
3	$\{5, 1, 4, -4\}$	$4, -4$
4	$\{3, -2, 5, 2\}$	$5, 2$

$$\boxed{y(n) = \{\underset{\uparrow}{1}, 1, 0, 3, 4, -4, 5, 2\}}$$

This equals the direct linear convolution $h * x$.

(d) [1] Block 1: $y_c(0) = b(0) + h(1)b(3) + h(2)b(2) = 0 - 2 + 2 = 0$ and $y_c(1) = b(1) - b(0) + 2b(3) = 4$. The true values are $y(-2) = y(-1) = 0$. The first $M - 1$ circular outputs wrap around and pick up samples from the **end** of the block (time-aliasing), so they are wrong and must be discarded. Here $y_c(0) = 0$ is right only by coincidence.

MARKS: (a) 1. (b) 1 for correct blocks with the 2-sample overlap. (c) 2 for the block outputs (½ each), 1 for the assembled $y(n)$. (d) 1 with a reason.

COMMON MISTAKES

- Confusing the two methods. Overlap-**save** overlaps the **inputs** and discards outputs. Overlap-**add** uses non-overlapping input blocks and adds the overlapping output tails.
- Forgetting the $M - 1$ leading zeros, so the first block has nothing to discard.
- Stopping after 3 blocks and losing the last two output samples (5, 2).
- Keeping the first L outputs instead of the last L.

REREAD: D6 (block convolution, overlap-save and overlap-add).

Question 6: Every possible sequence for each ROC [6]

(a) [2] Cover-up with $w = z^{-1}$. The numerator is $N(w) = 1 + 11w - 11w^2$.

- Pole 0.5 (set $w = 2$): $A = \dfrac{1 + 22 - 44}{(1-4)(1+6)} = \dfrac{-21}{-21} = 1.$
- Pole 2 (set $w = 0.5$): $B = \dfrac{1 + 5.5 - 2.75}{(1-0.25)(1+1.5)} = \dfrac{3.75}{1.875} = 2.$
- Pole -3 (set $w = -\frac{1}{3}$): $C = \dfrac{1 - \frac{11}{3} - \frac{11}{9}}{(1+\frac{1}{6})(1-\frac{2}{3})} = \dfrac{-35/9}{35/18} = -2.$

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{2}{1 - 2z^{-1}} - \frac{2}{1 + 3z^{-1}}$$

The degree of the numerator is less than that of the denominator, so there is no direct term.

(b) [3] Use the **given** expression, with coefficients 1, 1 and −1. The poles 0.5, 2 and -3 give four ROCs. Rule: a pole **inside** the ROC's inner radius gives a causal term $p^n u(n)$. A pole **outside** gives an anticausal term $-p^n u(-n-1)$.

ROC	$x(n)$	Type
$\lvert z\rvert < 0.5$	$[-(0.5)^n - 2^n + (-3)^n]u(-n-1)$	anticausal
$0.5 < \lvert z\rvert < 2$	$(0.5)^n u(n) + [-2^n + (-3)^n]u(-n-1)$	two-sided
$2 < \lvert z\rvert < 3$	$[(0.5)^n + 2^n]u(n) + (-3)^n u(-n-1)$	two-sided
$\lvert z\rvert > 3$	$[(0.5)^n + 2^n - (-3)^n]u(n)$	causal

(The anticausal term for $-\frac{1}{1+3z^{-1}}$ is $-\left[-(-3)^n u(-n-1)\right] = +(-3)^n u(-n-1)$.)

(c) [1] Only $0.5 < \lvert z\rvert < 2$ contains the unit circle, so only that sequence is absolutely summable and has a DTFT.

- $x(0) = (0.5)^0 = \mathbf{1}$
- $x(-1) = -2^{-1} + (-3)^{-1} = -0.5 - 0.3333 = \mathbf{-0.8333}$

MARKS: (a) ½ per coefficient, ½ for a correct method or form. (b) ½ per ROC sequence and 1 for the ROC list and classification. (c) ½ for the ROC, ½ for both values.

COMMON MISTAKES

- Listing only three ROCs. n distinct pole radii give $n + 1$ annular ROCs.

- Getting the sign of the anticausal term wrong: $\frac{1}{1-pz^{-1}} \leftrightarrow -p^n u(-n-1)$.
- Using $u(-n)$ instead of $u(-n-1)$.
- In (b), carrying on with your own (a) answer instead of the expression given. In an exam, the "assume" value is what earns the marks.
- Saying the causal sequence is stable. It has poles outside the unit circle.

REREAD: D7 (ROC rules), D8 (inverse by partial fractions; T7 Q1 is the two-pole version).

Question 7: Unilateral z-transform with initial conditions [6]

(a) [2] Use $y(n-1) \leftrightarrow z^{-1}Y^+(z) + y(-1)$ and $y(n-2) \leftrightarrow z^{-2}Y^+(z) + z^{-1}y(-1) + y(-2)$:

$$Y^+ - 0.75\left[z^{-1}Y^+ + 2\right] + 0.125\left[z^{-2}Y^+ + 2z^{-1} + 4\right] = \frac{3}{1-z^{-1}}.$$

The initial-condition terms are $-1.5 + 0.25z^{-1} + 0.5 = -1 + 0.25z^{-1}$. Move them to the right-hand side:

$$Y^+(z)\left(1 - 0.75z^{-1} + 0.125z^{-2}\right) = \frac{3}{1-z^{-1}} + 1 - 0.25z^{-1}.$$

The denominator factors as $(1-0.5z^{-1})(1-0.25z^{-1})$, so

$$Y^+(z) = \underbrace{\frac{3}{(1-z^{-1})(1-0.5z^{-1})(1-0.25z^{-1})}}_{\text{zero-state}} + \underbrace{\frac{1-0.25z^{-1}}{(1-0.5z^{-1})(1-0.25z^{-1})}}_{\text{zero-input}}.$$

(b) [3] The zero-input term cancels to $\dfrac{1}{1-0.5z^{-1}}$.

Zero-state partial fractions (cover-up with $w = z^{-1}$):

- Pole 1 ($w=1$): $\dfrac{3}{(0.5)(0.75)} = 8$
- Pole 0.5 ($w=2$): $\dfrac{3}{(-1)(0.5)} = -6$
- Pole 0.25 ($w=4$): $\dfrac{3}{(-3)(-1)} = 1$

$$Y^+(z) = \frac{8}{1-z^{-1}} - \frac{6}{1-0.5z^{-1}} + \frac{1}{1-0.25z^{-1}} + \frac{1}{1-0.5z^{-1}}$$

$$\boxed{y(n) = [8 - 5(0.5)^n + (0.25)^n]u(n)}$$

(c) [1]

- Zero-input: $(0.5)^n u(n)$. These initial conditions excite only the 0.5 mode.
- Zero-state: $[8 - 6(0.5)^n + (0.25)^n]u(n)$.

Recursion check: $y(0) = 3 + 0.75(2) - 0.125(4) = 4$, and the formula gives $8 - 5 + 1 = 4$. Next, $y(1) = 3 + 0.75(4) - 0.125(2) = 5.75$, and the formula gives $8 - 2.5 + 0.25 = 5.75$. Both match. As $n \to \infty$, $y \to 8 = 3H(1) = 3/0.375$.

MARKS: (a) 1 for the shift property with correct IC terms, 1 for $Y^+(z)$. (b) 1 for the partial fractions, 1 for combining, 1 for $y(n)$. (c) ½ for the split, ½ for the checks.

COMMON MISTAKES

- Sign errors in the IC terms. With $-0.75y(n-1)$ in the equation, the IC contribution is $-0.75y(-1)$, which becomes $+1.5$ on the right-hand side.
- Dropping the $z^{-1}y(-1)$ term from the $y(n-2)$ transform.
- Using $y(-1), y(-2)$ as if they were $y(0), y(1)$.
- Missing the cancellation in the zero-input term. It is not an error to keep it, but the partial fractions get longer.

REREAD: D9 (unilateral z-transform; the L08 slide 14 example is the course version).

Question 8: Digital resonator and allpass filter [6]

(a) [1] $\omega_0 = 2\pi(1000)/8000 = \pi/4$. The denominator is $1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$:

$$H(z) = \frac{b_0(1 - z^{-2})}{1 - 1.2728z^{-1} + 0.81z^{-2}}.$$

The zeros at $z = 1$ and $z = -1$ reject **0 Hz (DC)** and **4 kHz ($F_s/2$)** completely.

(b) [2] Evaluate at $e^{j\omega_0}$ with the factored form:

$$|H(\omega_0)| = b_0 \frac{|1 - e^{-j2\omega_0}|}{|1 - re^{j\omega_0}e^{-j\omega_0}|\,|1 - re^{-j\omega_0}e^{-j\omega_0}|} = b_0 \frac{2\sin\omega_0}{(1-r)\,|1 - re^{-j\pi/2}|}.$$

This gives $b_0 \dfrac{1.4142}{0.1 \times |1 + 0.9j|} = b_0 \dfrac{1.4142}{0.1 \times 1.3454} = 10.512\, b_0.$

So $b_0 = 1/10.512 = \mathbf{0.0951}$, and

$$y(n) = 1.2728\, y(n-1) - 0.81\, y(n-2) + 0.0951\,[x(n) - x(n-2)].$$

(The true peak is at 0.252π, slightly above ω_0, with $|H| = 10.53\, b_0$. The question asks for normalisation at ω_0.)

(c) [1] With $N = 1$, $A(z^{-1}) = 1 - 0.8z$:

$$H_{ap}(z) = z^{-1}\frac{1 - 0.8z}{1 - 0.8z^{-1}} = \frac{-0.8 + z^{-1}}{1 - 0.8z^{-1}}.$$

The pole is at $z = 0.8$ and the zero at $z = 1/0.8 = 1.25$ (reciprocals). On the unit circle,

$$|-0.8 + e^{-j\omega}| = |e^{-j\omega}|\,|1 - 0.8e^{j\omega}| = |1 - 0.8e^{-j\omega}|$$

(a number and its conjugate have equal modulus). So $|H_{ap}(\omega)| = 1$.

(d) [1] $H_{ap}(\pi/2) = \dfrac{-0.8 - j}{1 + 0.8j}$. Then

$$\angle H_{ap}(\pi/2) = \angle(-0.8 - j) - \angle(1 + 0.8j) = -128.66^\circ - 38.66^\circ = \mathbf{-167.3}^\circ\ (-2.920\ \text{rad}).$$

Equivalently, $\theta(\omega) = -\omega - 2\arctan\dfrac{0.8\sin\omega}{1 - 0.8\cos\omega}$. At $\pi/2$ this is $-\pi/2 - 2\arctan 0.8 = -2.920$ rad.

(e) [1] For small ω, $\theta \approx -\omega - 2\dfrac{0.8\omega}{0.2} = -9\omega$. So $\tau(0) = \mathbf{9}$ samples. In general $\tau(0) = (1+a)/(1-a)$. An allpass filter changes phase (delay) without changing magnitude, so it is used as a **phase equaliser**: it corrects the nonlinear phase of another filter.

MARKS: (a) ½ for $H(z)$, ½ for the frequencies. (b) 1 for b_0, 1 for the difference equation. (c) 1. (d) 1. (e) ½ for the delay, ½ for the use.

COMMON MISTAKES

- Using $\omega_0 = 2\pi f_0$ without dividing by F_s.
- Writing $+2r\cos\omega_0$ in the denominator. The sign is negative.
- Leaving $b_0 = 1$, or guessing a value. Compute $|H(\omega_0)|$ directly from the factored form.
- Assuming the peak is exactly at ω_0. It is close (for r near 1) but not exact.
- Building the allpass with pole and zero both at 0.8 (that gives $H = 1$). The zero is at the **reciprocal** of the pole.

REREAD: D10 (resonators, notch, comb and allpass; L09 slides 17 and 23).

Question 9: Chebyshev type I order and ϵ [5]

(a) [1]

- $\omega_p = 2\pi(1)/10 = 0.2\pi$ and $\omega_s = 2\pi(2.5)/10 = 0.5\pi$.
- Prewarp with $T = 1$: $\Omega_p = 2\tan(0.1\pi) = \mathbf{0.6498}$ and $\Omega_s = 2\tan(0.25\pi) = \mathbf{2.0000}$.

(b) [1]

- $\delta_p = 10^{-0.5/20} = 0.9441$ and $\delta_s = 10^{-40/20} = 0.01$.
- $\epsilon^2 = 1/\delta_p^2 - 1 = 0.1220$, so $\epsilon = \mathbf{0.3493}$.

(c) [2] The lecture's formula uses inverse **hyperbolic** cosine, because both arguments are greater than 1:

$$N = \frac{\cosh^{-1}\left(\frac{1}{\epsilon}\frac{\sqrt{1-\delta_s^2}}{\delta_s}\right)}{\cosh^{-1}(\Omega_s/\Omega_p)} = \frac{\cosh^{-1}(286.26)}{\cosh^{-1}(3.0777)} = \frac{6.3501}{1.7898} = 3.548.$$

Round up: $\boxed{N = 4}$. (Check: $N = 4$ gives 47.0 dB at Ω_s, and $N = 3$ gives only 31.5 dB.) A Butterworth filter for the same specification needs $N = 5.03 \to 6$, which shows why Chebyshev is "more efficient".

(d) [1] Use $N = 5$: $T_5(3.0777) = \cosh(5 \times 1.7898) = 3850.5$. Equivalently, $T_5(x) = 16x^5 - 20x^3 + 5x$.

$$A_s = 10\log_{10}\left(1 + \epsilon^2 T_5^2\right) = 10\log_{10}\left(1 + 0.1220 \times 3850.5^2\right) = \mathbf{62.6\ dB}.$$

MARKS: (a) ½ each for the ω pair and the Ω pair. (b) 1. (c) 1 for correct substitution, 1 for N rounded up. (d) 1.

COMMON MISTAKES

- Typing $\cos^{-1}$ into the calculator. The argument is greater than 1, so you get an error or a complex number. The slide's "$\cos^{-1}$" means $\cosh^{-1}$ here, as the T9 Q7 solution confirms.
- Not prewarping (see Q11(c)).
- Rounding N down.
- Using $\delta_s = 10^{-40/10}$. Amplitude specifications use $/20$.

REREAD: D11 (prewarping, bilinear design steps), D12 (Chebyshev formulas; T9 Q7 is the type II version).

Question 10: Completing a linear-phase FIR filter [5]

(a) [1] $M = 6$. The end samples are $h(0) = -1$ and $h(5) = 1$, which are negatives of each other, so the filter must be **antisymmetric**: $h(n) = -h(5-n)$. This is consistent with $h(1) = 2 = -h(4)$. Therefore

$$h(3) = -h(2) = \mathbf{-4}, \qquad h(n) = \{\underset{\uparrow}{-1}, 2, 4, -4, -2, 1\}.$$

(With M even there is no middle sample, so it is not forced to zero as in T9 Q8.)

(b) [2] Factor out the centre $e^{-j2.5\omega}$ and pair the samples:

$$H(\omega) = e^{-j2.5\omega}\left[-\left(e^{j2.5\omega} - e^{-j2.5\omega}\right) + 2\left(e^{j1.5\omega} - e^{-j1.5\omega}\right) + 4\left(e^{j0.5\omega} - e^{-j0.5\omega}\right)\right]$$

$$\boxed{H(\omega) = \underbrace{2\left[4\sin\tfrac{\omega}{2} + 2\sin\tfrac{3\omega}{2} - \sin\tfrac{5\omega}{2}\right]}_{A(\omega)} e^{-j(2.5\omega - \pi/2)}}$$

(c) [1] $A(\omega) \geq 0$ for $0 \leq \omega \leq \pi$ (checked numerically: it never changes sign). So

$$\theta(\omega) = -2.5\omega + \frac{\pi}{2}, \quad 0 < \omega \leq \pi,$$

which is linear, with group delay $\tau = (M-1)/2 = \mathbf{2.5}$ samples. (Spot check: $H(\pi/2) = -7 - 7j$, with angle $-3\pi/4 = -2.5(\pi/2) + \pi/2$.)

(d) [1]

- $H(0) = \sum h(n) = -1 + 2 + 4 - 4 - 2 + 1 = \mathbf{0}$
- $H(\pi) = \sum (-1)^n h(n) = -1 - 2 + 4 + 4 - 2 - 1 = \mathbf{2}$

An antisymmetric FIR filter always has $H(0) = 0$ (a zero at $z = 1$), so it **cannot** be a lowpass filter. This one passes high frequencies (highpass or bandpass behaviour).

MARKS: (a) ½ for antisymmetric with a reason, ½ for the value. (b) 1 for factoring out $e^{-j2.5\omega}$, 1 for $A(\omega)$ and the $\pi/2$. (c) 1. (d) ½ for the values, ½ for the conclusion.

COMMON MISTAKES

- Choosing symmetric because "most filters are". Check the first and last samples.
- Writing $h(3) = 0$ by copying T9 Q8. That rule is only for odd M.
- Losing the $+\pi/2$. It comes from $e^{jx} - e^{-jx} = 2j\sin x$ and $j = e^{j\pi/2}$.
- Using a group delay of $M/2 = 3$. It is $(M-1)/2$.

REREAD: D12 (linear-phase FIR, symmetric and antisymmetric cases).

Question 11: Short answers [4]

(a) [1] The ideal LPF has $h(n) = \dfrac{\sin \omega_c n}{\pi n}$ (T9 Q1). It is **noncausal** ($h(n) \neq 0$ for $n < 0$), and it is of **infinite duration**, so it cannot be computed with finite memory and operations. (It is also not absolutely summable.) ½ mark per reason.

(b) [2]

- Bin spacing: $F_s/N = 8000/1024 = \mathbf{7.81\ Hz}$.
- Resolution is set by the record length, not by N: $1/T_0 = F_s/L = 8000/64 = \mathbf{125\ Hz}$. This is greater than the 50 Hz spacing, so the tones **cannot be resolved**. Zero padding only interpolates the same smeared spectrum (a simulation shows one peak near 1026 Hz).
- Minimum record: $L \geq F_s/\Delta F = 8000/50 = \mathbf{160}$ samples, so $T_0 = 160/8000 = \mathbf{20\ ms}$.

Marks: ½ for the bin spacing, 1 for "no" with a reason, ½ for L and T_0.

(c) [1] The bilinear map gives $\omega = 2\arctan(\Omega T/2) = 2\arctan(\pi/4) = 1.3316$ rad $= 0.4239\pi$. That corresponds to $f = 0.4239 \times 5000 = \mathbf{2119\ Hz}$, not 2500 Hz. The map $\Omega \to \omega$ is nonlinear (frequency warping): the whole $j\Omega$ axis is squeezed onto $-\pi < \omega < \pi$, and $\Omega = \infty$ maps to $\omega = \pi$. That is why the critical edges must be prewarped with $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$.

COMMON MISTAKES

- Saying zero padding "improves resolution". It improves how finely you sample the DTFT, not your ability to separate tones. The lecture's wording ("better resolution for sampled DTFT") means the finer sampling only.
- Saying the ideal LPF is unstable or has infinite gain. The course's two reasons are causality and infinite length.

REREAD: D6 (frequency analysis with the DFT), D11 (warping), D12 (ideal LPF and windows).

Score table

Score	What it means two days out
50–60	Ready. Use tonight for the checklist only.
40–49	Solid. Redo the one or two questions you dropped most marks on, open book, tomorrow.
30–39	Reread the packs flagged on your weakest three questions. Redo those questions without looking.
below 30	Focus on method, not speed: work through D7–D9 and D11–D12 worked examples, then redo Q6, Q7 and Q9.

Final-evening checklist (for the night before the exam)

Do this in about 45 minutes. Do not start new material.

Pack your bag

- ☐ Printed lecture slides L02–L10, all tutorial solutions T1–T9, your Test 1 Reference, the L05–L10 reference sheet (REFD), and both mock marking guides.
- ☐ A calculator that does complex numbers and `cosh⁻¹`, plus a spare battery. Check it is in **radians** mode.
- ☐ Pens, pencil and ruler (for stem plots, pole-zero plots and sketches).

Tab these pages so you can find them in 10 seconds

- ☐ DTFT property table (L05) and Parseval.
- ☐ DFT properties and overlap-save/overlap-add slides (L06, slides 20–27).
- ☐ z-transform pairs and ROC rules (L07), the unilateral shift formulas (L08 slide 13).
- ☐ Resonator, notch, comb and allpass formulas (L09 slides 17–25).
- ☐ Butterworth and Chebyshev N formulas, prewarp formula, window table (L10 slides 7, 17, 24, 41).

Ten things to say out loud once

1. Correlation peak at lag d gives the delay. Range $r = dc/(2F_s)$.
2. Complex modes $\rho e^{\pm j\theta}$ give $\rho^n[A\cos\theta n + B\sin\theta n]$. Fit A and B from $h(0)$ and $h(1)$.
3. $X(0) = \sum x$, $X(\pi) = \sum(-1)^n x$, $\int X e^{j\omega n_0} = 2\pi x(n_0)$, $\int |X|^2 = 2\pi \sum |x|^2$. Find the origin first.
4. DFT convolution is linear only if $N \geq L + M - 1$. Deconvolution fails where $H(k) = 0$.
5. Overlap-save: overlap the inputs by $M - 1$ and throw away the first $M - 1$ outputs. Overlap-add: split the inputs and add the tails.

6. n pole radii give $n+1$ ROCs. A pole inside the ROC is causal, $p^n u(n)$. A pole outside is anticausal, $-p^n u(-n-1)$. The unit circle in the ROC means a DTFT exists (stable).
7. Unilateral shift: $y(n-1) \to z^{-1}Y^+ + y(-1)$ and $y(n-2) \to z^{-2}Y^+ + z^{-1}y(-1) + y(-2)$.
8. Resonator: $\omega_0 = 2\pi f_0/F_s$, denominator $1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2}$. Allpass zeros are the reciprocals of its poles.
9. Prewarp with $\Omega = 2\tan(\omega/2)$ ($T=1$). Amplitude in dB uses $/20$. Chebyshev N uses $\cosh^{-1}$. Always round N **up**.
10. Linear-phase FIR: compare the end samples to choose symmetric or antisymmetric. Odd-length antisymmetric means the middle sample is 0. Group delay is $(M-1)/2$. Antisymmetric means $H(0) = 0$.

SANITY CHECKS TO USE IN THE EXAM

- ☐ After any partial fraction expansion, put one value of z back in, or check $x(0)$ by long division or recursion.
- ☐ After any LCCDE answer, check $y(0)$ and $y(1)$ by recursion (costs 30 seconds).
- ☐ After any DFT, check that $X(0) = \sum x(n)$ and that $X(N-k) = \overline{X(k)}$ for real x.
- ☐ After any filter design, check that $|H|$ meets the specification at both edges.

EXAM TECHNIQUE

- ☐ Read the whole paper first (5 minutes). Do the questions you are surest of first.
- ☐ If you are stuck on a part, use the “assuming the answer is…” value and move on. Never leave a later part blank.
- ☐ Write the formula before the numbers. The method marks are there even if the arithmetic slips.
- ☐ Sleep. A rested 2 hours beats one more hour of revision tonight.